The uncomputable price of precision

Seth Douglas Nidhal Mghirbi

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Abstract

Which numbers can be the entanglement price of perfection for one fixed finite game? The positive dimension exponents of finite rational Boolean constraint-system games of quantum value one are exactly the positive Δ_2^0 reals. For every such α we construct one game, with no finite-dimensional perfect strategy, whose least local dimension satisfies $\log_2 D_G(\varepsilon) = \alpha \log_2(1/\varepsilon) + o(\log(1/\varepsilon))$. The infimum one-way distillable entanglement of the original mixed strategy state has the same coefficient. This is a realization theorem for usable entanglement; its full spectrum remains open. A finite stop-or-continue program combines finite perfect witnesses from MIP*=RE with a scalar tail verifier for an arbitrary computable convergent rational sequence. Per-gate data are loaded once and held fixed while checked. Matrix-approximable traces give the lower bound; finite threshold histories with sublinear logarithmic witness overhead give matching models. The left-c.e. construction, exact-trace obstruction and non-left-c.e. example of version 2.0.0 become steps toward the classification. In the reset model of the companion heat paper, local-reset heat has coefficient 2α, while joint-reset heat stays bounded: the order of communication and reset changes the asymptotic cost of one device. With recovery before reset, the critical quantum return rate is α in the companion block model. Every positive Δ_2^0 local-heat coefficient is thus realized; the full heat spectrum and the computability of the finite-precision heat function remain open. The version-1.0.0 halting-degree game and its proofs are retained. Meyerovitch's classification of entropy dimensions is the precedent in form; here the classified exponent is a nonlocal-game dimension cost.

The authors developed this work with current frontier AI models from more than one developer and take responsibility for every claim. How the work was done and checked, including refutation-first proof audits and finite computations, is described before the appendices.

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1 The price of perfection

Which numbers can be the entanglement price of perfection for one fixed finite game? The positive dimension exponents of finite rational Boolean constraint-system (BCS) games of quantum value one are exactly the positive Δ_2^0 reals. Every number in that class also occurs as the logarithmic coefficient of one-way distillable entanglement on the original mixed strategy state. The exponent measures how rapidly the least local dimension grows when one fixed device must play closer to perfection. It is the entanglement budget that precision forces on a device whose rules are fixed, so the class of possible exponents decides whether that budget can be read off from the finite rulebook. In general it cannot: a computable finite rulebook can have a noncomputable limiting resource rate. The full usable-entanglement spectrum remains open.

A real is *left-computably-enumerable*, or *left-c.e.*, if it is the limit of a computable nondecreasing sequence of rationals. Increasing approximations below are understood in this non-strict sense. All logarithms without a displayed base are base two, and entanglement is measured in ebits. For a finite game G of quantum value one, let $D_G(\varepsilon)$ be the least local dimension of a strategy with loss at most ε, padding the two local spaces equally. Define

$$d_G^{\rightarrow}(\varepsilon) = \inf_{\text{finite-dimensional strategies of loss } \le \varepsilon} E_D^{A\to B}(\rho_{AB}). \tag{1.1}$$

The state may be mixed. Its one-way distillable entanglement is the asymptotic Bell-pair rate using local operations and classical communication from Alice to Bob. A real is *right-c.e.* if its negative is left-c.e.; it is Δ_2^0 if it is the limit of a computable sequence of rationals, equivalently, by Shoenfield's limit lemma [38], if it is computable relative to the halting set. Let $\mathcal{B}_{\mathrm{BCS}}$ be the set of positive limits $\lim_{\varepsilon\downarrow 0} \log_2 D_G(\varepsilon)/\log_2(1/\varepsilon)$ for finite BCS games with rational question probabilities and quantum value one.

Theorem 1.1 (Complete dimension classification and matched realization)**.** *The positive dimension spectrum is*

$$\mathcal{B}_{\mathrm{BCS}} = \{\alpha > 0 : \alpha \text{ is } \Delta_2^0\}.$$

For every positive Δ_2^0 real α there is a fixed finite two-player Boolean constraint-system game G_α, with rational question probabilities, quantum value one and no finite-dimensional perfect strategy, such that, as $\varepsilon \downarrow 0$,

$$\log_2 D_{G_\alpha}(\varepsilon) = \alpha \log_2(1/\varepsilon) + o(\log(1/\varepsilon)), \tag{1.2}$$

$$d_{G_\alpha}^{\rightarrow}(\varepsilon) = \alpha \log_2(1/\varepsilon) + o(\log(1/\varepsilon)). \tag{1.3}$$

Conversely, every finite dimension exponent of a rational finite game is Δ_2^0 whenever its defining limit exists. The usable-entanglement assertion is an inclusion, not a classification of its full spectrum.

Section 16 proves the realization; Lemma 2.4, with the limit argument in the proof of Corollary 16.7, gives the converse (compare [12, Proposition 11.13]). Corollary 13.1 extends the order contrast to every positive Δ_2^0 coefficient: local-reset heat has coefficient 2α, while joint-reset heat stays bounded. This identifies limiting coefficients, without asserting computability or noncomputability of the finite-precision heat function.

1.1 The flagship halting game

Fix a standard effective enumeration of programs, with halting set $K \subseteq \mathbb{N} = \{0, 1, \ldots\}$.

Theorem 1.2 (The version 1.0.0 game)**.** *There is a fixed finite two-player Boolean constraint-system game G_K, with rational question probabilities, such that $\omega_q(G_K) = 1$, no finite-dimensional strategy wins perfectly, and*

$$D_{G_K}(\varepsilon) = \varepsilon^{-(3+\theta_K)/4+o(1)}, \qquad \theta_K = \sum_{e \in K} 4^{-(e+1)}. \tag{1.4}$$

The exponent is Turing-equivalent to K. The sequence $\log D_{G_K}(2^{-n})/n$ has no computable certified rate of convergence.

This is the original explicit degree example. The existence of increasing computable rational sequences with uncomputable limits is classical, beginning with Specker [43]; the construction places such a limit inside the resource law of a single finite game. Sections 2–10 retain the construction and proof of [11]. Sections 11–13 retain the extensions of version 2.0.0 [13]: the left-c.e. reader, extraction and heat; Sections 13.1 and 13.2 are new. Section 14 explains the exact-trace barrier and, new in this version, classifies matrix-approximable projection minima (Theorem 14.5); Section 15 gives the non-left-c.e. example of version 2.0.0. Section 16 completes the dimension classification, and Section 17 states what remains open.

1.2 Dependency map

The finite presentations, stop-or-continue proof, simultaneous upper models and matched resource estimates are proved here. The Boolean correspondence is used as stated from [12, Theorem 4.9], and the actual-state interface from [12, Proposition 6.1]; Section 12 records how they feed an explicit extraction on the original mixed state. The external inputs to Theorem 1.1 are the finite perfect-completeness reduction of MIP* = RE [20, Theorem 12.6], Holevo's bound [18] in the measurement count, and Devetak–Winter hashing [9]. Classical results used without proof are counter-machine universality [37], real quantifier elimination and Shoenfield's limit lemma [38]. The bounded tracial moment method [2, 23] enters only the exact-trace barrier of Section 14 and its uses in Proposition 15.9 and Section 16.6. The normal form [14, Theorem 5.1] enters only Theorem 14.5. The real simulation [27] and [12, Lemma 11.12] enter only the remark on real strategies and question weights. The heat paper [10] is needed only for Section 13 and the heat consequences. These dependencies are substantive: this is not a self-contained proof of the imported correspondences or complexity theorem.

1.3 Related work and credit

Mousavi, Nezhadi and Yuen prove that deciding whether the quantum value is exactly one is Π^0_2-complete [31, 32]. In particular, there is no total computable dimension bound, uniform over all value-one games and all precisions. Indeed, such a bound would make value one a universal integer quantification of decidable bounded-dimensional feasibility statements, placing that problem in Π^0_1. Their obstruction is uniform over games. Theorems 1.1 and 1.2 instead identify convergent coefficients for individual fixed games. The theorem MIP* = RE puts halting information in the value of a computable family of games [20]; it does not by itself identify a resource exponent of a fixed value-one game.

The algebraic connection between constraint games and operator assignments goes back to Cleve and Mittal [5] and, for linear systems, Cleve, Liu and Slofstra [4]. Slofstra established

an embedding theorem for solution groups [41] and the non-closure of quantum correlations [40]. Slofstra and Vidick introduced the hyperlinear profile as a quantitative bridge from approximate representations to dimension requirements [42]. Slofstra later obtained a finitely presented group with at least subexponential hyperlinear profile [39]. Mančinska and Vidick give a two-player game with two questions per player, but answers of unbounded length, whose optimal value one is reached only in the limit: some local-dimension-d strategies succeed with probability $1 - O(d^{-c})$, and none exceeds $1 - \Omega(d^{-2})$ [26]. Ji, Leung and Vidick construct a fixed finite three-player game of value one for which near-perfect play requires at least $\Omega(\varepsilon^{-c})$ qubits for a fixed $c > 0$ [19]. Coladangelo constructs a fixed finite two-player score game for which every ε-optimal strategy has local dimension $2^{\Omega(\varepsilon^{-1/8})}$ [6]. These results supply the framework and important growth examples; they do not identify the matched finite exponent in (1.4).

Mehta, Slofstra and Zhao construct halting-dependent Pauli systems and embed them in finite presentations with polynomial bounds on relation decompositions [28, Proposition 4.8]. Those bounds transfer lower estimates; they do not assert small simultaneous finite-model extensions. The construction here supplies its own upper models. The conjugacy gadgets used in our compiler are those of Fanizza, Kroell, Mehta, Paddock, Rochette, Slofstra and Zhao [15, Lemma 5.8 and Theorem 5.5]. Their credit is essential: the passage from nested conjugacy to ordinary Boolean contexts is not a new gadget of this paper. We give its matrices and converse calculation because quantitative completeness is needed here.

The state-preserving Boolean correspondence was established in *The price of precision* [12, Theorem 4.9]; its linear-system predecessor and further dimension examples appear in the companion paper [30]. We use that correspondence as an existing theorem. Section 10 records the elementary square-root estimate needed for the present local-dimension application and checks the constants' effect on the exponent. Theorem 1.1 proves and strengthens the computable-coefficient inclusion conjectured in the discussion of [12, Problem 18.3(i)], for its original usable-entanglement demand as well as dimension. The Δ_2^0 upper bound for dimension exponents is [12, Proposition 11.13]; Lemma 2.4 records the rational case.

Uncomputable asymptotic data in finite systems have other precedents. Hochman and Meyerovitch characterize the entropies of multidimensional shifts of finite type as exactly the nonnegative right-c.e. reals [17]; their construction passes through a shift whose maximal symbol frequency is a prescribed right-c.e. real. Their theorem is a template for identifying the possible values of a physical quantity of a fixed finite system with a computability class. The exact precedent in form is Meyerovitch's entropy-dimension classification [29, Theorem 5.1(3), arXiv version], for every $d \geq 2$:

> The class of entropy dimensions of $\mathbb{Z}^d$ SFT's is $[0, d] \cap \Delta_2$.

The same theorem gives upper entropy dimensions Π_3 and lower entropy dimensions Σ_2. Our classification concerns nonlocal-game dimension exponents and matched entanglement coefficients. It is not the first Δ_2^0 classification of an exponent. Sapir, Birget and Rips relate group isoperimetry and computation [36]; a Dehn exponent is not a dimension exponent, and no identification between them is used here. Cubitt, Perez-Garcia and Wolf prove undecidability of the spectral gap over families of local Hamiltonians [8]. Bausch, Cubitt and Watson obtain uncomputable phase diagrams for a continuous one-parameter family [1]. These are precedents for asymptotic noncomputability, rather than proofs of (1.4).

The binary adding machine used below is the classical odometer on a Cantor set; its interpretation through operator algebras is standard [16]. The new point needed here is a finite polynomial presentation and a carry estimate that remain quantitative in a non-flat state. The final information bound uses Holevo's theorem [18] and the random-access-code principle exemplified by Nayak [34]. Its branching estimate is proved explicitly.

Jockusch and Schupp prove that the densities of computable sets are exactly the Δ^0_2 reals in $[0,1]$ [21, Theorem 2.21], and Downey, Jockusch and Schupp classify the densities, lower densities and upper densities of computable and c.e. sets [14, Section 5]. The flag densities forced in Section 16 realize the same arithmetic class; what is new here is forcing such a density on every near-perfect strategy of one fixed game. The moving-window estimate in Section 11 uses the binary radical-inverse ordering of van der Corput [44], not a new discrepancy principle. Vidick's *Almost synchronous quantum correlations* [45] supplies the closely related spectral-compression mechanism; Section 14 gives the word-moment form needed here. Tracial moment optimization [2, 23] supplies the exact-trace lower hierarchy. Its distinction from general commuting-model NPA [33] is the distinction used to escape the barrier in Section 15.

Devetak–Winter hashing [9] is the operational last step in Section 12. The new ingredient is the inactive-outcome extension of the triangular extraction in [12, Theorem 6.2] and its cyclic finite-alphabet form [12, Lemma 6.3], with random targets and an original-mixed-state bridge. The heat conclusion invokes the existing reset sandwich of [10, Theorem 2.1 and Corollary 2.2]. Kolchinsky and Wolpert [24] study noncomputable heat functions for universal Turing machines. Corollary 13.1 and Theorem 15.7 identify uncomputable logarithmic precision coefficients for fixed finite nonlocal games and compare their local and joint reset schedules. The halting-degree dimension construction, its compiler, clocks and proofs appeared in version 1.0.0 [11]; they are retained here. The version-2 realization, extraction and consequences [13] are retained as steps toward the complete construction. Section 15 additionally uses the finite perfect-completeness form of Ji–Natarajan–Vidick–Wright–Yuen, states its exact theorem numbers, and credits the Connes embedding and commuting-complexity context, including Lin [25].

1.4 Proof outline

At source precision 4^{-N}, three compulsory clocks cost $3N-o(N)$ bits of logarithmic dimension. They hold an address, fuel and time. For G_K, a finite counter program recognizes halting inputs on separated base-four cylinders. Reset forces initialized computations at every finite fuel scale. Exact tracial models must flag every halting cylinder; compactness extends this to all weighted approximate models, without an effective modulus.

A conditional clock contributes $\theta_K N - o(N)$ further bits. Explicit finite histories are diagonal labels on the existing registers. A moving register of optional bits realizes the conditional clock in dimension $2^{\theta_N N+O(\log N)}$, with computable $\theta_N \uparrow \theta_K$. Source precision 4^{-N} becomes game loss of order 16^{-N}, giving the denominator four.

For an arbitrary left-c.e. $\theta \in [0,1]$, an address reader accepts exactly the binary values below θ. Radical-inverse window counts match its finite flag density. Extra compulsory clocks add to the numerator, and replacing polynomial residuals by fixed powers divides the exponent. These two operations cover every positive left-c.e. α. Finally a coherent extraction with an inactive outcome converts the expected active count into distillable Bell pairs on the original mixed state. The cited heat sandwich then matches both reset laws.

The complementary construction reverses the flag rule: an unflagged decoded cylinder must carry a finite perfect witness for its halting game. A fixed counter transducer with two operator registers tests every relation of that witness's projection presentation, packed into three data generators. Nonhalting unflagged corners have zero weight in matrix-approximable traces, and moment compression transfers that bound to the actual strategy state. Finite perfect witnesses can be admitted so slowly that their combined logarithmic dimension is at most $\sqrt{N}$, while the unflagged density tends to θ_K. The flag density is therefore $1-\theta_K$, giving exponent $(4-\theta_K)/4$. The same extraction and heat sandwich apply. The exact-trace minimum is strictly smaller, as the barrier forces it to be.

The stop-or-continue construction replaces monotone admission by a stopping parameter R_*. Each continuation has one loaded finite perfect witness; each stop runs a scalar verifier against the entire remaining approximation. A bad address can neither stop safely nor continue forever. For the upper bound, any fixed threshold below the limit needs only finitely many witnesses. Letting the threshold advance sufficiently slowly makes their logarithmic overhead sublinear. This handles every computable convergent approximation, without a variation bound. Within this mechanism, even though both quantum and commuting values equal one, the exact-versus-matrix-approximable trace gap associated with the failure of Connes' embedding conjecture changes the attainable exponent class beyond the exact-trace left-c.e. barrier.

The construction at a glance

The following pipeline separates the finite rulebook from the two asymptotic estimates. The imported Boolean correspondence is the final conversion, not a new complexity reduction.

Stage	Finite construction	Role in the estimates
Counter program	A program computes $a_s \to \theta$. At each depth it stops and verifies the tail, or loads one witness and continues.	Bad prefixes force flag mass at least $1 - \theta$ in matrix-approximable traces.
Polynomial presentation	Three compulsory clocks, one conditional clock, R_*, commutative workspace, per-gate data and two contraction registers.	Positive initialized corners test one tuple; fixed thresholds give exact new relations with a fixed finite witness factor.
Parameters	Add r clocks and take the qth power of every defining residual.	The resource coefficient is $(4 + r - \theta)/(4q)$; every positive Δ_2^0 real has this form.
Compiler	Rational bounded relations pass through conjugacy gadgets to finitely many Boolean contexts.	Upper residual $O(4^{-qN})$ gives loss $O(16^{-qN})$; loss ε gives unpowered precision $(C\sqrt{\varepsilon})^{1/q}$.
BCS game	A fixed finite rational question distribution tests those contexts.	Explicit flat models have $\log d = (4 + r - \theta)N + o(N)$; no game changes as precision improves.
Upper flow	Threshold $b < \theta \to$ finitely many witnesses $\to$ initialized histories $\to$ slowly advancing $b \uparrow \theta$.	Witness logarithmic cost is $o(N)$; window counts fix the optional-register cost.
Lower flow	Near-perfect strategy $\to$ actual-state interface $\to$ weighted flag bound $\to$ raw clock measurements.	Counting bounds dimension; coherent extraction and hashing bound one-way entanglement on the original mixed state.
Heat consequence	Apply the existing reset sandwich only after both bounds match.	Local reset costs $2\alpha \log_2(1/\varepsilon) + o(\log(1/\varepsilon))$; joint reset stays bounded.

2 Weighted models and elementary estimates

Definition 2.1. A bounded polynomial presentation consists of finitely many variables, each declared to be a positive contraction, a projection, a self-adjoint unitary, or a unitary, and finitely many rational real-coefficient noncommutative $*$-polynomial equations $r = 0$. Its weighted precision-δ model on $\mathbb{C}^d$ is an assignment in the declared domains and a density matrix ρ such that, with $\xi = \sqrt{\rho}$,

$$\|r\xi\|_2 \le \delta, \qquad \|[s, \xi]\|_2 \le \delta \tag{2.1}$$

for each defining relation r and each variable s. The norm is the unnormalized Hilbert–Schmidt norm. A flat model has $\rho = I/d$ and $\|r\|_f = d^{-1/2}\|r\|_2$. Write $W(\delta)$ and $H(\delta)$ for the least weighted and flat model dimensions, with value ∞ if no model exists.

We write $|D|_\rho = \|D\xi\|_2$. Constants for a fixed presentation may depend on its finite polynomials, but never on d, ρ, δ or a growing digit depth.

Lemma 2.2 (Right multiplication)**.** *For bounded A, B,*

$$|AB|_\rho \le \|A\||B|_\rho, \qquad |BA|_\rho \le \|A\||B|_\rho + \|B\|\|[A, \xi]\|_2.$$

A word w of length l in contraction variables and unitary adjoints satisfies $\|[w, \xi]\|_2 \le l\delta$. A fixed finite algebraic derivation from the defining relations has error at most $C\delta$ on the same state.

Proof. Use $BA\xi = B\xi A + B[A, \xi]$. Expand the commutator of a product one factor at a time. Adjoint unitaries have the same state-commutator norm as the unitary. For adjoint relations,

$$|r^*|_\rho = \|\xi r\|_2 \le |r|_\rho + \|[r, \xi]\|_2.$$

All variables are bounded, so a fixed polynomial has a fixed operator-norm bound and a commutator bound $C\delta$. Expand a finite ideal identity as a finite sum $a_j r_j b_j$ and apply the first two inequalities. This proves the last assertion. □

Lemma 2.3 (Hilbert–Schmidt functional calculus)**.** *Let A, B be self-adjoint, Z an arbitrary rectangular Hilbert–Schmidt matrix, and g an L-Lipschitz function on their spectra. Then*

$$\|g(A)Z - Zg(B)\|_2 \le L\|AZ - ZB\|_2, \tag{2.2}$$

$$|g(A) - g(B)|_\rho \le L\big(|A - B|_\rho + 2\|[B, \xi]\|_2\big), \tag{2.3}$$

$$|[Y, g(A)]|_\rho \le L\big(|[Y, A]|_\rho + 2\|[A, \xi]\|_2\big) \tag{2.4}$$

in the last line for a contraction Y. Also $\|[g(A), \xi]\|_2 \le L\|[A, \xi]\|_2$.

Proof. Diagonalize A on the left and B on the right. Corresponding entries in the first inequality are $(g(a) - g(b))Z_{ab}$ and $(a - b)Z_{ab}$. Square the scalar Lipschitz inequality and sum. The state-commutator estimate is its case $A = B, Z = \xi$. Insert and subtract $\xi g(B)$ to obtain (2.3). For (2.4), apply (2.2) with $A = B, Z = Y\xi$; then insert and subtract $Y\xi g(A)$. The two resulting state-commutator terms are each bounded by $L\|[A, \xi]\|_2$. These are Schatten-two estimates, not operator-norm Lipschitz statements. □

Lemma 2.4 (Finite-precision computability)**.** *For a rational-data game promised to have quantum value one, $D_G(\varepsilon)$ is computable at every positive rational ε. If $a_n = \log D_G(2^{-n})/n$ converges to β and $|a_n - \beta| \le r(n)$ eventually for a computable $r(n) \to 0$, then β is computable.*

Proof. In a fixed local dimension d, introduce real variables for the real and imaginary matrix entries of the state and both families of POVMs. Positivity is a finite list of principal-minor inequalities. Normalization and winning probability at least $1-\varepsilon$ are polynomial conditions with rational coefficients. Real quantifier elimination decides feasibility. Test $d = 1, 2, \ldots$; the value-one promise ensures termination. For the second assertion, hardcode a threshold beyond which the error bound holds. Given k, search above it for n with $r(n) < 2^{-k-1}$ and compute a_n within 2^{-k-1}. This gives a 2^{-k} approximation to β. □

3 From bounded polynomials to Boolean constraints

A BCS specifies finitely many Boolean variables and finitely many contexts. A context lists variables and its allowed sign assignments. In an operator model the context variables are commuting reflections and their joint spectrum is restricted to those assignments. Its game asks Alice for a valid context assignment and Bob for one sign, testing consistency on an incidence. We use the uniform distribution on the finite incidence set, hence rational probabilities. The algebraic interpretation is standard [5, 4].

Theorem 3.1 (Polynomial compiler)**.** *For each finite bounded polynomial presentation $\mathcal{P}$, there are a finite ordinary BCS $\mathcal{B}$ and constants $c, C > 0$ with the following properties.*

(a) Every weighted precision-η model of $\mathcal{B}$ reconstructs a precision-$C\eta$ model of $\mathcal{P}$ on the same space and with the same state.

(b) Every flat precision-δ model of $\mathcal{P}$ on $\mathbb{C}^d$ gives a strategy for its context-variable game in local dimension at most cd, with loss at most $C\delta^2$.

All constants depend only on the presentation. In particular, guarded equations such as $(V-W)(I-A)=0$ are compiled with their guard intact.

3.1 Domains and selectors

Every positive contraction is

$$r = \frac{2I + u + u^*}{4}, \qquad u = 2r - I + 2i\sqrt{r(I-r)}. \tag{3.1}$$

The real and imaginary parts in this expression commute and their squared sum is I, so u is unitary. Conversely the first formula is a positive contraction for any unitary. For upper models realify the whole assignment by

$$\mathcal{R}(M) = \begin{pmatrix} \Re M & -\Im M \\ \Im M & \Re M \end{pmatrix}.$$

This doubles dimension, preserves real-coefficient equations and flat norms, and takes unitaries to real orthogonal matrices. A real orthogonal matrix is a product of two symmetric orthogonal involutions: on a rotation plane use

$$\begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cos t & -\sin t \\ -\sin t & -\cos t \end{pmatrix},$$

and use scalar blocks on the remaining eigenspaces. Orthogonal conjugation and direct sums finish the factorization. Replace each unitary by a product of two reflections. Lower reconstructions are automatically unitary, and (3.1) is automatically positive. Their state

commutators cost a fixed factor. No commutation between the two factors is imposed. The upper models are flat, so no state estimate for the square root in (3.1) is needed.

A source equation is now $p = \sum_j a_j W_j = 0$, with rational a_j and reflection words W_j. Clear denominators with an integer c, duplicate signed terms, and add cancelling pairs until

$$\sum_{j=1}^{L} \widetilde{W}_j = 2cp, \qquad L = 2^q.$$

Introduce q Pauli pairs X_i, Z_i, commuting across pairs and with every source reflection. Let e_j be the joint Z spectral projections and E the joint $+1$ projection of the X_i. Then

$$Ee_jE = L^{-1}E, \qquad \sum_{v\in\{0,1\}^q} Z^v E Z^v = I. \tag{3.2}$$

For the first identity, expand e_j in Z monomials. Every nonconstant monomial anticommutes with some X_i and hence has zero compression by E. The second identity sums the joint X spectral projections.

The controlled reflection $(I - e_j) + e_j s$ is a Boolean truth table in commuting variables. Products of these controls and controlled signs form a reflection word whose value is the unitary

$$S = \sum_j e_j \widetilde{W}_j.$$

The projection E receives its own Boolean variable x_E. Its defining context is $\{X_1, \dots, X_q, x_E\}$, whose allowed assignments are those with $x_E = +1$ exactly when $X_1 = \cdots = X_q = +1$. In an operator model this truth table says $(I + x_E)/2 = \prod_i (I + X_i)/2 = E$. Introduce $x_F = S x_E S^*$ by successive reflection conjugacies and put $F = (I + x_F)/2 = SES^*$. Orthogonality is imposed on this pair alone: the context $\{x_E, x_F\}$ forbids exactly the assignment $(+1, +1)$, which in an operator model is the condition $EF = 0$. The exact identities are

$$EFS = ESE = \frac{2c}{L} pE, \qquad p = \frac{L}{2c} \sum_v Z^v EFS Z^v. \tag{3.3}$$

They show why a low-weight selector corner cannot conceal an equation. All of its translates are used.

3.2 The conjugacy gadgets

We use the gadgets of [15, Lemma 5.8]. Their formulas are included to establish the precise finite-dimensional extension needed here. For a general relation $x_1 x_2 x_1 = x_3$, introduce commuting parity contexts

$$x_i y_i z_i = I, \qquad x_i f w_i = I, \qquad g y_2 z_3 = I,$$

and the conjugacies $f y_i f = z_i$ and $w_1 y_2 w_1 = z_3$. The targets of these remaining conjugacies commute by the parity contexts. If the original reflections are X_i, an exact doubled extension is

$$x_i = \operatorname{diag}(X_i, X_i), \quad y_i = \operatorname{diag}(X_i, I), \quad z_i = \operatorname{diag}(I, X_i),$$
$$f = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}, \quad w_i = \begin{pmatrix} 0 & X_i \\ X_i & 0 \end{pmatrix}, \quad g = \operatorname{diag}(X_2, X_3).$$

Conversely $w_1 = x_1 f$, $[f, x_1] = 0$, and $f y_2 f = z_2$ give $x_1 z_2 x_1 = z_3$. Conjugation by f gives $x_1 y_2 x_1 = y_3$. Multiplying gives the required conjugacy.

For $aba = c$ with commuting targets b, c, use six commuting parity contexts:

$$ad_1d_2 = bd_2d_3 = d_3d_4d_5 = ad_5d_6 = cd_6d_7 = d_1d_4d_7 = I. \tag{3.4}$$

They imply

$$aba = d_1d_2d_2d_3d_5d_6 = d_1d_4d_6 = d_7d_6 = c.$$

Writing $M(a) = \left(\begin{smallmatrix} 0 & a \\ a & 0 \end{smallmatrix}\right)$, an exact extension, with old variables repeated diagonally, is

$$d_1 = M(a), \quad d_2 = M(I), \quad d_3 = M(b), \quad d_4 = \begin{pmatrix} 0 & ba \\ ab & 0 \end{pmatrix},$$
$$d_5 = \operatorname{diag}(bab, a), \quad d_6 = \operatorname{diag}(bc, I), \quad d_7 = \operatorname{diag}(b, c).$$

For clarity, $d_4^* = d_4$ and $d_4^2 = I$. The equality $aba = c$ implies that a commutes with bc, while bc is a reflection. These facts verify the fourth context; the other five follow by direct multiplication. Each listed triple consists of commuting reflections with product I. Sequentially applying the two gadgets has constant dimension cost for a fixed presentation. Pauli anticommutation is $XZX = -Z$, with a Boolean variable for the negation of Z.

3.3 Weighted soundness and flat completeness

Proof of Theorem 3.1. All identities above are finite algebraic calculations in the defining relations. A Boolean truth table gives finite algebraic relations: expand in its assignment projections, and any polynomial vanishing on allowed assignments is the finite sum of its values on forbidden assignments times their projections. Thus the controlled-variable definitions, both gadget converses and (3.3) are finite ideal derivations. In an approximate model put $E_{\mathrm{pol}} = \prod_i (I + X_i)/2$, $E_0 = (I + x_E)/2$ and $F_0 = (I + x_F)/2$. The defining AND context of x_E and the conjugacy gadgets give $|E_{\mathrm{pol}} - E_0|_\rho = O(\eta)$ and $|SE_{\mathrm{pol}}S^* - F_0|_\rho = O(\eta)$, while the orthogonality context gives $|E_0F_0|_\rho = O(\eta)$. Interpret (3.3) as a polynomial identity modulo the Pauli and controlled-variable relations. Substitution of these estimates, using Lemma 2.2, gives $|p|_\rho = O(\eta)$. Only E_0 and F_0 are automatically projections here; the selector polynomial need not be a projection in an approximate model. Lemma 2.2 proves linear weighted soundness. This statement is stronger than mere exact implication; the explicit finite derivations are what justify the linear error.

For upper models choose the selectors on an exact tensor factor and define every controlled variable and successive conjugate projection canonically. Put $x_E = 2E - I$ with $E = \prod_i (I + X_i)/2$. It commutes with every X_i, and the joint spectrum of $(X_1, \dots, X_q, x_E)$ lies in the allowed assignments, so the defining context of x_E is exact. The conjugacies defining x_F are exact even when p is nonzero. All gadget contexts extend exactly. The only inexact constraint is the final orthogonality, where

$$\|EF\|_f = \|ESE\|_f = \frac{2c}{L\sqrt{L}}\|p\|_f. \tag{3.5}$$

Let F' be the projection onto $(I - E)\operatorname{ran} F$. Then

$$EF' = 0, \qquad \|F - F'\|_f^2 \le 2\|EF\|_f^2. \tag{3.6}$$

To prove this, diagonalize FEF on $\operatorname{ran} F$ with eigenvalues λ_i and orthonormal eigenvectors v_i. For $\lambda_i < 1$, the vectors $(I - E)v_i/\sqrt{1 - \lambda_i}$ are orthonormal and span the new range. Those with $\lambda_i = 1$ disappear. Hence

$$\|F - F'\|_2^2 = \#\{i : \lambda_i = 1\} + 2\sum_{\lambda_i < 1} \lambda_i \le 2\sum_i \lambda_i = 2\|EF\|_2^2.$$

On the orthogonality context $\{x_E, x_F\}$, Alice answers with the PVM

$$E \mapsto (+1,-1), \qquad F' \mapsto (-1,+1), \qquad I-E-F' \mapsto (-1,-1).$$

These are orthogonal projections summing to I, because $F' \le I-E$, and every outcome is an allowed assignment. Her operator for x_E is therefore $2E-I$ and her operator for x_F is $2F'-I$. Bob uses $2E-I$ and $2F-I$. On a maximally entangled state there is no disagreement for x_E, and the disagreement probability for x_F is $\|F-F'\|_f^2$. This is why E carries its own variable: if the context listed $X_1,\dots,X_q$ together with x_F, the PVM above would not be a legal answer, since F' commutes with E but in general not with each X_i, and so does not fix their values. All other contexts, including the defining context of x_E, are exact. With incidence probability π, summing over the finitely many source equations and using (3.6) and (3.5) bounds the loss by

$$2\pi \sum \|EF\|_f^2 = 8\pi \sum \frac{c^2}{L^3}\|p\|_f^2 \le C\delta^2.$$

Here c and L are the integers of each source equation. The new variable adds no tensor factor. Realification, selectors and gadget extensions multiply the dimension by the fixed factor $2^{1+q_{\mathrm{tot}}+g}$, where q_{tot} is the total number of selector qubits and g the number of gadget doublings; this is the dimension constant in (b). This proves both assertions. □

4 A finite Cantor clock

The clock has a positive contraction A, a projection P, a unitary T and a self-adjoint unitary X, with relations

$$[A,P]=0, \quad TAT^* = 4A-3P, \quad XPX = I-P, \quad [X,TAT^*]=0. \tag{4.1}$$

For $j \in \mathbb{N}$ put $P_j = T^j P T^{-j}$ and $X_j = T^j X T^{-j}$. The Cantor set used here is

$$\mathcal{K} = \left\{ \sum_{j\ge 0} 3b_j 4^{-j-1} : b_j \in \{0,1\} \right\}.$$

Let p be its first-digit function, extended to $[0,1]$ by

$$p(a) = \begin{cases} 0 & a \le 1/4, \\ 2a-1/2 & 1/4 \le a \le 3/4, \\ 1 & a \ge 3/4, \end{cases} \qquad s(a) = 4a - 3p(a).$$

These functions are respectively 2- and 4-Lipschitz and map $[0,1]$ into itself.

Lemma 4.1 (Exact digits)**.** *In an exact model, A has spectrum in $\mathcal{K}$, all P_j are commuting spectral functions of A, and*

$$A = \sum_{j<M} 3\cdot 4^{-j-1} P_j + 4^{-M} T^M A T^{-M}. \tag{4.2}$$

The flip X_j exchanges P_j and $I-P_j$ and preserves all P_l with $l>j$. In every tracial state,

$$\tau\left(\prod_{j<M} P_j^{b_j} (I-P_j)^{1-b_j} \right) = 2^{-M}. \tag{4.3}$$

The spectral projection of A at 1 is zero in the faithful tracial representation.

Proof. Since A, P commute and $0 \le 4A - 3P \le I$, the $I - P$ block of A has spectrum in $[0, 1/4]$ and the P block in $[3/4, 1]$. Hence $P = p(A)$ and $TAT^* = s(A)$. Conjugating this statement repeatedly proves $P_j = p(s^j(A))$ and (4.2). Its remainder has norm at most 4^{-M}, which places the spectrum in $\mathcal{K}$. The tail coordinate TAT^* determines every later digit. Its commutation with X, and $XPX = I - P$, give the flip assertions, then their translates. A trace assigns equal mass to a cylinder and its earliest-bit flip. Add the two and repeat on the remaining suffix to obtain (4.3). The spectral projection at 1 lies under every all-ones cylinder and has trace at most 2^{-M} for every M. Faithfulness makes it zero. □

Lemma 4.2 (Linear spectral repair)**.** *There is an absolute constant C such that any weighted precision-δ clock has a self-adjoint $\widehat{A}$ with spectrum in $\mathcal{K}$ satisfying*

$$|\widehat{A} - A|_\rho + \|[\widehat{A}, \xi]\|_2 \le C\delta, \tag{4.4}$$

$$|P_j - p(s^j(\widehat{A}))|_\rho \le C4^j\delta. \tag{4.5}$$

The first M raw digit projections and triangular flips satisfy digit–digit commutations, own-bit flips, later-digit preservation and state commutations with error at most $C(M+1)^3 4^M\delta$. The spectral digit projections $p(s^j(\widehat{A}))$ commute exactly and differ from the raw ones by (4.5).

Proof. Set $B_0 = TAT^*$, $D_0 = PB_0P + (I-P)B_0(I-P)$ and $C_0 = (D_0 + 3P)/4$. The affine relation and $[A, P]$ give $|[B_0, P]|_\rho = O(\delta)$ by Lemma 2.2. Since $B_0 - D_0 = (2P - I)[P, B_0]$, it follows that $|D_0 - B_0|_\rho = O(\delta)$ and $|C_0 - A|_\rho = O(\delta)$. Their state commutators are $O(\delta)$ from their fixed polynomial definitions. All are positive contractions, and D_0 commutes with P.

Let $d_\mathcal{K}$ be distance to $\mathcal{K}$, a 1-Lipschitz function. Cantor self-similarity on the two P blocks gives $d_\mathcal{K}(C_0) = d_\mathcal{K}(D_0)/4$. Lemma 2.3 and unitary transport give

$$|d_\mathcal{K}(A)|_\rho \le \tfrac{1}{4}|d_\mathcal{K}(A)|_\rho + O(\delta).$$

Thus $|d_\mathcal{K}(A)|_\rho = O(\delta)$. Round each spectral value of A to a nearest point of $\mathcal{K}$. The resulting $\widehat{A}$ has $|\widehat{A} - A|_\rho = O(\delta)$. For any self-adjoint E, $\|[E, \xi]\|_2 \le 2|E|_\rho$; apply this to $E = \widehat{A} - A$ to prove (4.4). No continuity of the nearest-point map is needed.

Since $p(C_0) = P$, functional calculus gives $|P - p(\widehat{A})|_\rho = O(\delta)$, and then the affine relation gives $|T\widehat{A}T^* - s(\widehat{A})|_\rho = O(\delta)$. For $g = p \circ s^r$, the one-step error

$$|Tg(\widehat{A})T^* - g(s(\widehat{A}))|_\rho \le C4^r\delta$$

follows from (2.3). Telescope these identities to depth j. The sum is $O(4^j\delta)$. Every digit function has norm at most one, so transporting the errors through the outer powers of T adds only $O(j^2\delta)$, absorbed by $O(4^j\delta)$. This proves (4.5).

For the flips, translate $XPX = I - P$ and replace each P_j using (4.5). For $l > j$, use the tail coordinate $T^{j+1}AT^{-(j+1)}$. The translated last relation of (4.1) makes its commutator with X_j at most $C(j+1)\delta$ in weighted norm. Its digit at position $l - j - 1$ is a function with Lipschitz constant at most $2 \cdot 4^{l-j-1}$. Apply (2.4), replace the digit back, and use Lemma 2.2. The error is at most $C(M+1)^3 4^M\delta$. Raw digit commutators follow by replacing both with commuting functions of $\widehat{A}$; the two replacement costs add. Finally $\|[P_j, \xi]\|_2, \|[X_j, \xi]\|_2 \le (2j+1)\delta$. This proves all assertions. □

Remark 4.3. The spectral repair and digit estimate use only positivity, the affine equation, $[A, P] = 0$, and state commutators. They apply without a flip. This will be used for the conditional coordinates.

Theorem 4.4 (Clock profile)**.** *For the presentation (4.1),*

$$\log W(2^{-n}) = \frac{n}{2} + O(\log(n+2)), \qquad \log H(2^{-n}) = \frac{n}{2} + O(\log(n+2)).$$

There are flat models on N qubits of precision $3 \cdot 4^{-N}$.

Proof. On N qubits let P_j read bit j, let T rotate contents from slot j to slot $j+1$ cyclically, and let X flip bit zero. Put

$$A_N = \sum_{j=0}^{N-1} 3 \cdot 4^{-j-1} P_j. \tag{4.6}$$

Then $[A_N, P_0] = 0$, $XP_0X = I - P_0$, and

$$TA_NT^* - 4A_N + 3P_0 = 3 \cdot 4^{-N} P_0, \qquad [X, TA_NT^*] = 3 \cdot 4^{-N}[X, P_0].$$

All state commutators vanish in the flat state, proving the upper bound. For the lower bound, use M digits and flips from Lemma 4.2 in the binary case of the counting lemma, Theorem 8.1, proved below. Choose $M = \lfloor \log_4(1/\delta) - 20 \log_4(2 + \log_4(1/\delta)) \rfloor$. Its error terms tend to zero, so $\log d \geq M - o(1)$. At $\delta = 2^{-n}$ this is $n/2 - O(\log(n+2))$. Since $W \leq H$, both conclusions follow. This forward reference uses no property of the program constructed later. □

4.1 Guarded increment

Add a unitary V and the relation

$$(V - X[(I - P) + (TVT^*)P])(I - A) = 0. \tag{4.7}$$

Proposition 4.5. *The finite clocks have increment models of dimension 2^N whose guarded defect has operator norm $\sqrt{2}\, 4^{-N}$ and flat norm $\sqrt{2}\, 4^{-N} 2^{-N/2}$. In an exact faithful tracial model, V increments every finite binary digit prefix.*

Proof. Let V_N add one modulo 2^N, with bit zero least significant. If that bit is zero, the right-hand side before the guard is the correct first-bit flip. If it is one and some later bit is zero, TV_NT^* carries through slots $1, 2, \ldots$ to the first zero and X then clears slot zero. Only the all-ones input fails:

$$R_N := V_N - X[(I - P_0) + (TV_NT^*)P_0] = (|0^N\rangle - |10^{N-1}\rangle)\langle 1^N|.$$

The guard equals 4^{-N} there. The two norm statements follow from the rank-one formula. For exact tracial models, Lemma 4.1 makes $I - A$ have dense range, so its bounded left factor in (4.7) vanishes. Put $C = TVT^*$ and $D = (I - P) + CP$. Since $V = XD$, D is unitary. It fixes $\ker P$ pointwise, so unitarity forces $\operatorname{ran} P$ onto itself; there D is C. Hence $[C, P] = 0$. The recursion now reads binary increment on the first bit followed by increment on the tail on its carry branch. Induction on prefix length proves the assertion. □

Proposition 4.6 (Mass near the vanishing guard)**.** *There is an absolute constant C such that every weighted precision-δ clock and every integer $M \geq 1$ satisfy*

$$\operatorname{Tr}(\rho\, 1_{[1-4^{-M},1]}(A)) \leq 2^{-M} + C(M+1)^5 4^M \delta. \tag{4.8}$$

If $\|R\| \leq L$ and $|R(I - A)|_\rho \leq \delta$, then

$$|R|_\rho \leq 4^M(1 + L)\delta + L\sqrt{2^{-M} + C(M+1)^5 4^M \delta}. \tag{4.9}$$

In particular $|R|_\rho = O((1 + \log(1/\delta))^{5/2} \delta^{1/6})$ for fixed L.

Proof. Here it is useful to avoid rounding A first. From the proof of Lemma 4.2, $|P - p(A)|_\rho = O(\delta)$ and $|TAT^* - s(A)|_\rho = O(\delta)$. The same telescoping gives $|P_j - p(s^j(A))|_\rho \leq C4^j\delta$. Threshold $p(s^j(A))$ at $1/2$ to commuting projections R_j. Against an exact projection the cross-spectrum Lipschitz bound for this threshold is two: for $b \in \{0, 1\}$ and $a \in [0, 1]$,

$|1_{[1/2,1]}(a)-b| \le 2|a-b|$. The eigenbasis proof of Lemma 2.3, followed by its state-commutator insertion, therefore gives $|R_j-P_j|_\rho \le C(j+1)4^j\delta$. Their commutators with ξ and the triangular flip defects are at most $\eta = C(M+1)^3 4^M\delta$.

For completeness, an approximately invariant suffix law differs from its earliest-bit swap by at most $CM^2\eta$. Pinch ξ in the commuting R_j to ζ; each pinching costs at most η, so $\|\xi-\zeta\|_2 \le M\eta$. Intertwining a flip with one recorded digit has error $O((M+1)\eta)$ on ζ. Inserting previous joint spectral projections does not enlarge the sum of squared branch errors, since they commute with ζ. Telescoping at most M digits and returning to ξ costs $CM^2\eta$. The initial state changes by trace norm at most 2η under its flip. Purification distance now bounds the claimed total variation.

Let q_i be the probability that $R_i,\dots,R_{M-1}$ are all one, with $q_M = 1$. The swap bound gives $|2q_i - q_{i+1}| \le CM^2\eta$, hence $q_0 \le 2^{-M} + CM^2\eta$. If $a \ge 1-4^{-M}$, then $s^j(a) = 1-4^j(1-a) \ge 3/4$ for $j < M$. Thus all R_j are one on that spectral interval. This proves (4.8).

Put $G = I - A$ and $E = 1_{[0,4^{-M}]}(G)$. Since $R\xi G = RG\xi + R[\xi, G]$, its norm is at most $(1+L)\delta$. On $I-E$ the inverse of G has norm at most 4^M, while $\|R\xi E\|_2 \le L\sqrt{\mathrm{Tr}(\rho E)}$. Add the two column bounds and use (4.8). Finally choose $M = \lfloor \frac{1}{3}\log_2(1/\delta)\rfloor$. Both 2^{-M} and $4^M\delta$ are of order $\delta^{1/3}$, proving the last assertion. The relaxed polynomial factor here is harmless; only exact guard cancellation is needed in the flag proof. □

5 The finite programmed presentation

Fix a finite deterministic counter program recognizing K. Counters take nonnegative integer values and instructions increment, decrement on a nonzero branch, or test zero. Such a recognizer follows from the classical universality of counter machines [37]. One direct construction stores the two halves of a Turing tape as integer-coded stacks. Pushing multiplies a stack code by a fixed base and adds a digit; popping computes quotient and remainder by that base. Increment, decrement and zero-test loops perform these operations using finitely many scratch counters. A unary input loop writes the integer input on the simulated tape. This gives one fixed finite instruction table. Add absorbing idle and halt controls.

The variables introduced below have the domains stated explicitly. Projection inclusions $Q \le F$ mean the polynomial equation $(I-F)Q = 0$. A precision model satisfies every listed commutator and every displayed defining equation to precision δ in Definition 2.1.

5.1 Three compulsory clocks and a carry coordinate

Use three copies of (4.1), with indices a, f, t for address, fuel and time. Complete variable sets of different clocks commute. Add an address increment V and a time increment U, satisfying (4.7) for their own clocks. The address and fuel sets commute with U, and the time and fuel sets commute with V. Add a positive contraction H with

$$H = (I - P_a) + \tfrac{1}{4}P_a T_a H T_a^*, \tag{5.1}$$

$$[P_a, T_a H T_a^*] = [H, A_a] = 0, \qquad V^* A_a V = A_a - I + \tfrac{7}{4}H. \tag{5.2}$$

Include H in the address set, commuting with all fuel and time variables. The last equality will control many address translations without an exponential precision cost.

5.2 Address decoding and reset

Put $P_{a,1} = T_a P_a T_a^*$, $E_{00} = (I - P_a)(I - P_{a,1})$ and $E_1 = P_a(I - P_{a,1})$. Add a projection L and a positive contraction R with

$$L = E_1 + E_{00} T_a^2 L T_a^{-2}, \tag{5.3}$$

$$R = (I - E_{00}) + \tfrac{1}{2} E_{00} T_a^2 R T_a^{-2}. \tag{5.4}$$

The products are polynomials even before the approximate digits commute. In exact models the cylinder

$$B_e = \left(\prod_{j<e} T_a^{2j} E_{00} T_a^{-2j} \right) T_a^{2e} E_1 T_a^{-2e} \tag{5.5}$$

selects first nonzero base-four address digit one in position e. Here each base-four address digit groups two successive binary clock bits, with the first bit least significant.

Add a reset projection Z, set $S = T_f T_t$, and impose

$$Z = (I - P_f) + P_f (I - P_t) S Z S^*. \tag{5.6}$$

In exact models write $Q_j^f = T_f^j P_f T_f^{-j}$, $P_{t,j} = T_t^j P_t T_t^{-j}$ and

$$C_k = \left(\prod_{j<k} Q_j^f \right) (I - Q_k^f), \qquad D_k = \prod_{j<k} (I - P_{t,j}). \tag{5.7}$$

The product defining C_k means $(\prod_{j<k} Q_j^f)(I - Q_k^f)$, with empty product I.

5.3 Counters and transitions

Encode counter value n by $r = 2^{-n}$. For each counter add a positive contraction r_i, a zero-test projection z_i and a positive contraction s_i with

$$r_i = z_i + \tfrac{1}{2}(I - z_i) s_i. \tag{5.8}$$

All instantaneous workspace variables commute. For each control state σ, add a projection q_σ, with $q_\sigma q_\tau = 0$ for $\sigma \neq \tau$ and $\sum_\sigma q_\sigma = I$. Each program branch selector is q_σ, $q_\sigma z_i$ or $q_\sigma(I - z_i)$; these selectors partition I. The next control is the sum of selectors entering that state. The next counter is the sum, over selectors, of that selector times $r_i/2$, r_i or $2r_i$ as its instruction dictates. The factor two is used only on a nonzero decrement branch. These prescriptions define a fixed rational polynomial map Step on counters and controls.

Initialization is

$$\mathrm{Init}(q_{\mathrm{start}}) = L, \quad \mathrm{Init}(q_{\mathrm{idle}}) = I - L, \quad \mathrm{Init}(r_1) = R, \quad \mathrm{Init}(r_i) = I \ (i > 1),$$

with other controls zero. For each counter or control W, impose

$$ZW = Z\,\mathrm{Init}(W), \tag{5.9}$$

$$U^* W U = Z_+\,\mathrm{Init}(W) + (I - Z_+)\,\mathrm{Step}(W), \qquad Z_+ = U^* Z U. \tag{5.10}$$

The witnesses and zero-tests satisfy (5.8) at every time; separate update equations for them are unnecessary. Add a flag projection F and require

$$q_{\mathrm{halt}} \leq F. \tag{5.11}$$

5.4 Commutations and the conditional clock

The following is the complete additional commutation list.

- L, R, F commute with each other, with A_a, and with every fuel and time clock variable, including U.
- Z and all instantaneous workspace variables commute with one another and with A_a, A_f, A_t, L, R, F.
- The address set, including V, H, commutes with the complete fuel and time sets.

Workspace is not required to commute with U or the address shifts and flips.

For the conditional clock add positive contractions B, C, a projection Q and a self-adjoint unitary Y. Impose

$$Q \le F, \qquad [B,Q] = [B,F] = [C,Q] = [C,F] = [B,C] = 0, \tag{5.12}$$

$$VBV^* = 4B - 3Q, \qquad VCV^* = 4C - 3F, \tag{5.13}$$

$$YFY = F, \qquad YQY = F - Q, \tag{5.14}$$

$$[Y, VBV^*] = 0, \qquad [Y, C] = 0. \tag{5.15}$$

Each of B, C, Q, Y commutes with A_a and every fuel and time clock variable, including U. Define

$$F_i = V^i F V^{-i}, \qquad Q_i = V^i Q V^{-i}, \qquad Y_i = V^i Y V^{-i}. \tag{5.16}$$

The intended ternary measurement is $(I - F_i, F_i - Q_i, Q_i)$. Its inactive outcome contributes no required bit; its two active outcomes are exchanged by Y_i. The coordinate C records the flags. It is needed to control commutation of their translates, which commutation with A_a alone would not imply.

This finishes the finite presentation, denoted $\mathcal{P}_K$. Every branch and every variable used above occurs in a finite list; e, k, i in later arguments index consequences, not additional defining relations.

6 Every tracial limit pays for halting

Lemma 6.1 (Forced finite initialization)**.** *In an exact model of $\mathcal{P}_K$,*

$$B_e L = B_e, \quad B_e R = 2^{-e} B_e, \qquad C_k Z = C_k D_k. \tag{6.1}$$

The zero-test equation forces $z_i v = v$ when $r_i v = v$, and $z_i v = 0$ when $r_i v = 2^{-n} v$ for an integer $n > 0$.

Proof. By Lemma 4.1, every digit is a spectral function of its clock coordinate. The required commutations imply that L, R, Z commute with the relevant digits. Expand (5.3) and (5.4) along the e successive E_{00} factors of B_e. The nonrecursive terms vanish there. At the first E_1 factor, the nonrecursive term is I. The coefficient accumulated in the second expansion is 2^{-e}. These finite expansions prove the first two identities.

In (5.6), on C_k the first k fuel bits force the recursive branch. Each supplies the next factor $I - P_{t,j}$. At the next fuel bit the constant branch is forced. Thus $C_k Z = C_k D_k$. In particular the $k = 0$ identity forces reset on $I - P_f$, so $Z = 0$ cannot be a tracial solution.

For the zero-test, z_i commutes with s_i. On its range $r_i = I$; on its kernel $0 \le r_i \le I/2$. If $r_i v = v$, the latter component of v must vanish; if $r_i v = 2^{-n} v$ with $n > 0$, the former component must vanish. The argument applies to every vector in a given subspace, including after unitary conjugation. Although $r_i = 0$ is permitted, no finite sequence of the prescribed updates starting at an ordinary encoded integer reaches it. □

Theorem 6.2 (Flag density)**.** *Every exact tracial model of $\mathcal{P}_K$ satisfies $\tau(F) \geq \theta_K$. There is a function $\omega(\delta) \to 0$ such that every weighted precision-δ model satisfies*

$$\mathrm{Tr}(\rho F) \geq \theta_K - \omega(\delta). \tag{6.2}$$

No effective modulus is asserted.

Proof. Work first in the faithful tracial representation. The clock variables satisfy Lemma 4.1. By Proposition 4.5, U increments each finite time prefix; it commutes with fuel and address variables. Fix $e \in K$ with halting time h and choose k with $2^k > h$. The operator

$$Q_0 = B_e C_k D_k (I - F)$$

is a projection. Lemma 6.1 and (5.9) set all counters and controls on its range to the initialized values. Input counter one has value $r_1 = 2^{-e}$, while other counters encode zero.

Write $W_l = U^{-l} W U^l$ and $Z_l = U^{-l} Z U^l$. On Q_0, the first k time bits are initially zero. Increment does not return them to zero before step 2^k, so

$$Z_0 Q_0 = Q_0, \qquad Z_l Q_0 = 0 \quad (1 \leq l \leq h). \tag{6.3}$$

We prove by induction that every counter and control at time $l \leq h$ has its classical scalar right action on Q_0. The assertion holds at zero by initialization. Given these right actions, the zero-test part of Lemma 6.1, applied to the conjugated workspace, forces the correct zero-test right actions as well. Every branch selector and every update polynomial therefore has its prescribed scalar right action. If $AQ_0 = aQ_0$ and $BQ_0 = bQ_0$, then $ABQ_0 = abQ_0$; no inter-time commutation is needed.

Conjugate (5.10) by U^{-l} and multiply by Q_0 on the right. The initialized polynomials are functions of L, R and constants, so retain their scalar actions. By (6.3) the reset term vanishes and the step term yields exactly the next configuration. This proves the induction. At time h,

$$U^{-h} q_{\mathrm{halt}} U^h Q_0 = Q_0.$$

Since $q_{\mathrm{halt}} \leq F$ and $[F, U] = 0$, the left side is killed by $I - F$. Thus $Q_0 = 0$.

It remains to remove the auxiliary conditions, which must not be assumed to have uniform conditional mass. The operator $B_e(I - F)$ commutes with all fuel and time flips. Conjugate $B_e(I - F)C_k D_k = 0$ by the flip of the earliest specified fuel digit. It preserves every later specified digit. Adding the two equations removes that digit. Repeat in increasing order to remove all fuel conditions, then do the same for time. The result is $B_e(I - F) = 0$. The address cylinders B_e are orthogonal and have trace $4^{-(e+1)}$ by (4.3). Taking finite sums over halting inputs and then their supremum proves $\tau(F) \geq \theta_K$.

For the uniform approximate statement, suppose instead that a sequence of models has $\delta_j \to 0$ but $\mathrm{Tr}(\rho_j F) \leq \theta_K - \gamma$ for some $\gamma > 0$. Form their states on the universal C^*-algebra of the finitely many bounded-domain variables before imposing equations. A weak-* limit of moment states exists. It is tracial, since

$$\|[\rho_j, s]\|_1 \leq 2 \|[\sqrt{\rho_j}, s]\|_2 \longrightarrow 0$$

for each generator and hence each fixed word. Here $\|\cdot\|_1$ denotes trace norm; the inequality follows by expanding $[\xi^2, s]$ and applying Hilbert–Schmidt Cauchy–Schwarz. For every relation r, the limit has $\tau(r^* r) = 0$. Traciality gives, for bounded a,

$$\tau(a^* r^* r a) = \tau(r^* r a a^*) \leq \|a\|^2 \tau(r^* r) = 0.$$

Thus all relations are zero operators in the tracial GNS representation. This exact model has $\tau(F) \leq \theta_K - \gamma$, contradicting the first part. Taking the supremum of the positive shortfall over precision-δ models defines a function $\omega(\delta) \to 0$ and proves (6.2). □

The proof treats only finite computations before taking compactness. It never tries to quantify how far a given finite precision sees into the halting set. This compactness argument supplies no effective modulus for the uniform flag-density estimate.

7 Simultaneous finite models

Theorem 7.1. *Let $h(e)$ be the recognizer's halting time, or ∞ if it does not halt. For every even integer $N \geq 2$, define*

$$K_N = \{e < N/2 : h(e) < 2^N\}, \qquad \theta_N = \sum_{e \in K_N} 4^{-(e+1)}.$$

There is a flat model of $\mathcal{P}_K$ on dimension

$$d_N = 2^{3N+k_N}, \qquad k_N = \theta_N N + O(\log(N+2)), \tag{7.1}$$

with every defining defect at most $3 \cdot 4^{-N}$ in operator norm. The coefficients θ_N increase to θ_K.

Proof. Take N-bit address, fuel and time registers, with the finite coordinate (4.6). Their bit zero is least significant. Let the address integer be $x \in \mathbb{Z}/2^N\mathbb{Z}$ and put

$$f_N(x) = \sum_{e \in K_N} 1_{\{x \equiv 4^e \pmod{4^{e+1}}\}}. \tag{7.2}$$

The residue classes are disjoint and their periods divide 2^N, so this is a Boolean function of mean θ_N.

Every interval of N consecutive addresses has

$$\sum_{j=1}^{N} f_N(x-j) = N\theta_N + O(\log(N+2)) \tag{7.3}$$

uniformly in x, including cyclic intervals. To prove it, put $m = \lfloor \log_4 N \rfloor$. For each $e \leq m$, one residue class has a count differing from its mean by at most one. Every flagged address from $e > m$ is a multiple of 4^{m+1}, so the whole remaining family contributes at most one address to such an interval. Its expected count is at most $N/(3 \cdot 4^{m+1})$. Summing proves (7.3).

Set

$$k_N = 1 + \max_x \sum_{j=1}^{N} f_N(x-j) \tag{7.4}$$

and add k_N optional qubits. Let R_0 rotate their contents from slot i to slot $i+1$ cyclically. Define

$$V|x,y\rangle = |x+1, R_0^{f_N(x)} y\rangle, \qquad F = f_N(x), \quad Q = FP_{\text{slot }0}, \quad Y = (I-F) + FX_{\text{slot }0}. \tag{7.5}$$

Define the translates F_j, Q_j by (5.16) and set

$$B = \sum_{j \geq 0} 3 \cdot 4^{-j-1} Q_j, \qquad C = \sum_{j \geq 0} 3 \cdot 4^{-j-1} F_j. \tag{7.6}$$

These are explicit finite diagonal matrices. Indeed V is a permutation of a finite basis, so the series are periodic geometric sums. They are positive contractions. Their terms commute, so (5.12) is exact. Shifting the sums proves (5.13) exactly; the controlled flip gives (5.14) exactly.

At an active Q_j and a fixed current address x, its occupied slot is

$$\sum_{r=x-j}^{x-1} f_N(r) \pmod{k_N}.$$

If $1 \le j \le N$, this sum is positive because $f_N(x-j) = 1$, and is at most $k_N - 1$ by (7.4). Thus Y preserves Q_j for those j. It always commutes with C, and the uncontrolled remainder satisfies

$$\|[Y, VBV^*]\| \le 2 \sum_{j>N} 3 \cdot 4^{-j} = 2 \cdot 4^{-N}. \tag{7.7}$$

This verifies (5.15).

To couple V to the address clock, let T_0, X_0 be its ordinary cyclic rotation and first-bit flip and put

$$h_N(x) = \sum_{0 \le r < x} f_N(r), \quad D = \bigoplus_x R_0^{h_N(x)}, \quad T_a = D(T_0 \otimes I)D^*, \quad X_a = D(X_0 \otimes I)D^*. \tag{7.8}$$

The coordinate A_a and digit P_a commute with D, so all address clock defect bounds from Theorem 4.4 are unchanged. In the D^* gauge, V is ordinary increment except at the all-ones address, where it carries a fixed optional-register permutation. The recursive increment agrees away from that address. At all ones its two output addresses are distinct, with the same optional permutation. The guard is 4^{-N} there. Hence its operator defect is $\sqrt{2}\,4^{-N}$ and its flat defect is $\sqrt{2}\,4^{-N}2^{-N/2}$.

For $H(x)$ use 4^{-k} when the address has $k < N$ initial ones and then a zero, and use zero at all ones. Rotation removes one initial one, so (5.1) is exact, including its all-ones case. All commutations in (5.2) are exact, and off all ones

$$A_a(x+1) - A_a(x) = -1 + \tfrac{7}{4}\,4^{-k}.$$

At all ones the defect of the last equation is 4^{-N}. Its flat norm is $4^{-N}2^{-N/2}$. Let U be ordinary cyclic time increment and use ordinary clocks on fuel and time. Their defects have the same bounds as before.

It remains to verify the whole program on these same registers. For a nonzero address let e be the position of its first nonzero base-four digit. Set $R(x) = 2^{-e}$ and $L(x) = 1$ precisely when that digit is one. At the all-zero address set $R = L = 0$. Since N is even, rotation by two bits removes one whole base-four digit. Thus (5.3)–(5.4) hold exactly, including at the all-zero address.

For fuel with k initial ones then a zero, put

$$Z(q,t) = 1_{\{t \equiv 0 \pmod{2^k}\}}.$$

On all-ones fuel use $k = N$. When the first fuel bit is zero, $Z = 1$. Otherwise the factor $I - P_t$ in (5.6) requires the first time bit to be zero and the rotated equation tests the remaining $k-1$ bits. On all-ones fuel, the recursive test is true precisely at the all-zero time word. This proves (5.6) in every case.

At a basis point (x, q, t) assign every workspace counter and control the deterministic configuration

$$\mathrm{Step}^{t \bmod 2^k}(\mathrm{Init}(x)). \tag{7.9}$$

These are diagonal entries on existing registers; they add no quantum tensor factor. For ordinary counter entries set $z_i = 1, s_i = 0$ when $r_i = 1$ and $z_i = 0, s_i = 2r_i$ otherwise; the idle $r_i = 0$ case uses $z_i = s_i = 0$. These are contractions and give (5.8) exactly. The next

time point is initialized if its residue modulo 2^k is zero, and is otherwise one step of the old configuration. This is exactly (5.9)–(5.10), also at the cyclic time boundary. A displayed halt occurs in fewer than 2^N steps on an input $e < N/2$, so its address is flagged in (7.2). Hence (5.11) is exact.

All workspace and reset variables are diagonal in (x, q, t), and the optional register carries no workspace labels. Every specified remaining commutation therefore holds. The only nonzero defining defects are the clock affine and tail-flip defects, the guarded increments, the carry step and (7.7). Their operator norms are bounded by the theorem's constant. The three compulsory registers and the optional register have dimension 2^{3N+k_N}, and (7.3) gives (7.1). Finally each halting input eventually meets both cutoffs in K_N, while the address weights have a summable geometric tail. This proves $\theta_N \uparrow \theta_K$. □

8 Counting measurements on the actual state

The count must tolerate measurements that commute only approximately and a density matrix that is not flat. Its proof keeps the measurement branches together. Its last step is the usual random-access-code argument [34], using Holevo's information bound [18].

Theorem 8.1 (Triangular measurement count)**.** *Let m PVMs on $\mathbb{C}^d$ have two or three outcomes. Let ρ be a density matrix and $\xi = \sqrt{\rho}$. Assume each outcome has state commutator at most η, and outcomes of different measurements have weighted commutator at most η. For each measurement i, assume a unitary U_i has state commutator at most η, exchanges its own two binary outcomes, or its two active ternary outcomes while fixing the inactive one, and preserves every later measurement. Each of these intertwining equations is required in weighted norm with error at most η.*

If b measurements are binary and F_i is the active projection of a ternary measurement, then for an absolute constant C, with $e = Cm^4\eta^2 \le 1/2$,

$$\log d \ge b + \sum_{i \text{ ternary}} \operatorname{Tr}(\rho F_i) - Cm^3\eta - m\big[h_2(e) + e\big]. \tag{8.1}$$

Here h_2 is binary entropy, with $h_2(0) = 0$.

Proof. For a PVM (P_a) define left and right Hilbert–Schmidt isometries

$$L_P(Z) = (P_a Z)_a, \qquad R_P(Z) = (Z P_a)_a.$$

They differ on ξ by at most $\sqrt{3}\eta$. Every left multiplication commutes with every right multiplication. Moving a product of s left recording maps through ξ to right maps in reverse order therefore changes the recording vector by at most $s\sqrt{3}\eta$, by telescoping isometries.

Suppose a column of left operators $D = (D_c)_c$ has column operator norm at most L and $\|D\xi\|_2 \le \alpha$. It commutes with all right recording maps. After any prefix of s measurements, the preceding comparison gives

$$\left(\sum_{\mathbf{a},c} \|D_c P_{s,a_s} \cdots P_{1,a_1}\xi\|_2^2\right)^{1/2} \le \alpha + Ls\sqrt{3}\eta. \tag{8.2}$$

This estimates all branches together. In particular, no inverse of a small branch probability appears.

Measure in reverse index order. For the suffix starting at i, move U_i through its recorded measurements. It exchanges its own two labels and preserves later ones. A single intertwining column is a difference of two isometries, has column operator norm at most two and weighted

norm $O(\eta)$ on ξ. Equation (8.2) bounds it after any preceding measurements by $O(m\eta)$. At most m intertwiners, followed by the initial state-invariance error, give total vector error $O(m^2\eta)$. Distances of purifications bound total variation of the classical records. Thus the suffix distribution differs from its own i-th outcome swap by at most $Cm^2\eta$.

For a binary outcome with conditional law $(p, 1-p)$, concavity gives $h_2(p) \geq 1 - |2p-1|$. The suffix swap variation is exactly the average of $|2p-1|$ over the later record. Hence its conditional entropy is at least one minus that variation. For ternary probabilities,

$$H(p_0, p_1, p_2) \geq (p_1 + p_2) - |p_1 - p_2|.$$

Indeed split off the active event and apply the binary inequality within it. The ternary swap variation is the corresponding average of $|p_1 - p_2|$. The chain rule now bounds the full record entropy below by b plus recorded active probabilities minus $Cm^3\eta$.

Those recorded active probabilities can be replaced by their values on ρ. For one PVM, $(P_a\xi)_a$ and $(\xi P_a)_a$ purify, respectively, its nonselectively measured state and ρ on the original system. Their distance is at most $\sqrt{3}\eta$, so those states differ by trace norm at most $C\eta$. If Φ_j are the nonselective measurement channels, expand

$$\Phi_s \cdots \Phi_1(\rho) - \rho = \sum_{j=1}^{s} \Phi_s \cdots \Phi_{j+1}(\Phi_j(\rho) - \rho).$$

Trace-norm contractivity gives an $O(m\eta)$ bound for every prefix. Thus all active expectations change by at most $Cm^2\eta$ in total, already covered by the preceding error term.

We next bound record entropy above by dimension. Fix one recorded variable. Move its measurement to the end of the actual sequence by adjacent swaps. An elementary swap column has components $[P_a, Q_b]$, weighted norm $O(\eta)$ and column operator norm at most two: it is the difference of the two recording isometries, with record factors reordered. Equation (8.2) bounds its norm after a prefix by $O(m\eta)$. After at most m swaps the total recording vectors differ by $O(m^2\eta)$.

In the reordered process this PVM is last. Remeasuring it on the final system recovers its recorded value with zero error, since equal projective measurements are adjacent. In the original process, apply the same failure projection to the difference of recording vectors. Its squared norm bounds the decoding error by $e = Cm^4\eta^2$.

Let Z be the classical record and Q_{fin} the final system, of dimension d. The decoder just described acts only on this system, not on the purification reference. Holevo's bound, conditional entropy subadditivity and the error-indicator bound for at most three outcomes give

$$\begin{aligned} H(Z) &= I(Z : Q_{\mathrm{fin}}) + H(Z \mid Q_{\mathrm{fin}}) \\ &\leq \log d + \sum_i H(Z_i \mid Q_{\mathrm{fin}}) \\ &\leq \log d + m[h_2(e) + e]. \end{aligned}$$

For the last line, measuring the proposed decoder can only increase conditional entropy; the classical conditional entropy is at most $h_2(e) + e\log(3-1)$. Combining the upper and lower record bounds proves (8.1). □

9 The matched lower bound

Theorem 9.1. *Every weighted precision-δ model of $\mathcal{P}_K$ on $\mathbb{C}^d$ satisfies*

$$\log d \geq (3 + \theta_K)\log_4(1/\delta) - o(\log(1/\delta)). \tag{9.1}$$

The error is uniform over all states and dimensions.

9.1 The carry estimate

Identify $\mathcal{K}$ with binary sequences viewed as 2-adic integers, and let g_i add the nonnegative integer i. If two sequences first differ at digit j, their Cantor coordinates differ by between $\frac{1}{2}4^{-j}$ and 4^{-j}. Addition preserves that first differing digit. Consequently every g_i is 2-Lipschitz on $\mathcal{K}$, uniformly in i.

Let $h(a) = 4^{-k}$ if the sequence starts with exactly k ones followed by a zero, and put $h(1) = 0$. It is also 2-Lipschitz: equal first-zero positions give equal values; otherwise the smaller first-zero position j is the first differing digit, and the value difference is at most 4^{-j}. Its identities are

$$h = 1 - p + \tfrac{1}{4}p(h \circ s), \qquad g_1(a) = a - 1 + \tfrac{7}{4}h(a). \tag{9.2}$$

The latter follows by summing the coordinate change when k initial ones become zero and the next zero becomes one. The all-ones case follows by the limit $k \to \infty$. Lipschitz functions on $\mathcal{K}$ can be extended linearly across its gaps with the same constant.

Lemma 9.2 (Translations with polynomial cost)**.** *For an absolute constant C, every model of the address relations satisfies*

$$|H - h(\widehat{A}_a)|_\rho \le C\delta, \tag{9.3}$$

$$|V^{-i}\widehat{A}_a V^i - g_i(\widehat{A}_a)|_\rho \le C(i+1)\delta, \tag{9.4}$$

$$|[V^i J V^{-i}, P_{a,j}]|_\rho \le C(i+1)4^j\delta \quad (J \in \{F, Q, Y\}). \tag{9.5}$$

Proof. Let $p = p(\widehat{A}_a)$ and $E = H - h(\widehat{A}_a)$. Subtract the first identity of (9.2) from (5.1), retaining its error r. The exact subtraction is

$$\begin{aligned} E = \; & -(P_a - p) + \tfrac{1}{4}P_a T_a E T_a^* \\ & + \tfrac{1}{4}P_a(T_a h(\widehat{A}_a)T_a^* - h(s(\widehat{A}_a))) \\ & + \tfrac{1}{4}(P_a - p)h(s(\widehat{A}_a)) + r. \end{aligned}$$

The first, third, fourth and last terms have weighted size $O(\delta)$ by Lemmas 2.3 and 4.2. In the fourth term, moving the bounded right factor past ξ costs $O(\delta)$, since $h \circ s$ is Lipschitz with fixed constant. The second term is bounded by $\frac{1}{4}|E|_\rho + O(\delta)$ using $\|E\| \le 1$. Absorb it to prove (9.3). The last equation of (5.2), together with spectral repair and unitary transport, gives

$$|V^*\widehat{A}_a V - g_1(\widehat{A}_a)|_\rho \le C_1\delta.$$

Put $e_i = |V^{-i}\widehat{A}_a V^i - g_i(\widehat{A}_a)|_\rho$, so $e_0 = 0$. Since $g_i = g_{i-1} \circ g_1$ on $\mathcal{K}$ and functional calculus commutes with unitary conjugation, for $i \ge 1$

$$\begin{aligned} V^{-i}\widehat{A}_a V^i - g_i(\widehat{A}_a) = {} & V^*(V^{-(i-1)}\widehat{A}_a V^{i-1} - g_{i-1}(\widehat{A}_a))V \\ & + (g_{i-1}(V^*\widehat{A}_a V) - g_{i-1}(g_1(\widehat{A}_a))). \end{aligned}$$

The bracket in the first term is a difference of positive contractions, so it has norm at most one, and Lemma 2.2 bounds the first term by $e_{i-1} + \|[V, \xi]\|_2 \le e_{i-1} + \delta$. Both $V^*\widehat{A}_a V$ and $g_1(\widehat{A}_a)$ have spectrum in $\mathcal{K}$, where g_{i-1} is 2-Lipschitz. Hence (2.3), the bound $\|[g_1(\widehat{A}_a), \xi]\|_2 \le 2\|[\widehat{A}_a, \xi]\|_2$ and (4.4) bound the second term by $2(C_1 + 4C)\delta$. Thus $e_i \le e_{i-1} + C_2\delta$ with C_2 independent of i, and induction gives $e_i \le C_2 i\delta$. This proves (9.4). Each step pays one transport of the accumulated error and one functional-calculus loss with the uniform constant 2; no iterated factor four occurs.

Put $f_j = p \circ s^j$. Replace $P_{a,j}$ by $f_j(\widehat{A}_a)$, at cost $C4^j\delta$ by (4.5), and conjugate by V^{-i}, at transport cost $i\delta$. Then apply (2.3) to (9.4), with the Lipschitz constant $2 \cdot 4^j$ of f_j. The result is

$$|V^{-i}P_{a,j}V^i - f_j(g_i(\widehat{A}_a))|_\rho \le C(i+1)4^j\delta.$$

The composition $f_j \circ g_i$ has Lipschitz constant at most $4 \cdot 4^j$ on $\mathcal{K}$. Each $J \in \{F, Q, Y\}$ commutes with A_a to precision δ, and hence with $\widehat{A}_a$ to precision $C\delta$. Equation (2.4) bounds its commutator with this composition by $C4^j\delta$. Replace back at cost $C(i+1)4^j\delta$: for the self-adjoint difference Δ just estimated, $|[J,\Delta]|_\rho \le 2|\Delta|_\rho + \|\Delta\|\|[J,\xi]\|_2$. Conjugate by V^i, at transport cost $2i\delta$. This gives (9.5). □

9.2 Joint digits and conditional measurements

Lemma 9.3. *Fix $M \ge 1$. The $3M$ compulsory binary measurements are the raw digit PVMs $(P_{c,j}, I - P_{c,j})$, where $P_{c,j} = T_c^j P_c T_c^{-j}$ for $c \in \{a, f, t\}$ and $j < M$, with flips $X_{c,j} = T_c^j X_c T_c^{-j}$. The M conditional ternary measurements are $(I - F_i, F_i - Q_i', Q_i')$ for $i < M$, with flips Y_i, where Q_i' is the repaired data projection constructed in the proof. Index the conditional measurements first, then the address, fuel and time digits, each family in increasing digit order. All commutations between outcomes of different measurements, all flip equations required in this order, and all state commutations have error at most*

$$\eta = C(M+1)^6 4^M \delta. \tag{9.6}$$

Each Q_i' differs from Q_i by weighted size at most $C\eta$, and the conditional measurements are exact PVMs on the same state.

Proof. Two elementary rules are used throughout. First, if D is self-adjoint and Y is a contraction, then $DY\xi = D\xi Y + D[Y,\xi]$ gives

$$|[D,Y]|_\rho \le 2|D|_\rho + \|D\|\,\|[Y,\xi]\|_2. \tag{9.7}$$

Hence, if $X = X_0 + D_X$ and $Y = Y_0 + D_Y$ with self-adjoint corrections, the identity

$$[X,Y] = [D_X, Y] + [X_0, D_Y] + [X_0, Y_0]$$

shows that replacing both sides of a commutator by nearby operators adds the two replacement errors; it never multiplies them. Second, suppose every letter of a word w commutes to precision δ with every letter of a word w', adjoint letters included. Moving letters one at a time and applying Lemma 2.2 to each interchange gives $|[w,w']|_\rho \le C|w|\,|w'|(|w| + |w'| + 1)\delta$. We call this the word-interchange bound.

The required estimates are the following.

(i) *One compulsory clock.* Lemma 4.2 gives the digit commutations, own-bit flips, later-digit preservation and state commutators within each clock, with error $C(M+1)^3 4^M\delta$. Its raw digit commutators already replace both digits by commuting spectral digits, at additive cost.

(ii) *Different compulsory clocks.* For $c \ne c'$, $P_{c,j}$ and $P_{c',k}$ are words of lengths $2j+1$ and $2k+1$ in two complete clock sets, which commute by hypothesis. The word-interchange bound gives $|[P_{c,j}, P_{c',k}]|_\rho \le C(M+1)^3\delta$. The same bound shows that $X_{c,j}$ preserves every $P_{c',k}$. These estimates must go through the raw conjugated digits. Comparing the spectral digits $p(s^j(\widehat{A}_c))$ and $p(s^k(\widehat{A}_{c'}))$ by two applications of (2.4) would multiply their Lipschitz constants. That costs order $4^{j+k}\delta$, which for j, k near M exceeds (9.6) by a factor of order $4^M/(M+1)^6$ and would defeat the choice (9.8) below. If spectral

digits were measured instead, put $D_{c,j} = p(s^j(\widehat{A}_c)) - P_{c,j}$. It is self-adjoint, with norm at most one and $|D_{c,j}|_\rho \le C4^j\delta$ by (4.5). Then

$$[p(s^j(\widehat{A}_c)), p(s^k(\widehat{A}_{c'}))] = [D_{c,j}, p(s^k(\widehat{A}_{c'}))] + [P_{c,j}, D_{c',k}] + [P_{c,j}, P_{c',k}].$$

By (9.7), with $\|[p(s^k(\widehat{A}_{c'})), \xi]\|_2 \le C4^k\delta$ from Lemma 2.3 and $\|[P_{c,j}, \xi]\|_2 \le (2j+1)\delta$, its weighted norm is at most $C4^M\delta + C(M+1)^3\delta$. Thus either choice satisfies (9.6); we fix the raw digits.

(iii) *Conditional against conditional.* Use the affine-only part of the spectral repair on (B, Q, V) and (C, F, V). If $i \le j$,

$$[Q_i, F_j] = V^i[Q, F_{j-i}]V^{-i}.$$

Recover F_{j-i} from the rounded flag coordinate and apply (2.4) with $[Q, C] = 0$. The error is $O(4^{j-i}\delta)$ plus polynomial transport. If $i > j$, use $[F, B] = 0$ instead. The same argument with $[Q, B] = 0$ controls $[Q_i, Q_j]$, and with $[F, C] = 0$ controls $[F_i, F_j]$. Only one side is replaced by a coordinate function here, so there is a single depth loss.

(iv) *Conditional against address digits.* By (9.5) with $J \in \{F, Q\}$, the commutators of F_i and Q_i with $P_{a,j}$ are at most $C(M+1)4^M\delta$. With $J = Y$, the same bound shows that Y_i preserves every address digit.

(v) *Conditional against fuel and time digits.* The operators F_i, Q_i and Y_i are words of length $2i+1$ in V, V^* and one of F, Q, Y. Each of these letters commutes with every fuel and time clock variable by hypothesis. The digits $P_{f,k}$ and $P_{t,k}$ are words of length $2k+1$ in the fuel and time sets. The word-interchange bound gives commutators at most $C(M+1)^3\delta$, and shows that Y_i preserves every fuel and time digit. These bounds are again taken through the raw conjugated words. Writing F_i or Q_i as a digit of a conditional coordinate, and a fuel or time digit as a spectral digit, would stack two depth losses.

(vi) *Flips.* The own conditional flip is the V^i-translate of (5.14). For later flags, $[Y, C] = 0$ and digit recovery imply preservation by Y_i. For later data, use $[Y, VBV^*] = 0$ and digit recovery on the tail coordinate, with depth $j - i - 1$ when $j > i$. This again costs at most one factor 4^{j-i}. Conditional flips preserve all compulsory digits by (iv) and (v). A compulsory flip $X_{c,j}$ exchanges $P_{c,j}$ and preserves later digits of its own clock by (i), and every digit of a later clock family by (ii).

(vii) *State commutators.* State commutators of translated raw variables are at most $C(M+1)\delta$.

All the estimates just listed fit (9.6): cubic word-interchange costs and linear address costs are dominated by its sixth power. No factor 4^{i+j} is used.

We finally repair inclusion. For projections F, Q with $|(I-F)Q|_\rho \le \eta$ and state commutators at most η, put

$$A_0 = FQF, \quad L_0 = (I-F)QF, \quad Q' = 1_{[1/2,1]}(A_0) \le F.$$

The weighted error rule and adjoints give $|Q - A_0|_\rho \le 4\eta$ and $|L_0|_\rho \le 2\eta$. Since $A_0 - A_0^2 = L_0^*L_0$ and

$$|1_{[1/2,1]}(t) - t|^2 \le t(1-t) \quad (0 \le t \le 1),$$

we get $|Q' - A_0|_\rho \le 2\eta$ and thus $|Q' - Q|_\rho \le 6\eta$. This difference is self-adjoint, so its state commutator is at most twice its weighted norm. Replacing Q by Q' in a commutator or a flip equation therefore costs only $C\eta$, by Lemma 2.2. Apply this independently to the raw pairs F_i, Q_i, after absorbing their polynomial transport constants into (9.6). The outcomes $(I - F_i, F_i - Q_i', Q_i')$ are now exact PVMs. All previous estimates survive with a fixed factor. □

Proof of Theorem 9.1. Index the optional measurements first, then the three compulsory clock families, in increasing digit order within each family. Optional flips preserve all compulsory measurements. A compulsory flip needs to preserve only later compulsory measurements, not the earlier optional family. Thus this ordering meets the hypotheses of Theorem 8.1, with $m = 4M$ and $b = 3M$.

Put $t = \log_4(1/\delta)$ and, for sufficiently small δ, choose

$$M = \lfloor t - 30 \log_4(2+t) \rfloor . \tag{9.8}$$

Then $4^M \delta \le (2+t)^{-30}$, so (9.6) gives $\eta \le C(1+t)^{-24}$. The error terms of (8.1) are $o(1)$. Consequently

$$\log d \ge 3M + \sum_{i<M} \operatorname{Tr}(\rho F_i) - o(1).$$

Unitary state transport gives

$$|\operatorname{Tr}(\rho F_i) - \operatorname{Tr}(\rho F)| \le \|V^{-i} \rho V^i - \rho\|_1 \le 2i\delta.$$

Apply Theorem 6.2 to the same ρ. The result is

$$\log d \ge (3 + \theta_K - \omega(\delta))M - o(M).$$

Since $M = t - O(\log(t+2))$ and $\omega(\delta) \to 0$, this is (9.1). Every estimate used the actual model state. □

Remark 9.4 (Where the guards are used). Two structural facts about the proof follow from tracing where each relation enters.

(a) The guarded relations (4.7) are used only in the exact tracial limit. In the proof of Theorem 1.2 they enter only through Proposition 4.5, in the exact part of Theorem 6.2, and the approximate part of that theorem reaches the exact case by compactness, not by a quantitative estimate. The quantitative lower bound of this section uses instead the unguarded carry equation, the last relation of (5.2), whose defect in the finite models is 4^{-N} (Theorem 7.1). Consequently Proposition 4.6 is not used in the proof of Theorem 1.2, and its guard-removal loss, of order $\delta^{1/6}$ up to logarithms, is never paid.

(b) The guarded increment (4.7) for the address increment V is used by no proof. Section 6 applies Proposition 4.5 only to U, and V acts on the address digits only through the carry relations (5.1)–(5.2), in Lemma 9.2. That relation is therefore redundant. It is harmless: the finite models satisfy it with operator defect $\sqrt{2}\,4^{-N}$ (Theorem 7.1).

10 The game and its exponent

Corollary 10.1 (Matched source profiles)**.** *The presentation $\mathcal{P}_K$ satisfies*

$$\log W(4^{-N}) = (3 + \theta_K)N + o(N) = \log H(4^{-N}). \tag{10.1}$$

One computable comparison sequence is $A_N = 3N + \lfloor N\theta_N \rfloor$, and both logarithmic profiles equal $A_N + o(N)$.

Proof. Theorem 9.1 is the lower bound for W, and $W \le H$. Theorem 7.1 is the matching upper bound for H at a fixed multiple of the requested precision. Replacing N by an even integer at a fixed greater distance absorbs that multiple. Such a change costs $O(1)$ depth, and the upper and lower limiting slopes are both $3 + \theta_K$ since $\theta_N \to \theta_K$. Monotonicity interpolates intermediate depths. Finally $N\theta_N = N\theta_K + o(N)$, giving the comparison sequence. □

The Boolean correspondence theorem [12, Theorem 4.9] supplies both game directions with the actual marginal state kept. We do not reprove that general correspondence or its return theorem. To make the precision accounting for this construction transparent, we give the elementary matrix estimate responsible for its square-root extraction and explain how it applies here.

Lemma 10.2 (Consistency and the square root of the state)**.** *Let A, B be reflections and Ξ a rectangular amplitude matrix with $\rho_B = \Xi\Xi^*$. If $\|B\Xi - \Xi A\|_2 \le \eta$, then*

$$\|[B, \sqrt{\rho_B}]\|_2 \le \sqrt{2}\,\eta. \tag{10.2}$$

Proof. Set $Y = (\Xi + B\Xi A)/2$. Then $BY = YA$, so B commutes with YY^* and its square root. Also $\|\Xi - Y\|_2 \le \eta/2$. For any rectangular matrices M, N,

$$\|\sqrt{MM^*} - \sqrt{NN^*}\|_2 \le \sqrt{2}\,\|M - N\|_2. \tag{10.3}$$

Indeed apply the eigenbasis proof of (2.2) to $f(t) = |t|$ and the self-adjoint block matrices $\left(\begin{smallmatrix} 0 & M \\ M^* & 0 \end{smallmatrix}\right)$ and its N analogue, with $Z = I$. Their absolute values are block diagonal with square-root blocks. The right-hand squared norm is $2\|M - N\|_2^2$. Discard the second left-hand block to obtain (10.3). Since B commutes with $\sqrt{YY^*}$, its commutator with $\sqrt{\rho_B}$ is bounded by twice their difference, proving (10.2). □

Proposition 10.3 (The two transfers needed here)**.** *Let G be the fixed context-variable game supplied by Theorem 3.1 for $\mathcal{P}_K$. There are constants $c, C > 0$ such that a loss-ε strategy with local dimensions at most d gives a weighted source model of precision $C\sqrt{\varepsilon}$ in dimension at most cd. A flat source model of precision δ in dimension d gives a strategy of dimension at most cd and loss at most $C\delta^2$.*

Proof. The upper direction is precisely the explicit flat strategy in Theorem 3.1. For the lower direction apply the established Boolean correspondence, followed by that theorem's weighted soundness. The correspondence is applicable because every compiled context is an ordinary Boolean predicate with a nonempty set of valid answers and every tested incidence has fixed positive probability.

Here is also a direct check of the precision ingredients in its extraction, without invoking its more general return theorem. A mixed strategy may first be replaced by a pure component of at least the same winning probability. Compressing to Schmidt supports gives local dimension at most d. Alice's context POVMs have simultaneous fixed-size ancilla dilations, and Bob's binary contractions have reflection dilations on a common extra qubit. The dimensions grow by a fixed factor; the initial ancilla state is included in the marginal. Invalid Alice answers can be reassigned to a fixed valid answer, since an invalid answer was already a certain loss. After these reductions, consistency at a tested incidence gives

$$\|(A \otimes I - I \otimes B)\psi\| \le 2\sqrt{\varepsilon/\pi_{\min}}.$$

Pull a fixed Bob context word across to Alice in reverse order. Every replacement costs this bound, and Alice's context observables commute and satisfy their truth table exactly. Thus each Bob context commutator and forbidden assignment polynomial has error $C\sqrt{\varepsilon}$ on the marginal. Lemma 10.2 gives the same order for its state commutators, with no additional square root. These are exactly the Boolean-model conditions used by the compiler. Hence the composed source precision is $C\sqrt{\varepsilon}$ on that state. □

Proof of Theorem 1.2. Apply Proposition 10.3 and Corollary 10.1. For $\varepsilon = 2^{-n}$, precision $C\sqrt{\varepsilon}$ has

$$\log_4(1/(C\sqrt{\varepsilon})) = n/4 + O(1).$$

The weighted lower profile therefore gives

$$\log D_G(2^{-n}) \geq \tfrac{3+\theta_K}{4} n - o(n).$$

The flat models at precision 4^{-N} give loss at most $C16^{-N}$ and logarithmic dimension $(3+\theta_K)N + o(N)$. Choosing an even $N = n/4 + O(1)$ at or above the required precision gives the matching upper bound. Fixed dimension multipliers add only $O(1)$ to the logarithm. Monotonicity also gives (1.4) for arbitrary $\varepsilon \downarrow 0$. These strategies have losses tending to zero, so $\omega_q(G) = 1$. A finite-dimensional perfect strategy would extract a source model at every positive precision in one fixed dimension, contradicting Theorem 9.1.

An oracle for K computes θ_K by finite summation and a geometric tail bound. Conversely, suppose membership below e is known and subtract those contributions. With $w_e = 4^{-(e+1)}$, the residual is at most $w_e/3$ when $e \notin K$, and at least w_e when $e \in K$. An approximation of error less than $w_e/6$, compared with threshold $2w_e/3$, distinguishes these cases. Induction computes K from a real oracle for θ_K. Thus $\theta_K \equiv_T K$, and the same holds for $\beta = (3+\theta_K)/4$. Lemma 2.4 now excludes a computable certified convergence rate. □

11 Every positive left-c.e. exponent

The halting game uses separated address cylinders so that its coefficient has an explicit Turing degree. This first realization step, retained from version 2.0.0, treats arbitrary left-c.e. reals by reading the whole address. The clocks, reset, carry and conditional measurement estimates stay the same. Section 16 will replace monotone acceptance by stop-or-continue verification to reach all positive Δ^0_2 exponents.

11.1 A finite reader

Fix a computable increasing sequence of rationals $0 \leq q_s \leq 1$ tending to $\theta \in [0,1]$. For a binary address $x = (b_0, b_1, \ldots)$ put

$$v(x) = \sum_{j \geq 0} b_j 2^{-j-1}, \qquad p_s = \sum_{j<s} b_j 2^{s-1-j}.$$

At stage $s \geq 1$, a fixed program computes q_s, reads the next bit, updates p_s, and halts if

$$(p_s + 1)/2^s < q_s. \tag{11.1}$$

Counter instructions implement the integer arithmetic and strict rational comparison. Each stage takes at least one instruction. Acceptance implies $v(x) \leq (p_s+1)/2^s < \theta$; conversely, if $v(x) < \theta$, the two sequences $(p_s+1)/2^s \to v(x)$ and $q_s \to \theta$ eventually satisfy (11.1). Thus the program halts exactly on $\{v < \theta\}$. Only the finite program for (q_s) occurs in the presentation; θ is not an operator coefficient.

Here is the full change to Section 5. Delete L, R, equations (5.3)–(5.4), and their occurrences in initialization and commutation lists. Retain all other clock, reset, carry and conditional relations. Add positive contractions $A_{\mathrm{r}}, C_{\mathrm{r}}$ and a projection P_{r} to the instantaneous commuting workspace, with

$$4A_{\mathrm{r}} = 3P_{\mathrm{r}} + C_{\mathrm{r}}, \qquad [P_{\mathrm{r}}, C_{\mathrm{r}}] = 0. \tag{11.2}$$

The workspace commutes with Z, A_a, A_f, A_t, F as before. The flag still commutes with A_a and the complete fuel and time sets, including U; all the conditional-clock commutations are retained. Initialize every ordinary counter to I, the start control to I, the other controls to

zero, and A_{r} to A_a. A READ control q_σ has selectors $q_\sigma P_{\mathrm{r}}$ and $q_\sigma(I-P_{\mathrm{r}})$. If Q_{r} is the sum of READ controls, set

$$\mathrm{Step}(A_{\mathrm{r}}) = Q_{\mathrm{r}} C_{\mathrm{r}} + (I - Q_{\mathrm{r}}) A_{\mathrm{r}}. \tag{11.3}$$

Include A_{r}, together with every ordinary counter and control, in (5.9)–(5.10). In particular,

$$Z A_{\mathrm{r}} = Z A_a, \qquad U^* A_{\mathrm{r}} U = Z_+ A_a + (I - Z_+)\,\mathrm{Step}(A_{\mathrm{r}}).$$

The bit and tail are witnesses, like the zero-test witnesses; they need no separate update equations. Call this finite rational presentation $\mathcal{P}_\theta$.

Lemma 11.1 (Reader forcing)**.** *Every exact tracial model of $\mathcal{P}_\theta$ has $\tau(F) \geq \theta$. Uniformly over its weighted precision-δ models,*

$$\mathrm{Tr}(\rho F) \geq \theta - \omega_\theta(\delta), \qquad \omega_\theta(\delta) \longrightarrow 0. \tag{11.4}$$

Proof. On the kernel and range of P_{r}, equation (11.2) places the spectrum of A_{r} in $[0, 1/4]$ and $[3/4, 1]$, respectively. Therefore

$$P_{\mathrm{r}} = p(A_{\mathrm{r}}), \qquad C_{\mathrm{r}} = s(A_{\mathrm{r}}).$$

This conclusion does not require Cantor spectrum in advance.

Let B_w specify an s-bit address word w satisfying (11.1). Every extension follows the same computation to a halt, at some time h. It may halt at an earlier stage; in either case it halts by stage s after at most s reads. Choose $2^k > h$ and put $Q_0 = B_w C_k D_k (I - F)$. These factors commute. The reset part of Lemma 6.1 is independent of the deleted decoder and initializes this subspace.

The reader requires an operator induction in addition to the scalar counter induction of Theorem 6.2. After j reads, track

$$(A_{\mathrm{r}})_\ell Q_0 = s^j(A_a) Q_0. \tag{11.5}$$

The right side commutes with Q_0. Taking adjoints shows that Q_0 reduces the self-adjoint left side. Functional calculus on that reducing subspace gives

$$(P_{\mathrm{r}})_\ell Q_0 = b_j Q_0, \qquad (C_{\mathrm{r}})_\ell Q_0 = s^{j+1}(A_a) Q_0$$

whenever the next bit is read. All bits read before the halt belong to w. On other instructions the reader remains fixed. The ordinary counters and controls have their prescribed scalar right actions, with zero tests forced by Lemma 6.1. There is no reset before h. Thus (5.10) proves the initialized history through its halt. As in Theorem 6.2, $q_{\mathrm{halt}} \leq F$ and $[F, U] = 0$ force $Q_0 = 0$. Conjugating by successive fuel and time flips and adding removes C_k, D_k, giving $B_w \leq F$.

The length-s address cylinders have trace 2^{-s}. The number accepted at stage s is

$$a_s = \max\{0, \lceil 2^s q_s \rceil - 1\}, \qquad 2^{-s} a_s \geq q_s - 2^{-s}.$$

Summing accepted cylinders gives $\tau(F) \geq q_s - 2^{-s}$, hence $\tau(F) \geq \theta$. This includes $\theta = 0$ and $\theta = 1$: the possibly unaccepted all-ones address has trace zero. Finally the bounded moment-limit and tracial GNS argument in Theorem 6.2 applies to all the added bounded variables and proves (11.4). □

Lemma 11.2 (Finite reader models)**.** *For even $N \geq 2$, $\mathcal{P}_\theta$ has flat models of dimension*

$$d_N = 2^{3N + k_N}, \qquad k_N = \theta N + o(N), \tag{11.6}$$

with every defining residual at most $3 \cdot 4^{-N}$ in operator norm. Every new program and reader equation is exact.

Proof. Use the three N-bit registers of Theorem 7.1, with bit zero least significant for increment. Write

$$v_N(x) = \sum_{j<N} b_j(x) 2^{-j-1}, \quad \vartheta_N = q_{2^N}, \quad f_N(x) = 1_{\{v_N(x)<\vartheta_N\}}.$$

Read the N bits followed by zeros. The Cantor reader and its witnesses remain exact after any number of reads. Assign all workspace entries the history (7.9), with this initialization. The next history point is reset precisely when its time residue is zero, including the cyclic boundary and all-ones fuel. Thus all transitions are exact. A displayed halt has accepting stage $s \le 2^N$, so $v_N(x) < q_s \le q_{2^N}$; hence $q_{\mathrm{halt}} \le F$ is exact. Specifying q_{2^N} may take arbitrarily long classical computation. It does not lengthen a displayed history or add a tensor factor.

The moving-window estimate holds for this new mask. On an aligned block of 2^r consecutive address integers, the radical-inverse values v_N are $(j+u)/2^r$, $0 \le j < 2^r$, for a fixed $u \in [0,1)$. Their count below ϑ_N differs from $2^r\vartheta_N$ by at most one. Decompose an ordinary interval of length L into at most two aligned dyadic blocks of each size, and a cyclic interval into at most two ordinary intervals. Uniformly in the starting point,

$$\sum_{j=1}^{N} f_N(x-j) = N\vartheta_N + O(\log(N+2)). \tag{11.7}$$

This is the elementary radical-inverse discrepancy argument associated with van der Corput [44]. It is proved here for the moving windows used by the construction.

Use this f_N and $k_N = 1 + \max_x \sum_{j=1}^{N} f_N(x-j)$ in (7.5)–(7.8). The slot noncollision, geometric series and dressing arguments in Theorem 7.1 use only a Boolean mask and this slot bound. Their residual estimates therefore still hold. Since $\vartheta_N \to \theta$, equation (11.7) gives (11.6). The full address mean differs from ϑ_N by at most 2^{-N}, so these flat models also have $\mathrm{Tr}(\rho F) \to \theta$. □

11.2 Adding clocks and powering equations

Proposition 11.3. *For every $r \in \mathbb{N}$, integer $q \ge 1$ and left-c.e. $\theta \in [0,1]$, there is a fixed finite rational BCS game $G_{\theta,r,q}$ of quantum value one, with no finite-dimensional perfect strategy, and*

$$\log D_{G_{\theta,r,q}}(\varepsilon) = \frac{3+r+\theta}{4q} \log(1/\varepsilon) + o(\log(1/\varepsilon)). \tag{11.8}$$

Proof. Append r copies of the compulsory clock, requiring each complete new generator set to commute with all generators outside that clock. Tensoring the upper models adds rN to logarithmic dimension. For the lower bound, append the new compulsory digit families to the order in Lemma 9.3. Its word-interchange argument bounds all new cross terms by a polynomial in M times δ. The same error form $C_r(M+1)^6 4^M \delta$ holds, with $m = (4+r)M$ and $b = (3+r)M$. Neither the carry argument nor any other quantitative step uses the removed decoder. Applying Theorem 8.1 and Lemma 11.1, at depth (9.8), gives

$$\log W(\delta) \ge (3+r+\theta)\log_4(1/\delta) - o(\log(1/\delta)). \tag{11.9}$$

Indeed the intermediate count is $(3+r)M + \sum_{i<M} \mathrm{Tr}(\rho F_i) - o(1)$, and state transport bounds the sum below by $M(\theta - \omega_\theta(\delta)) - O(M^2\delta)$.

Replace every defining polynomial equation $p = 0$, including specified cross-commutators, by $h_q(p) = 0$, where

$$h_q(p) = \begin{cases} (p^*p)^{q/2}, & q \text{ even}, \\ p(p^*p)^{(q-1)/2}, & q \text{ odd}. \end{cases} \tag{11.10}$$

Keep the exact variable domains and the weighted state-commutator conditions. Since $h_q(p)^* h_q(p) = (p^*p)^q$, scalar Jensen applied to the probability measure $B \mapsto \mathrm{Tr}(\rho 1_B(p^*p))$ gives

$$\|p\sqrt{\rho}\|_2 \le \|h_q(p)\sqrt{\rho}\|_2^{1/q}. \tag{11.11}$$

No commutation between ρ and p^*p is needed. A powered precision-η model is consequently an original precision-$\eta^{1/q}$ model, also for its state commutators when $\eta \le 1$. Conversely the operator-norm bounds of Lemma 11.2 give

$$\|h_q(p)\| = \|p\|^q \le 3^q 4^{-qN},$$

and the flat state commutators vanish. All powered equations are fixed rational $*$-polynomials. Theorem 3.1 and the game transfer give original precision $(C\sqrt{\varepsilon})^{1/q}$ for the lower bound, and upper loss $C16^{-qN}$. Taking an even $N = \log(1/\varepsilon)/(4q) + O(1)$ above the required precision proves (11.8). All compiler dimension multipliers are constant for each fixed presentation. The upper losses tend to zero; the lower bound diverges, excluding a finite-dimensional perfect strategy. □

The left-c.e. dimension case of Theorem 1.1. Given a positive left-c.e. α, choose integers $q \ge 1$ and $m \ge 3$ with $m \le 4q\alpha < m + 1$. Put $r = m - 3$ and $\theta = 4q\alpha - m \in [0, 1)$. This is left-c.e.; an increasing rational name is $\max(0, 4q\alpha_s - m)$. Proposition 11.3 gives exponent α. The integers q, m are chosen once for the given α. The usable-entanglement assertion is proved in Section 12. □

Corollary 11.4. *Uncomputable dimension exponents of fixed finite BCS games accumulate at zero. In fact powering the original halting presentation gives exponents $(3 + \theta_K)/(4q)$, each Turing-equivalent to K.*

Proof. Apply (11.11) and the operator upper estimate of Theorem 7.1 to $\mathcal{P}_K$. The invertible rational affine map preserves Turing degree, and these positive exponents tend to zero. □

12 Usable entanglement of the original state

Dimension alone does not give a distillation protocol. The separable state $d^{-1}\sum_x |xx\rangle\langle xx|$ has marginal I/d and no distillable entanglement. The measurement count of Section 8 must therefore be supplemented by agreement between the players' actual operators. This section extends the binary triangular extraction of [12, Theorem 6.2] to measurements with an inactive outcome. Keeping the random targets preserves the mean number of active Bell pairs.

12.1 Triangular extraction with an inactive outcome

For operators D_ℓ and a vector ψ, write

$$\mathcal{R}_\psi((D_\ell)_\ell) = \left(\sum_\ell \|D_\ell\psi\|^2\right)^{1/2}.$$

Opposite-party tensor factors are suppressed below.

Lemma 12.1 (Extraction on a mixed state)**.** *Let ψ_{ABE} purify a finite-dimensional state ρ_{AB}. For $1 \le j \le m$, each party $D = A, B$ has an exact three-outcome PVM $(P^D_{j,\perp}, P^D_{j,0}, P^D_{j,1})$. Bob has a reflection U^B_j and Alice a unitary U^A_j. Put $F^D_j = P^D_{j,0} + P^D_{j,1}$. Suppose each of the following quantities is at most η:*

(i) For every j, $\mathcal{R}_\psi((P^A_{j,a} - P^B_{j,a})_a)$ and $\|(U^A_j - U^B_j)\psi\|$.

(ii) For each $s < j$, with $a, b \in \{\perp, 0, 1\}$, the two separate quantities $\mathcal{R}_\psi(([P^B_{j,a}, P^B_{s,b}])_{a,b})$ and $\mathcal{R}_\psi(([U^B_j, P^B_{s,b}])_b)$.

(iii) For every j, with $b \in \{0, 1\}$, $\mathcal{R}_\psi((U^B_j P^B_{j,b} U^B_j - P^B_{j,1-b})_b)$.

Set $\kappa = 7m^3\eta$ and $g(t) = (1+t)h_2(t/(1+t))$. If $\kappa \le 1/2$, then

$$E_D^{A\to B}(\rho_{AB}) \ge \sum_{j=1}^{m} \mathrm{Tr}(\rho_B F^B_j) - \tfrac{1}{2}m(m-1)\eta - 6m\kappa - 2g(\kappa). \tag{12.1}$$

A compulsory binary stage is included by $P^D_{j,\perp} = 0$. Different controls need not commute.

Proof. One stage. Alice samples a uniform bit a and sends it to Bob. Each party applies the local isometry

$$W^a_D = |0,0\rangle P^D_\perp + \sum_{b=0,1} |1,b\rangle (U_D)^{a\oplus b} P^D_b. \tag{12.2}$$

The first recorded bit is activity and the second is a coherent output qubit. Orthogonality of the three labels gives $(W^a_D)^* W^a_D = I$. A purification of the actual random operation is

$$\mathcal{A}\zeta = 2^{-1/2} \sum_a |a,a,a\rangle (W^a_A \otimes W^a_B)\zeta.$$

The third target copy records classical randomness in a new environment. It does not act on E. Compare this with the isometry

$$\mathcal{I}\zeta = \sum_a |a,a,a\rangle \left[2^{-1/2}|00\rangle_{\text{flags}}|00\rangle_{\text{qubits}} P^B_\perp \zeta + |11\rangle_{\text{flags}}|\Phi_2\rangle P^B_a \zeta\right]. \tag{12.3}$$

Its squared branch norms sum to $\|\zeta\|^2$.

For an arbitrary input ζ, let p, u, c be its projection-mirror aggregate, control-mirror norm and active-swap aggregate. Then

$$\|(\mathcal{A} - \mathcal{I})\zeta\| \le p + (u+c)/\sqrt{2}. \tag{12.4}$$

To check this, put $V_{a,\perp} = I$, $V_{a,b} = U^{a\oplus b}$ and use the exact identity

$$V^A_{a,r} P^A_r = P^B_r V^B_{a,r} + V^A_{a,r}(P^A_r - P^B_r) + P^B_r (V^A_{a,r} - V^B_{a,r}).$$

Multiplication by Bob's $V^B_{a,o} P^B_o$ kills the first term unless $o = r$. Equal inactive outcomes give $P^B_\perp$; equal active outcomes give P^B_a, apart from one swap defect when $r = 1 - a$. Summing squares over all records bounds the three error families by p, $u/\sqrt{2}$, and $c/\sqrt{2}$. For example $\sum_o \|V^B_{a,o} P^B_o z\|^2 = \|z\|^2$, and the control error survives only at $r = 1 - a$. Minkowski proves (12.4). No swap condition on Alice is used.

All histories. The ideal residual Kraus operators on either party are

$$R_{j,a,0} = P_{j,\perp}/\sqrt{2}, \qquad R_{j,a,1} = P_{j,a}.$$

They are self-adjoint and $\sum_{a,f} R^2_{j,a,f} = I$. For a history of j stages let K^B_h be the chronological product on Bob and Q^A_h the reversed product on Alice. Summing the innermost outcome repeatedly, without interchanging measurements, gives

$$\sum_h K^{B*}_h K^B_h = \sum_h K^B_h K^{B*}_h = I, \tag{12.5}$$

and both analogous identities for Q_h^A. The recursion

$$R^B K_h^B - Q_h^A R^A = R^B(K_h^B - Q_h^A) + Q_h^A(R^B - R^A)$$

therefore implies

$$\Delta_j := \mathcal{R}_\psi((K_h^B - Q_h^A)_{|h|=j}) \le j\eta.$$

For any family of Bob operators D_ℓ, opposite-party commutation and completeness give the useful estimate

$$\left(\sum_{h,\ell} \|D_\ell K_h^B \psi\|^2\right)^{1/2} \le \mathcal{R}_\psi((D_\ell)_\ell) + \left(\sum_\ell \|D_\ell\|^2\right)^{1/2} \Delta_j. \tag{12.6}$$

Indeed write $D_\ell K_h^B \psi = Q_h^A D_\ell \psi + D_\ell(K_h^B - Q_h^A)\psi$ and sum squares.

Expand a current operator's commutator with a history one factor at a time. Later outcomes disappear by completeness. At a prefix of length s, equation (12.6) bounds each commutator column by $(1 + Ls)\eta$. For the three current projections one can take $L = 3/2$, since commutators of two projections have norm at most $1/2$. For the current control take $L = \sqrt{3}$, since a unitary's commutator with a projection has norm at most one. The two scaled inactive copies count as one in the squared sum. Thus, at a stage preceded by j stages $(0 \le j < m)$, their history commutator aggregates are at most

$$(j + \tfrac{3}{4}j(j-1))\eta \le 3j^2\eta, \qquad (j + \tfrac{\sqrt{3}}{2}j(j-1))\eta \le 2j^2\eta.$$

For a mirror use $(G_A - G_B)K_h^B = K_h^B(G_A - G_B) - [G_B, K_h^B]$, rather than inserting a bipartite defect into (12.6). Its two aggregates obey

$$p_j \le (3j^2 + 1)\eta, \qquad u_j \le (2j^2 + 1)\eta.$$

The two Bob swap defects have operator root norm at most $2\sqrt{2}$, so (12.6) gives $c_j \le (1 + 2\sqrt{2}j)\eta$.

The ideal prefix is an orthogonal sum of histories, with normalized Bell or inactive output factors. Apply (12.4) to these histories together. The error of the stage preceded by j stages is at most

$$[(3 + \sqrt{2})j^2 + 2j + 1 + \sqrt{2}]\eta \le 7(j+1)^2\eta.$$

Telescoping the compositions, with actual isometries on the left and ideal prefixes on the right, bounds the final vector difference by $\sum_{j=0}^{m-1} 7(j+1)^2\eta \le 7m^3\eta = \kappa$. Completeness, rather than the number of histories, controls every sum.

Mean activation. Discarding the ideal stage's records gives Bob the channel $\Lambda_j(X) = \sum_r P_{j,r}^B X P_{j,r}^B$. The two recording vectors $(P_{j,r}^B\psi)_r$ and $(P_{j,r}^A\psi)_r$ are unit vectors at distance at most η. Their Bob marginals are $\Lambda_j(\rho_B)$ and ρ_B, respectively. Hence $\|\Lambda_j(\rho_B) - \rho_B\|_1 \le 2\eta$. Contractivity and telescoping bound a prefix disturbance by $2(j-1)\eta$. For a traceless self-adjoint X and $0 \le F \le I$, $|\operatorname{Tr}(FX)| \le \|X\|_1/2$. The ideal number N_{act} of active outputs therefore has

$$\mathbb{E}N_{\mathrm{act}} \ge \sum_j \operatorname{Tr}(\rho_B F_j^B) - \tfrac{1}{2}m(m-1)\eta. \tag{12.7}$$

The targets remain random throughout this argument. Selecting one fixed target could select a different activation density.

Hashing the actual output. Discard the original systems and targets and locally dephase the activity flags. Keep the output qubits coherent. The ideal output is

$$\sigma = \sum_f p_f |f,f\rangle\langle f,f| \otimes \bigotimes_{j=1}^m \begin{cases} \Phi_2, & f_j = 1, \\ |00\rangle\langle 00|, & f_j = 0. \end{cases}$$

Its coherent information is $I(A\rangle B)_\sigma = \mathbb{E}N_{\mathrm{act}}$: $S(B) = H(p) + \mathbb{E}N_{\mathrm{act}}$ and $S(AB) = H(p)$. The actual output ω has half trace distance at most κ from σ. Each local output space has dimension 4^m.

For states on dimension D at half trace distance t,

$$|S(\alpha) - S(\beta)| \le t \log D + g(t). \tag{12.8}$$

For a direct proof write $\alpha - \beta = \Delta_+ - \Delta_-$ with $\operatorname{Tr}\Delta_\pm = t$ and compare the two binary-mixture decompositions of $(\alpha + \Delta_-)/(1+t) = (\beta + \Delta_+)/(1+t)$. Concavity and the binary-mixture upper bound give (12.8) in both directions. Also g is increasing. Apply the bound to B and AB, of dimensions 4^m and 16^m. Then

$$I(A\rangle B)_\omega \ge \mathbb{E}N_{\mathrm{act}} - 6m\kappa - 2g(\kappa).$$

The hashing inequality [9, Theorem 10] makes this an achievable Alice-to-Bob rate. All preprocessing was one-way LOCC on ρ_{AB}, so composing it with hashing lower-bounds $E_D^{A\to B}(\rho_{AB})$. Together with (12.7) this proves the lemma. □

12.2 The bridge from an actual mixed strategy

Proposition 12.2. *For a loss-ε strategy of the compiled game of any presentation containing the clock, carry and conditional-clock relations of Section 5 and r extra compulsory clocks, powered by q (in particular G_K, the games of Proposition 11.3 and those of Theorem 15.7), the hypotheses of Lemma 12.1 hold on its original mixed state, after adjoining pure local ancillas, with $m = (4+r)M$ and*

$$\eta = C_{\mathcal{P},r,q}(M+1)^6 4^M \delta, \qquad \delta = (C_{\mathcal{P},r,q}\sqrt{\varepsilon})^{1/q}. \tag{12.9}$$

The active expectations are $(3+r)M + \sum_{i<M} \operatorname{Tr}(\rho_B F_i)$ for the reconstructed source model on the actual Bob marginal.

Proof. The actual-state interface is [12, Proposition 6.1]; the relevant precision ingredients are also visible in the compiler and Lemma 10.2. Here they must be applied before any pure-component selection. Relabel illegal Alice answers to a valid answer and dilate the finitely many POVMs with pure local ancillas. On a purification ψ_{ABE} of the original state, fix for each variable one incident Alice context. Its reflection A_v and Bob's reflection B_v satisfy

$$\|(A_v - B_v)\psi\| \le 2\sqrt{\varepsilon/\pi_{\min}} =: \mu.$$

This statement is valid with the inaccessible E retained. Moving a Bob word to Alice reverses its order and costs at most its length times μ. For a fixed polynomial certificate, transfer its right factors, then its context relation, which vanishes exactly on Alice. This gives all Boolean residuals $O(\mu)$. Lemma 10.2, with the rectangular amplitude matrix for AE against B, gives the square-root-state commutators at the same scale. Theorem 3.1(a) reconstructs the powered source on this actual marginal. Equation (11.11) gives the unpowered precision δ in (12.9). Pure local ancillas do not change distillable entanglement.

The raw measurements and controls in Lemma 9.3 are word polynomials of length $O(M)$ in Bob's reflections. Alice uses their opposite word polynomials. These are exact projections and reflections for the raw translated observables: for example, the opposite of $T_B^j P_B T_B^{-j}$ is $T_A^{-j} P_A T_A^j$, with T_A the opposite word for T_B. Their mirrors cost $O(M\mu) \le O(M\delta)$. Applying Lemma 10.2 with parties exchanged gives their Alice square-root-state commutators as well.

Repair each raw pair F_i, Q_i locally on both sides by $Q'_i = 1_{[1/2,1]}(F_i Q_i F_i)$. Bob's error is $O(\eta)$ by Lemma 9.3. On Alice, transferring $(I - F_i)Q_i$ reverses its order, giving the adjoint of her inclusion defect. For any operator R,

$$\|R^*\sqrt{\rho}\|_2 \le \|R\sqrt{\rho}\|_2 + \|[R, \sqrt{\rho}]\|_2.$$

The raw mirrors and state commutators therefore give Alice's inclusion defect $O(\eta)$. The repair proof in Lemma 9.3, using $FQF - (FQF)^2 = ((I - F)QF)^*((I - F)QF)$, gives $\|(Q'_i - Q_i)\sqrt{\rho}\|_2 = O(\eta)$ on each party. The triangle inequality now yields $\|(Q_i'^A - Q_i'^B)\psi\| = O(\eta)$. Thus the ternary PVMs have actual mirrors at the same scale.

Use the reverse of the count order: the last compulsory family first, each digit family reversed, then the optional indices reversed. A flip's preservation of later count measurements is exactly preservation of earlier extraction measurements. Bob's reflection converts the conjugation-form estimates into commutators at cost $O(\eta)$ by right multiplication and its state commutator. Lemma 9.3, with the extra clocks as in Proposition 11.3, supplies every swap and triangular relation. A fixed enlargement of $C_{\mathcal{P},r,q}$ changes individual errors into the finite root aggregates. Only Q_i was repaired, so the activation flags are the original F_i. This proves every assertion. □

Theorem 12.3. *The games $G_{\theta,r,q}$ satisfy*

$$d^{\rightarrow}_{G_{\theta,r,q}}(\varepsilon) = \frac{3 + r + \theta}{4q} \log(1/\varepsilon) + o(\log(1/\varepsilon)). \tag{12.10}$$

The original halting game satisfies the same formula with coefficient $c_K = (3 + \theta_K)/4$.

Proof. Put $t = \log_4(1/\delta)$ and choose M by (9.8). Then $\eta = O(t^{-24})$, $\kappa = O(t^{-21})$, $m\kappa = O(t^{-20})$, and every extraction error in (12.1) tends to zero. On every original strategy state,

$$E_D^{A\to B}(\rho_{AB}) \ge (3 + r)M + \sum_{i<M} \operatorname{Tr}(\rho_B F_i) - o(1).$$

Lemma 11.1 and $|\operatorname{Tr}(\rho_B F_i) - \operatorname{Tr}(\rho_B F)| \le 2i\delta$ bound this below by $(3 + r + \theta - \omega_\theta(\delta))M - O(M^2\delta) - o(1)$. Since

$$M = \frac{1}{4q} \log(1/\varepsilon) - O(\log\log(1/\varepsilon)),$$

taking the infimum proves the lower bound. For the upper bound, $E_D^{A\to B}(\rho) \le \log d_B$ for any state on Bob dimension d_B, so $d^{\rightarrow}_G(\varepsilon) \le \log D_G(\varepsilon)$. Proposition 11.3 matches the lower coefficient. The explicit upper strategies are in fact maximally entangled, with only constant compiler overhead. For G_K, use Theorems 6.2 and 7.1 in the same argument; its retained raw measurements are identical. This also proves the left-c.e. usable-entanglement assertion of Theorem 1.1. □

These are equalities for the infimum demand. A particular successful strategy may carry arbitrary unused extra entanglement. The non-effective remainder comes from flag-density convergence; the extraction error above tends to zero absolutely.

Corollary 12.4. *Every positive left-c.e. real belongs to both coefficient sets $\mathcal{A}_{\log}$ and $\mathcal{A}_{\dim}$ of [12, Section 11.6].*

Proof. In that notation $\mathcal{A}_{\log}$ consists of the finite limits $d^{\rightarrow}_G(\varepsilon)/\log_2(1/\varepsilon)$ for finite value-one games, and $\mathcal{A}_{\dim}$ uses $\log_2 D_G(\varepsilon)$ instead. Theorem 1.1 uses these exact conventions. The present section proves the left-c.e. usable case; Section 16 strengthens the inclusion to every positive Δ^0_2 real, including the positive-computable-real inclusion conjectured in the discussion of [12, Problem 18.3(i)]. It is consistent with that paper's Theorem 11.14, which bounds $\mathcal{A}_{\log}$ by the reals left-c.e. relative to $0''$. □

13 Heat depends on the order of reset and communication

Use the thermodynamic model of *The heat price of precision* [10]: degenerate energy endpoints, asymptotic reversible preparation and erasure, and all working systems included in the reset. Work and bath heat coincide in its dimensionless units, with one unit equal to $k_B T \ln 2$. Write $K_{\mathrm{ans}} = |\mathcal{A}||\mathcal{B}|$ for the product of the two answer-alphabet sizes; it is unrelated to the halting set K.

The local schedule allows no communication until both working devices are blank. The joint schedule permits quantum reunion after answer release and before reset. Theorem 2.1 of [10] gives, for every attainable loss,

$$2d_G^{\rightarrow}(\varepsilon) \le C_G(\varepsilon) \le w_{\mathrm{local},G}(\varepsilon) \le C_G(\varepsilon) + 2\log K_{\mathrm{ans}} \le 2\log D_G(\varepsilon) + 2\log K_{\mathrm{ans}}, \tag{13.1}$$

where C_G is the infimum mutual information of a successful state. The same theorem gives $0 \le w_{\mathrm{joint},G}(\varepsilon) \le 2\log K_{\mathrm{ans}}$. These are cited results, not new erasure claims. The matching-sandwich argument of its Corollary 2.2 applies here with $o(\log(1/\varepsilon))$ remainders rather than its sharper hypothesis for the earlier coefficient family.

Corollary 13.1 (Order of operations)**.** *For every positive Δ_2^0 real α, the same game G_α of Theorem 1.1 satisfies, with $L = \log_2(1/\varepsilon)$,*

$$w_{\mathrm{local},G_\alpha}(\varepsilon) = 2\alpha L + o(L), \qquad 0 \le w_{\mathrm{joint},G_\alpha}(\varepsilon) \le 2\log_2 K_{\mathrm{ans}}. \tag{13.2}$$

For the original halting game the local coefficient is $(3+\theta_K)/2$, Turing-equivalent to K, while joint-reset heat stays bounded.

Proof. Substitute $d_{G_\alpha}^{\rightarrow}(\varepsilon) = \alpha L + o(L)$ and $\log D_{G_\alpha}(\varepsilon) = \alpha L + o(L)$ into (13.1). The fixed answer term is $O(1)$. The joint bound is the cited theorem. The halting coefficient is $2c_K$, an invertible rational affine image of θ_K. □

Thus one fixed device realizes any prescribed positive Δ_2^0 local-heat coefficient (replace α by half that coefficient), while its joint-reset cost stays bounded. The order of communication and reset alone selects between these regimes. These are limiting coefficients; whether the finite-precision heat function is computable is not decided here.

13.1 Recovery before the blank deadline

Use exactly the block model of [10, Section 4, Theorem 4.1 and Lemma 4.2]. Fix a finite strategy and loss first, then process independent copies after answer release, allowing coherent local compression. Per play, recovery transfers at most q qubits and c physically moved fully dephased bits, up to sublinear block corrections, with an Alice-to-Bob route available. The blank deadline follows recovery. Degenerate endpoints, independent local baths, thermal answer ports and reversible preparation/erasure are retained; private workspace, communication hardware, copied messages and failed branches all enter reset. There is no uncharged catalyst. Here q is a communication budget, unrelated to a residual power.

Let $E_G(\varepsilon)$ be the infimum of entanglement entropy of successful *pure* strategies, not entanglement cost, and set $B_G = \log_2(|\mathcal{X}||\mathcal{Y}|K_{\mathrm{ans}})$. Write $x_+ = \max\{x, 0\}$ and

$$\mathcal{F}(e; q, c) = (e - q)_+ + (e - q - c)_+.$$

The cited theorem gives the general bound

$$\mathcal{F}(d_G^{\rightarrow}(\varepsilon); q, c) \le w_{q,c,G}(\varepsilon) \le \mathcal{F}(E_G(\varepsilon); q, c) + 4B_G. \tag{13.3}$$

Its proof supplies an arbitrary-original-state lower bound; no new recovery inequality is assumed here.

Corollary 13.2 (Programmable quantum-recovery threshold)**.** *For each matched game of Theorem 1.1, in the stated block model,*

$$w_{q,c,G_\alpha}(\varepsilon) = \mathcal{F}(\alpha L; q, c) + o(L), \qquad L = \log_2(1/\varepsilon), \tag{13.4}$$

uniformly over $q, c \geq 0$. In particular, for $q/L \to \kappa$ and $c/L \to \chi$ with $\kappa, \chi \geq 0$, the heat coefficient is

$$(\alpha - \kappa)_+ + (\alpha - \kappa - \chi)_+. \tag{13.5}$$

The critical quantum-recovery rate that removes the leading heat term is α, even with unlimited classical recovery. It can be any positive Δ_2^0 real.

Proof. For fixed measurements the score is affine in the state, so a successful mixed strategy has a successful pure component in the same local spaces. Pure-state distillable entanglement equals its marginal entropy. Consequently

$$d_G^{\rightarrow}(\varepsilon) \leq E_G(\varepsilon) \leq \log_2 D_G(\varepsilon).$$

Theorem 1.1 squeezes both endpoints to $\alpha L + o(L)$. The function $\mathcal{F}$ is nondecreasing and 2-Lipschitz in its first argument, with constant independent of q, c. Apply (13.3) to obtain (13.4); homogeneity and continuity give (13.5). Since $\mathcal{F}(e; q, c) \geq (e - q)_+$, a rate $\kappa < \alpha$ cannot remove the leading cost for any classical budget, whereas $\kappa \geq \alpha$ does so even with $c = 0$. □

The imported inequality applies to these games; with the present $o(L)$ demand laws it gives $o(L)$ heat at the critical rate, which need not be bounded.

Proposition 13.3 (Undecidable critical rates under a two-value promise)**.** *There is a computably generated family of finite rational matched games whose critical recovery rates lie in $\{15/16, 1\}$, on which no uniform estimator has a guaranteed fixed error less than $1/32$. On this promised family neither strict side of $31/32$ has a uniform semidecision procedure.*

Proof. Given a machine index e, the left-c.e. reader of Section 11, with no extra clock or residual power, uniformly realizes

$$\theta_e = \tfrac{3}{4} + \tfrac{1}{4}1_{\{e \in K\}}, \qquad \alpha_e = \frac{3 + \theta_e}{4} = \tfrac{15}{16} + \tfrac{1}{16}1_{\{e \in K\}}.$$

Its increasing rational approximation simply simulates machine e for longer finite times. An estimate within any fixed error less than $1/32$, compared with $31/32$, decides halting.

Include also a tagged complementary family from Section 15: let $A_e = \{0\}$ if $e \in K$ and $A_e = \varnothing$ otherwise. Its uniform c.e. program enumerates zero exactly on halting. Theorem 15.7 gives the matched rate

$$\alpha'_e = \frac{4 - \theta_{A_e}}{4} = 1 - \tfrac{1}{16}1_{\{e \in K\}}.$$

Corollary 13.2's proof uses only matched demands, so these are the respective critical rates. A semidecision for the low side on the union, restricted to the reader family, would enumerate nonhalting indices. A semidecision for the high side, restricted to the complementary family, would do the same. Both are impossible. Each individual rate is a rational; the obstruction is choosing it uniformly from a finite game description. □

13.2 An arithmetic upper bound for heat coefficients

Proposition 13.4. *For finite rational games of quantum value one, $C_G(\varepsilon)$ is uniformly right-c.e. at positive rational ε. Every existing finite local-reset heat coefficient in the stated thermodynamic model is left-c.e. relative to $0'$.*

Proof. At a fixed local dimension d, density matrices and POVMs with loss at most ε form a rational semialgebraic feasible set, after splitting complex entries into real and imaginary parts as in Lemma 2.4. Its algebraic points are dense even when it has empty interior. Indeed, intersect it with any rational open ball meeting it. This is a nonempty rational semialgebraic set, so real-closed-field transfer gives a point with real algebraic coordinates. Quantifier elimination decides nonemptiness and can supply such a point. Enumerating dimensions and rational balls therefore enumerates a dense algebraic feasible family in each dimension.

Spectra of algebraic density matrices, and hence their entropies and mutual informations, are computable. Continuity at fixed dimension, including $-t\log_2 t = 0$ at zero, shows that the infimum over this family is exactly $C_G(\varepsilon)$. Running minima of certified rational upper approximations to the enumerated mutual informations decrease to that infimum. A first feasible algebraic strategy can be found by dimension search under the value-one promise. Thus the approximation is uniform and total on that promise; no bound on the required dimension is assumed.

Choose total computable rationals $c_{n,s} \downarrow C_G(2^{-n})$ for $n \geq 1$. If the finite local-heat coefficient is h, the sandwich $C_G \leq w_{\mathrm{local},G} \leq C_G + 2\log K_{\mathrm{ans}}$ implies $C_G(2^{-n})/n \to h$. For rational a,

$$a < h \quad \Longleftrightarrow \quad \exists r \in \mathbb{Q}\ \exists N \quad r > a \ \text{ and } \ \forall n \geq N\ \forall s \quad c_{n,s} \geq rn. \tag{13.6}$$

If $a < h$, choose $a < r < h$ and use convergence. Conversely the right side gives $C_G(2^{-n}) \geq rn$ on that tail, hence $h \geq r > a$. The rational comparisons are decidable, so the lower cut is Σ^0_2, equivalently c.e. in $0'$. □

If $\mathcal{H}^+_{\mathrm{local}}$ denotes the positive finite local-heat coefficients of rational value-one games in this model, then

$$\{\alpha > 0 : \alpha \in \Delta^0_2\} \ \subseteq \ \mathcal{H}^+_{\mathrm{local}} \ \subseteq \ \{h > 0 : h \text{ is left-c.e. in } 0'\}. \tag{13.7}$$

For the first inclusion choose the matched dimension coefficient to be half the desired heat coefficient and use Corollary 13.1. The second is Proposition 13.4. Equality for the full heat spectrum remains open. The right-c.e. assertion concerns finite-precision mutual information, not distillable entanglement.

14 The exact-trace barrier and matrix approximation

Let $\mathcal{T}(\mathcal{P})$ be the exact tracial states of a finite bounded rational presentation, and let $\mathcal{T}_{\mathrm{mat}}(\mathcal{P})$ be those obtained as moment limits of flat finite-dimensional models with defects tending to zero. Both spaces are viewed in the compact space of bounded word moments. A diagonal choice of approximants shows that $\mathcal{T}_{\mathrm{mat}}$ is closed. In this section the relevant space is assumed nonempty and F is a fixed rational self-adjoint polynomial. These results explain why the construction of Sections 11–12 reaches only left-c.e. targets; they do not classify all games.

14.1 The exact-trace barrier

Proposition 14.1. *Suppose every exact trace has $\tau(F) \geq \theta$, and some sequence of flat approximate models with defects tending to zero has F expectations tending to θ. Then*

$$\theta = \min_{\tau \in \mathcal{T}(\mathcal{P})} \tau(F) \tag{14.1}$$

is left-c.e. The matched fixed-observable mechanism used here realizes exactly the left-c.e. targets in $[0,1]$.

Proof. A moment limit of the flat models is tracial and has $\tau(p^*p) = 0$ for every defining relation p. The GNS argument in Theorem 6.2 makes it an exact model. Its F expectation is θ, proving (14.1).

Here is an effective lower hierarchy for that minimum. At degree $2j$ use complex word moments, impose normalization, adjoints, cyclic invariance, a positive moment matrix, the domain localizing constraints, and every relation insertion visible at that degree. Add $|y_w| \le 1$ for contraction words. These bounded constraints express all declared domains. Alternatively use (3.1) for positive contractions and $P = (I+x)/2$, $x = x^* = x^{-1}$, for projections. The exact encodings are finite and rational and preserve tracial models and the substituted observable's trace. The finite feasible sets are compact rational semialgebraic sets in real and imaginary coordinates. Their lower minima ℓ_j increase. A diagonal limit of minimizers is a positive tracial functional with all relations and bounded domains, so its GNS tuple is an exact model. Consequently $\ell_j \uparrow \theta$.

Real quantifier elimination decides whether $\ell_j > r$ for a rational r. Enumerate these certified lower rationals and take running maxima, starting with a rational norm lower bound for F. They converge to θ. This proves left semicomputability without an assumption about numerical SDP duality. This is the bounded tracial moment method [2, 23]; it optimizes over all exact traces. It does not identify matrix-approximable traces, just as general NPA convergence concerns commuting models [33].

Lemma 11.1 and Lemma 11.2 give the converse realization for every left-c.e. $\theta \in [0,1]$. Fixed positive rational affine changes, extra clocks and residual powering preserve this class. □

Complementing F changes a minimum to one minus a maximum. It therefore does not reverse this one-sided forcing argument. If all exact traces gave one common value, the lower hierarchy for F and for $-F$ would make that value computable. Similarly, an irreversible acceptance program presented with a nonmonotone approximation keeps earlier accepted cylinders: their union is effectively open and its measure remains left-c.e.

14.2 Weighted limits are matrix-approximable

The next compression uses Connes' spectral-cut inequality [7], in the form Vidick uses for almost synchronous correlations [45]. Its role here is to identify possible limiting traces. Its dimension cost is never charged to the actual-state measurement bound.

Lemma 14.2 (Moment compression)**.** *Let contraction generators* $X_i \in M_d(\mathbb{C})$ *lie in the declared domains of Definition 2.1, and suppose* $\max_i \|[X_i, \sqrt{\rho}]\|_2 \le \eta$. *For each integer* $l \ge 1$ *and* $\zeta > 0$ *there is a finite-dimensional tuple* Y *in the same domains such that every* $*$*-word* w *of length at most* l *satisfies*

$$|\mathrm{tr}_D(w(Y)) - \mathrm{Tr}(\rho w(X))| \le 2l\sqrt{2\eta} + \zeta. \tag{14.2}$$

No bound on D *is asserted.*

Proof. Put $P_t = 1_{(t,\infty)}(\rho)$, $0 \le t \le 1$. Spectral integration gives $\rho = \int_0^1 P_t\,dt$ and $\int_0^1 \mathrm{rank}\,P_t\,dt = 1$. In an eigenbasis of ρ,

$$\begin{aligned}\int_0^1 \|[X_i, P_t]\|_2^2\,dt &= \sum_{a,b} |(X_i)_{ab}|^2 |\lambda_a - \lambda_b| \\ &\le \|[X_i, \sqrt{\rho}]\|_2 \left(2\,\mathrm{Tr}(\rho X_i X_i^*) + 2\,\mathrm{Tr}(\rho X_i^* X_i)\right)^{1/2} \le 2\eta.\end{aligned}$$

The middle inequality factors $|\lambda_a - \lambda_b|$ into the difference and sum of their square roots and applies Cauchy–Schwarz.

On $\operatorname{ran} P_t$ compress to $B_i = P_t X_i P_t$. Insert P_t between consecutive letters of a word of length k. Each of the $k-1$ trace errors is at most $\sqrt{\operatorname{rank} P_t}\,\|[X_i, P_t]\|_2$. Integrating and applying Cauchy–Schwarz bounds the total by $(k-1)\sqrt{2\eta}$.

Restore the domains inside each compressed space. Positive contractions need no repair. Round a projection at $1/2$, a reflection by spectral sign, and a unitary by polar decomposition, completing the partial isometry on its kernel. Each resulting Z_i satisfies

$$\|Z_i - B_i\|_2^2 \le \|[X_i, P_t]\|_2^2.$$

For projections use $\min(s^2, (1-s)^2) \le s(1-s)$ and $\operatorname{Tr}(B_i - B_i^2) = \|(I-P_t)X_iP_t\|_2^2$. For unitaries use $(1-s)^2 \le 1 - s^2$ and $\operatorname{Tr}(I - B_i^* B_i) = \|(I-P_t)X_iP_t\|_2^2$; spectral sign is the self-adjoint case. Telescoping the k repaired letters adds at most $k\sqrt{2\eta}$ after integration.

There are finitely many spectral cuts. The integrated repaired moments form a convex combination of normalized traces of the compressed tuples, with weights $a_h = (\operatorname{rank} P_h)\,\Delta t_h$ summing to one. If their dimensions are r_h, choose integer multiplicities n_h with $n_h r_h / \sum_i n_i r_i$ approximating a_h in ℓ^1 within ζ. The direct sum realizes those weights and changes every contraction-word moment by at most ζ. Combining both errors proves (14.2). □

Corollary 14.3. *Moment limits of weighted approximate models are exactly $\mathcal{T}_{\mathrm{mat}}(\mathcal{P})$. In particular, if $\lambda = \min_{\mathcal{T}_{\mathrm{mat}}} \tau(F)$, then uniformly over weighted precision-δ models,*

$$\operatorname{Tr}(\rho F) \ge \lambda - \omega(\delta), \qquad \omega(\delta) \to 0. \tag{14.3}$$

Proof. In a convergent weighted sequence pass to a subsequence with commutator errors $\eta_j \le j^{-4}$. Apply Lemma 14.2 to length j with $\zeta_j \to 0$. The resulting flat tuples have the same limiting moments. Each defining p^*p moment tends to zero, so their flat relation defects tend to zero. The reverse inclusion takes $\rho = I/d$. Compactness and the closedness of $\mathcal{T}_{\mathrm{mat}}$ give attainment of λ. A sequence violating (14.3) would have a weighted moment limit below λ, a contradiction. □

14.3 Arithmetic and the necessary trace gap

Proposition 14.4. *Set*

$$m_n = \inf\{\operatorname{tr}_d F(X) : d \ge 1,\ X \text{ a flat precision-}2^{-n} \text{ model}\}.$$

Then $\lambda = \sup_n m_n$ is left-c.e. relative to $0'$. If $a = \min_{\mathcal{T}} \tau(F)$ and λ is not left-c.e., then $a < \lambda$.

Proof. The feasible sets shrink, so m_n increases. Approximants to any trace in $\mathcal{T}_{\mathrm{mat}}$ show $m_n \le \lambda$. Conversely choose models with costs at most $m_n + 1/n$. A moment subsequence gives a trace in $\mathcal{T}_{\mathrm{mat}}$ of cost $\lim_n m_n$, proving equality.

For fixed n, d, all domains, squared flat residuals and the cost are rational semialgebraic data; real quantifier elimination decides whether there is a feasible tuple of cost below a rational r. Enumerating d makes m_n uniformly right-c.e. More explicitly,

$$\begin{aligned} s < \lambda \iff \exists n\ \exists r \in \mathbb{Q} \quad & s < r \text{ and} \\ & \text{every dimension lacks a precision-}2^{-n} \text{ model of cost } < r. \end{aligned} \tag{14.4}$$

The last finite-dimensional predicate is decidable. The displayed lower cut is therefore Σ_2^0, or c.e. in $0'$. Proposition 14.1's hierarchy proves that a is left-c.e., and $a \le \lambda$. A non-left-c.e. λ must consequently satisfy $a < \lambda$.

For any rational r between them there is an exact trace of cost below r, but some fixed positive precision admits no flat finite-dimensional model of cost at most r. Otherwise a sequence of such models would limit to an approximable trace below λ. This is the necessary trace gap. □

If all approximable traces have one common value, apply the same arithmetic argument to $-F$. Both cuts are then c.e. in $0'$, so the value is Δ^0_2. An arbitrary minimum above is only asserted to be left-c.e. in $0'$; that statement alone does not make it Δ^0_2.

14.4 Sharpness of the approximable-minimum range

Theorem 14.5 (All approximable projection minima)**.** *The values* $\min_{\tau\in\mathcal{T}_{\mathrm{mat}}(\mathcal{P})}\tau(F)$*, over finite bounded rational presentations with nonempty* $\mathcal{T}_{\mathrm{mat}}(\mathcal{P})$ *and a declared projection* F*, are exactly the reals in* $[0,1]$ *left-c.e. relative to* $0'$*. More precisely, for every total computable sequence* $a_s \in \mathbb{Q}\cap[0,1]$ *there is such a presentation* $\mathcal{Q}[a]$ *with*

$$\min_{\mathcal{T}_{\mathrm{mat}}(\mathcal{Q}[a])}\tau(F) = 1 - \limsup_s a_s. \tag{14.5}$$

Proof. Use the dynamic presentation of Section 16.1, but remove R_*, the stop counter with its zero test and witness, the stop branch with its scalar verifier, the REJECT control and their equations. Section 16.1 and Lemma 16.2 do not use Theorem 14.5 or Corollary 14.6, so this forward reference is not circular. At every depth the program continues: compute the packed list for $C(j,w)$ in (16.3), execute exactly one LOAD, test the entire list, READ the next bit and increment j. Keep every other scalar, clock, register and data relation, deleting R_* only from the commutation lists. No decoded-region restriction is imposed. The list computation terminates without searching for a perfect witness. This is one finite rational instruction table.

Put $u = \limsup_s a_s$. If $v(x) > u$, it has a prefix w of some length J such that $\ell(w) > u$ and $a_s < \ell(w)$ for all $s \ge J$. The gate $C(J,w)$ is false. Its complete formal execution has a finite runtime independent of its data. If $\tau(B_w(I-F)) > 0$, choose 2^k above that runtime. The retained fuel/time fairness gives

$$\tau(B_w C_k D_k(I-F)) = 2^{-(2k+1)}\tau(B_w(I-F)) > 0.$$

Lemma 16.2 applies: its proof uses the finite initialized interval and one LOAD, not convergence or a stop counter. Every shifted variable reduces this corner; the same tuple is tested throughout; its positive packed projection corner would give flat approximants of the false gate. This is impossible. Fuel/time flips establish only the displayed scalar mass identity, not transport of dynamic data. Every such bad cylinder has zero unflagged mass. Cover the compact set $\{v \ge u+\zeta\}$ by finitely many bad cylinders and refine to a common depth. Fair address digits then bound the remaining unflagged mass by $u+\zeta$ plus a vanishing dyadic endpoint error. Letting the error and ζ decrease proves $\tau(F) \ge 1-u$.

For the upper bound fix a dyadic $0 < b < u$ and an even N beyond its denominator depth. Set $I-F = 1_{\{v_N<b\}}$. Every prefix of each unflagged finite address has $\ell(w) \le v_N < b$, including after all N bits have been read and the reader supplies zeros. Arbitrarily late a_s exceed b, so every continuation gate on that address is true. In the finitely many address/fuel/time histories at this fixed N, only finitely many gates reach LOAD. Choose an exact witness for each whole packed list, even if reset truncates its test, and tensor these finitely many factors. There is no bound on this data dimension.

Assign the actual initialized scalar and register histories blockwise at time t mod 2^k, with $k = N$ for all-ones fuel. Load the assigned tuple on each outgoing LOAD edge, keep it until the next LOAD or reset, use $(0,I,I)$ from a reset until the first LOAD after it, and

use constant default data $(0, I, I)$ on flagged addresses. The contraction registers have their blockwise two-unitary representations without a new factor. Thus all new equations and declared coordinate/workspace commutations are exact: ordinary steps hold data, LOAD permits replacement, reset removes the holding guard, and cyclic wrap is a reset because 2^k divides 2^N. Every completed unflagged CHECK uses a true witness. The address dressing commutes with the new assignments, so the old residuals remain at most $3 \cdot 4^{-N}$. The flag mean is exactly $1 - b$.

Let $N \to \infty$ at fixed b and take a moment subsequence. It gives a trace in $\mathcal{T}_{\mathrm{mat}}$ with cost $1 - b$. Let $b \uparrow u$ and use compactness and closedness to attain $1 - u$. For $u = 0$, fully flagged histories with constant default data give cost one directly; the argument with $b < 1$ includes $u = 1$. This proves (14.5) and nonemptiness.

Downey–Jockusch–Schupp [14, Theorem 5.1(ii)–(iii)] characterize reals in $[0, 1]$ with Σ^0_2 lower cuts as the liminf of computable rational sequences. These are precisely the reals left-c.e. in $0'$. Clipping the sequence to $[0, 1]$ preserves its liminf. For $\lambda = \liminf_s c_s$, apply (14.5) to $a_s = 1 - c_s$. Proposition 14.4 supplies the reverse inclusion. □

Corollary 14.6 (An unavoidable witness-budget obstruction)**.** *Let $\mathcal{P}$ be a finite bounded rational presentation containing the relations used in Lemma 9.3, with flag F and $\lambda = \min_{\mathcal{T}_{\mathrm{mat}}(\mathcal{P})} \tau(F)$, for example $\mathcal{P} = \mathcal{Q}[a]$. If λ is not Δ^0_2, there are no flat models at every sufficiently large even N with operator residuals $O(4^{-N})$ and*

$$\log_2 d_N = (3 + \lambda)N + o(N). \tag{14.6}$$

Proof. Corollary 14.3 gives $\mathrm{Tr}(\rho F) \geq \lambda - \omega(\delta)$ on the actual precision-δ state, with $\omega(\delta) \to 0$. Set $t = \log_4(1/\delta)$ and $M = \lfloor t - 30 \log_4(2 + t) \rfloor$. Lemma 9.3 bounds the retained measurements' joint error by $C(M + 1)^6 4^M \delta = O(t^{-24})$. The raw clock count of Theorem 8.1 and flag transport give

$$\log_2 d \geq 3M + \sum_{i<M} \mathrm{Tr}(\rho F_i) - o(M) \geq (3 + \lambda - \omega(\delta))M - O(M^2\delta) - o(M).$$

The new data do not enter these measurements. The compiler gives $\delta = C\sqrt{\varepsilon}$, hence lower game exponent $(3 + \lambda)/4$. Hypothetical models (14.6) compile to loss $O(16^{-N})$ with constant dimension overhead. Neighboring even depths have bounded logarithmic-precision gaps, so they give the matching upper exponent at every small loss. The rational finite game has value one. Lemma 2.4 and the limit argument of Corollary 16.7 would make $(3 + \lambda)/4$, hence λ, Δ^0_2, a contradiction. □

For example, $\lambda = \sum_{e \in K^{0'}} 4^{-(e+1)}$ is left-c.e. in $0'$ and has degree $0''$: the tail after each digit is at most one third of its weight, so successive accurate approximations recover membership in $K^{0'}$, and that set computes the sum. It is not Δ^0_2. The theorem realizes this minimum, but the corollary forbids its matched budget. This obstruction concerns all sufficiently large even depths; it asserts neither a fixed positive overhead gap nor a bound on isolated subsequences.

Projection charges in $[0, 1]$	Exact arithmetic range
Exact-trace minima	Left-c.e.
Matrix-approximable minima	Left-c.e. relative to $0'$
Charges with a matched three-clock budget	Δ^0_2

The first row is Proposition 14.1 and the reader realization; the second is Theorem 14.5. In the third, "matched" means the retained clock lower count and the all-scale bound (14.6). Corollary 14.6 gives necessity, and Theorem 16.1, with $\theta = 1 - \lambda$, realizes every Δ^0_2 charge. Feasibility and feasibility within that budget are distinct.

Context: why a uniform lower hierarchy is different. Use the finite-dimensional *synchronous* value in Mousavi–Nezhadi–Yuen's hardness reduction [32, Lemma 6.7], which directly makes synchronous value one Π^0_2-hard. Place the game projection relations under a central projection p and set $F = I - p$. The scalar branch $p = 0$ always exists. The synchronous/tracial correspondence [35, 22] shows that synchronous value one gives approximable traces with $p = I$, hence minimum zero. Otherwise a positive-trace p corner would give finite synchronous strategies approaching one: compress to the representing projections, normalize by their positive limiting trace, and repair each finite PVM. For that last step, successively compress and round its first outcomes into the complement of earlier repaired projections, then use the remaining complement as the last outcome. All errors tend to zero for a fixed finite family. Thus a uniform ordinary lower hierarchy would enumerate the Σ^0_2-complete complement of synchronous value one. This contextual reduction produces only the individual minima zero and one. It uses the synchronous-value hardness itself, without a conversion from unrestricted strategies that would require extra assumptions on diagonal question weights.

Towards the next construction. The exact-trace barrier leaves open a different matched mechanism: a non-left-c.e. minimum over matrix-approximable traces, together with precision-4^{-N} flat models of dimension $2^{(3+\lambda)N+o(N)}$. The trace gap alone supplies no such upper models. Section 15 constructs them using finite perfect witnesses and an elementary slow-admission schedule. It thereby proves that the dimension spectrum strictly extends beyond the left-c.e. reals. Section 16 completes the positive classification; the remaining questions are in Section 17.

15 Escaping the barrier: a complementary halting flag

The barrier in Section 14 concerns the minimum over *all* exact traces. We now construct a presentation whose minimum over matrix-approximable traces is

$$\lambda_A = 1 - \theta_A, \qquad \theta_A = \sum_{e \in A} 4^{-(e+1)},$$

for any c.e. set A. For $A = K$ this minimum is not left-c.e. The clocks retain their matched dimension budget. Every statement in this section is proved here, except the MIP* = RE input of Section 15.1 and, in Theorem 15.7, the cited Boolean correspondence, actual-state interface and heat sandwich.

15.1 Finite perfect witnesses as an external input

We use Ji–Natarajan–Vidick–Wright–Yuen's MIP* = RE theorem (arXiv version 4, accepted to Annals of Mathematics) in its finite perfect-completeness form [20, Theorem 12.6, pp. 193–194]. The theorem gives a computable map from machines to finite rational games: a halting machine yields a finite-dimensional perfect PCC strategy, and a nonhalting machine yields quantum value at most 1/2. Its proof starts with the trivial perfect PCC strategy of Lemma 12.3 and applies Claim 12.7 in finitely many downward steps, using the compression completeness of Theorem 12.1(1), p. 184. Only the finite-dimensional perfect PCC strategy of Theorem 12.6(1) is used: a state ψ on $H \otimes H$ and projective measurements, each player's measurements consistent on ψ in the sense of [20, Definition 5.10]. PCC means projective, consistent and commuting in that paper's sense; no symmetry between the players is assumed. The informal synchronous wording of Section 1.2 of [20] is not needed; mere quantum value one would not suffice.

Apply the map to the machine which halts precisely when $e \in A$, and denote the resulting game by $H_{A,e}$, with question sets $X_{\mathrm{A}}, X_{\mathrm{B}}$. Introduce projections p_{xa} ($x \in X_{\mathrm{A}}$) and q_{yb} ($y \in X_{\mathrm{B}}$), impose the PVM equations of both families, and impose $p_{xa}q_{yb} = 0$ for each tested losing pair. Untested pairs impose no zero-product condition. Call this finite rational projection presentation $\mathcal{A}_{A,e}$.

Lemma 15.1. *If $e \in A$, then $\mathcal{A}_{A,e}$ has an exact unital matrix model. If $e \notin A$, it has no sequence of flat matrix models whose defining defects tend to zero. Exact matrix certificates, without repeated indices, can be enumerated effectively, and their indices are exactly A.*

Proof. Let (A, B, ψ) be a finite-dimensional perfect PCC strategy and write $\psi = \mathrm{vec}(S)$. Consistency of each player's measurements reads $A_a^x S = S(A_a^x)^{\mathsf{T}}$ and $S(B_b^y)^{\mathsf{T}} = B_b^y S$; taking adjoints shows that every A_a^x and B_b^y commutes with Alice's reduced state $\rho = SS^*$. Write $\rho = \sum_c cP_c$. Its eigenspaces reduce both projection families. For a tested losing pair, consistency of Bob's measurement and perfect play give

$$0 = \langle \psi, A_a^x \otimes B_b^y \, \psi \rangle = \mathrm{Tr}(\rho A_a^x B_b^y) = \sum_{c>0} c \, \|(P_c B_b^y P_c)(P_c A_a^x P_c)\|_2^2.$$

Every summand is nonnegative. On each nonzero positive-eigenvalue eigenspace the two compressed families are PVMs and every tested losing product vanishes, including its adjoint. Restrict to any such eigenspace to obtain the unital matrix model. The commuting clause of PCC is not needed for this argument.

Conversely, approximate projection partitions can be repaired to PVMs in the same dimension. For projections $P_1, \ldots, P_k$ with $S_0 = \sum_i P_i$ and $\|S_0 - I\|_{2,d} \to 0$, where $\|X\|_{2,d}^2 = \mathrm{tr}_d(X^*X)$,

$$\sum_{i \neq j} \mathrm{tr}_d(P_i P_j) = \mathrm{tr}_d(S_0^2) - \mathrm{tr}_d(S_0) \longrightarrow 0.$$

Each term is nonnegative and equals $\|P_i P_j\|_{2,d}^2$. Keep P_1, successively compress P_i to the complement R of the previously repaired outcomes and round at $1/2$, and use the last complement as the final outcome. If $\alpha = \|(I - R)P_i\|_{2,d}$, then

$$\|P_i - RP_iR\|_{2,d} \leq 2\alpha, \qquad \|1_{[1/2,1]}(RP_iR) - RP_iR\|_{2,d}^2 \leq \mathrm{tr}_d(RP_iR - (RP_iR)^2) \leq \alpha^2.$$

Finite induction and the approximate sum equation give corrections tending to zero. Bounded-product telescoping preserves every tested losing-product defect. Using the repaired PVMs on a maximally entangled state, with Alice using the transposed p_{xa}^{T} and Bob q_{yb}, gives losing probabilities $\mathrm{tr}_d(p_{xa}q_{yb}) = \|q_{yb}p_{xa}\|_{2,d}^2 \to 0$. This contradicts the soundness bound when $e \notin A$.

For fixed e, d, real quantifier elimination decides the rational semialgebraic matrix equations, after splitting complex entries into real and imaginary parts. A nonempty set has an effectively obtainable algebraic point. Dovetail over e, d and suppress repeated indices. The first two assertions make its output exactly A. No dimension bound computable from e is asserted. □

15.2 Packing and a fixed polynomial-testing program

For a projection presentation $\mathcal{A}$ with $m \geq 1$ generators, use two projections D, E and a unitary W. Set $G_j = W^{-j}DW^j$, $0 \leq j < m$, and impose

$$\sum_{j<m} W^j E W^{-j} = I, \qquad [E, G_j] = 0. \tag{15.1}$$

Substitute EG_jE for its generators in every equation of $\mathcal{A}$, replacing the constant identity by E. Clear denominators, and call the finite integer-coefficient list consisting of (15.1) and the substituted equations $\mathcal{R}(\mathcal{A})$. In $\widehat{\mathcal{P}}_A$ these are not defining relations; the program tests them.

Lemma 15.2 (Three data generators)**.** *A d-dimensional model of $\mathcal{A}$ extends to an md-dimensional model of $\mathcal{R}(\mathcal{A})$. Conversely the E corner of every tracial model of the latter carries $\mathcal{A}$, and $\tau(E) = 1/m$.*

Proof. List the projection generators as $g_0, \dots, g_{m-1}$, including both players' families for $\mathcal{A}_{A,e}$. Take $D = \operatorname{diag}(g_0, \dots, g_{m-1})$, $E = |0\rangle\langle 0| \otimes I_d$, and let W cyclically shift the m blocks forward. Then $W^{-j} D W^j$ restricts to g_j on block zero, proving the extension, also when $m = 1$. Conversely $E G_j E$ is a projection in the unital E corner and satisfies every substituted relation. Tracing (15.1) gives $m\tau(E) = 1$. Arbitrary models need not satisfy $W^m = I$. □

Write $m_e = \#\{p_{xa}\} + \#\{q_{yb}\}$ for the total number of projections in both families of $\mathcal{A}_{A,e}$, adding an unused projection if needed, and $\mathcal{R}_e = \mathcal{R}(\mathcal{A}_{A,e})$. The map $e \mapsto \mathcal{R}_e$ is total computable. Retain the decoder L, R, reset Z, clocks, carry and conditional-clock relations of Section 5. Replace the halting recognizer by the following fixed counter transducer, delete (5.11), and add

$$(I - L)(I - F) = 0. \tag{15.2}$$

Thus an unflagged address must be in the decoded region. Keep the scalar counter domains, zero tests, controls, initialization and transition rules; their start and input are still determined by L, R.

Add the data generators D, E, W. Require them to commute with A_a, L, R, F, Z, every fuel and time generator including U, and every instantaneous scalar workspace variable. Do not require them to commute with the address shift or flip, or with one another.

The transducer has two operator registers, a monomial M and an accumulator Σ. Represent each $S \in \{M, \Sigma\}$ by two unitary generators:

$$S = (u_S + v_S)/2. \tag{15.3}$$

Every such S is a contraction, and every finite square contraction has this form. Indeed extend its polar partial isometry to a unitary V_0, put $J = |S| + i(I - |S|^2)^{1/2}$, and take $u_S = V_0 J$, $v_S = V_0 J^*$. This model assignment does not add irrational coefficients to (15.3).

On input e, compute the finite list $\mathcal{R}_e$. For each polynomial r, express its integer coefficients as repeated signed monomials, and compute an integer $b \geq 0$ with 2^b at least their total number. Clear Σ, set $M = I$ for each monomial, build it by successive right multiplications, and add its signed value times 2^{-b} to Σ. Enter CHECK, then process the next polynomial and finally idle. The extra instruction alphabet is

$$\Sigma := 0, \quad M := I, \quad M := Md \ (d \in \{D, E, W, W^*\}), \quad \Sigma := \Sigma \pm r_b M,$$

and holding a register fixed. The ordinary counter encoding supplies $r_b = 2^{-b}$. The final idle branch clears the registers to their initialized values. Every partial monomial and accumulator is a contraction, including on flagged addresses with arbitrary data.

Counter-machine universality compiles this algorithm into one fixed finite instruction table. Its finite branch selectors q_α and updates define rational polynomials Φ_S. The polynomial lists and word lengths are program outputs, rather than additional defining equations. Put $\operatorname{Init}(M) = I$, $\operatorname{Init}(\Sigma) = 0$, and impose

$$ZS = Z \operatorname{Init}(S), \quad U^* S U = Z_+ \operatorname{Init}(S) + (I - Z_+)\Phi_S, \quad Z_+ = U^* Z U, \tag{15.4}$$

$$q_{\mathrm{CHECK}} \Sigma (I - F) = 0. \tag{15.5}$$

For each of the four representing unitaries in (15.3), impose commutation with the instantaneous scalar workspace and with $A_a, A_f, A_t, L, R, F, Z$. Impose no commutation with other operator registers, data, time shifts, or fuel/time flips. Scalar workspace here excludes M, Σ. This completes a finite bounded rational presentation, denoted $\widehat{\mathcal{P}}_A$.

Lemma 15.3 (Programmed testing)**.** *Every faithful exact tracial model satisfies, for every e and $r \in \mathcal{R}_e$,*

$$B_e(I-F)r(D,E,W) = 0. \tag{15.6}$$

Proof. Fix the finite CHECK time h for this relation and choose $2^k > h$. Put $Q_0 = B_e C_k D_k (I - F)$, using (5.7). The scalar initialization lemma gives $ZQ_0 = Q_0$, with no reset before h, and forces the scalar controls and counters throughout this finite history. Non-self-adjoint registers require an extra justification: $[S, Z] = 0$ gives

$$SQ_0 = SZQ_0 = ZSQ_0 = Z\,\mathrm{Init}(S)Q_0.$$

Consequently $MQ_0 = Q_0$ and $\Sigma Q_0 = 0$ initially.

Every data word commutes with Q_0; the address digits are spectral functions of A_a in an exact clock. The data also commute with U. If $S_l = U^{-l}SU^l$ has right action $S_l Q_0 = wQ_0$, then $S_l dQ_0 = wdQ_0$ for a data letter d. A conjugated scalar selector commutes with every data word, since the data commute with that selector and with U. Its forced scalar right action therefore holds on wQ_0 as well. Induction in (15.4) proves the initialized operator history. Products do not require commuting computed registers: if $AQ_0 = aQ_0$, $BQ_0 = bQ_0$, and $[b, Q_0] = 0$, then $ABQ_0 = AQ_0 bQ_0 = abQ_0$. Before h the next-reset factor vanishes on Q_0, whose range also contains every computed right action. At CHECK,

$$\Sigma_h Q_0 = 2^{-b} r(D,E,W)Q_0, \qquad (q_{\mathrm{CHECK}})_h Q_0 = Q_0.$$

Conjugate (15.5) by U^{-h} and apply it to Q_0. The flag is time invariant, and the conjugated selector commutes with data words, so $r(D,E,W)Q_0 = 0$.

The operator $r(D,E,W)B_e(I-F)$ commutes with all fuel/time flips. Conjugate its zero product with $C_k D_k$ by the earliest specified fuel flip and add: that flip exchanges the two outcomes of its digit and preserves later digits. Repeat through the final zero of C_k, then through D_k. This removes both cylinders and proves (15.6). No conditional trace independence is needed. Since $\mathcal{R}_e$ includes the packed relations for both families, the E corner of every nonzero unflagged cylinder therefore carries both the p_{xa} and q_{yb} PVMs and all tested losing products. □

15.3 The approximable flag minimum

Proposition 15.4. *Every $\tau \in \mathcal{T}_{\mathrm{mat}}(\widehat{\mathcal{P}}_A)$ satisfies $\tau(F) \geq \lambda_A$.*

Proof. Let $Q_e = B_e(I-F)$. It commutes with the data and, by Lemma 15.3, its corner satisfies $\mathcal{R}_e$. Tracing the first packed equation gives $\tau(Q_e E) = \tau(Q_e)/m_e$. If $\tau(Q_e) > 0$, the $Q_e E$ corner carries $\mathcal{A}_{A,e}$, with generators given by fixed bounded polynomials $Q_e E W^{-j} D W^j E$.

This positive-trace corner inherits flat approximation. In the flat approximants, self-adjointize the polynomial for $Q_e E$ and round at $1/2$. Its projection defect tends to zero, and for every real t,

$$|1_{[1/2,\infty)}(t) - t| \leq 2|t^2 - t|.$$

The correction tends to zero in normalized Hilbert–Schmidt norm, and the rounded projection's rank fraction tends to $\tau(Q_e E) > 0$. Compress and self-adjointize the finitely many corner generators, and round them by the same estimate. Fixed polynomial relations survive by bounded-product telescoping. Normalizing the corner trace multiplies norms by a bounded factor tending to $\tau(Q_e E)^{-1/2}$. These are flat approximate models of $\mathcal{A}_{A,e}$. Lemma 15.1 therefore gives $\tau(B_e(I-F)) = 0$ for $e \notin A$.

The decoder exhausts L in trace. Iterating (5.3) yields

$$L = \sum_{e<s} B_e + R_s, \qquad 0 \leq R_s \leq \prod_{j<s} T_a^{2j} E_{00} T_a^{-2j}, \qquad \tau(R_s) \leq 4^{-s}.$$

Indeed $L - \sum_{e<s} B_e$ is a projection whose range lies in the all-00 prefix; no commutation between rotated L and all earlier digits is assumed. Since $I - F \le L$, the disjoint cylinders of trace $4^{-(e+1)}$ give

$$\tau(I - F) = \sum_e \tau(B_e(I - F)) \le \sum_{e \in A} 4^{-(e+1)} = \theta_A.$$

Only the passage excluding nonmember corners uses matrix approximability. The testing identity holds for every exact trace. □

15.4 Slow admission and simultaneous finite models

Lemma 15.5 (Elementary witness scheduling)**.** *Suppose finite witnesses with positive integer costs a_j arrive in a computable stream without repeated indices. At even depth N admit its longest prefix arrived by stage N, with every index $e_j < N/2$ and total cost at most $\lfloor \sqrt{N} \rfloor$. These prefixes are nested and eventually contain every fixed witness. Their total cost is $o(N)$.*

Proof. For a fixed prefix its arrival time, largest index and total cost are finite constants. All three cutoffs eventually exceed them, and none decreases with N. The cost bound is $\sqrt{N} = o(N)$. An enormous early witness may cause an enormous finite delay; no convergence rate is claimed. More generally any nondecreasing computable unbounded budget $o(N)$ has the same property. □

Apply this lemma to the certificates $(e_j, d_j, \mathrm{model}_j)$ of Lemma 15.1, with $a_j = \lceil \log_2 \max(2, m_{e_j} d_j) \rceil$. Use a dovetailing with finitely many outputs per stage. Let S_N be the admitted indices. Then $S_N \uparrow A$, and the common data factor

$$\mathcal{H}_{\mathrm{data},N} = \bigotimes_{e_j \in S_N} (\mathbb{C}^{m_{e_j}} \otimes \mathbb{C}^{d_j}), \qquad \ell_N = \dim \mathcal{H}_{\mathrm{data},N}$$

has $\log \ell_N \le \sqrt{N}$; the empty product has dimension one.

Lemma 15.6 (Complementary finite models)**.** *For even $N \ge 2$, $\widehat{\mathcal{P}}_A$ has flat models with all defining residuals at most $3 \cdot 4^{-N}$ in operator norm and*

$$d_N = 2^{3N + k_N} \ell_N, \qquad k_N = \lambda_N N + O(\log(N+2)), \qquad \lambda_N = 1 - \sum_{e \in S_N} 4^{-(e+1)} \downarrow \lambda_A. \tag{15.7}$$

Every new defining equation is exact, and $\mathrm{tr}_{d_N}(F) = \lambda_N$.

Proof. On the N-bit address cycle set

$$f_N(x) = 1 - \sum_{e \in S_N} 1_{\{x \equiv 4^e \pmod{4^{e+1}}\}}, \qquad k_N = 1 + \max_x \sum_{j=1}^{N} f_N(x - j). \tag{15.8}$$

The classes are disjoint, and their periods divide 2^N. Bit zero is least significant for increment, so these are precisely the cylinders B_e. The mask is Boolean, has mean λ_N, and satisfies (15.2). For $s = \lfloor \log_4 N \rfloor$, each selected class with $e \le s$ has interval-count error at most one. The disjoint union of the remaining classes lies in the multiples of 4^{s+1} and has at most one point in an interval of length $N < 4^{s+1}$; its expected count is at most $N/(3 \cdot 4^{s+1})$. Hence, uniformly in the subset and starting point,

$$\left| \sum_{j=1}^{N} f_N(x - j) - N\lambda_N \right| \le s + 2 + \frac{N}{3 \cdot 4^{s+1}}. \tag{15.9}$$

Cyclic intervals lift to integer intervals because all periods divide the cycle.

Use the three compulsory N-qubit registers, the k_N optional qubits, and the common data factor. Theorem 7.1's conditional permutation, series and dressing require only a Boolean mask and that k_N exceed every N-window count. They therefore apply to (15.8). On address $x \in B_{e_j}$ for an admitted index, assign the packed witness to factor j and tensor identities on the others. On other addresses take $(D, E, W) = (0, I, I)$. These address-block data are independent of fuel, time and optional slots. They commute with A_a, L, R, F, Z and all scalar workspace, and with the complete fuel/time sets. Address dressing acts on address and optional slots and leaves these commutations intact. The data need not commute with the dressed address shift or flip.

For fuel with k initial ones, use the actual initialized scalar and operator history at time $t \bmod 2^k$; all-ones fuel uses $k = N$. On each (x, q, t) block the scalar workspace is a scalar multiple of the data identity. Every M, Σ is a contraction by the signed-term bound. Choose their two representing unitaries within each block and take direct sums. All required representing-unitary commutators hold; naming these matrices adds no tensor factor.

When the next time residue is zero, the transition prescribes initialization; otherwise it prescribes the next instruction. Since $2^k \mid 2^N$, the cyclic transition from $2^N - 1$ to zero is also a reset. Thus (15.4) holds exactly at every ordinary and reset boundary. On an unflagged block every completed CHECK tests the packed witness belonging to that address, so it is zero. An unfinished CHECK imposes no test there; on flagged blocks $I - F = 0$. Therefore (15.5) is exact regardless of program speed.

The address wraparound is handled as in Theorem 7.1. In the gauge of the dressing (7.8), V is ordinary increment except at the all-ones address, where it also applies the fixed optional permutation $R_0^{\sum_x f_N(x)}$. There its output address differs from that of the recursive increment, so the guarded defect is $\sqrt{2}\,4^{-N}$ in operator norm, as before.

All new equations are exact simultaneously. The remaining clock, guarded-increment, carry and conditional-tail residuals have their old bounds, at most $3 \cdot 4^{-N}$. Counting the registers gives (15.7), and

$$\log d_N = (3 + \lambda_N)N + O(\log(N + 2)) + O(\sqrt{N}) = (3 + \lambda_A)N + o(N).$$

A fixed increase of even depth absorbs the constant three. Bounded gaps between even depths suffice for every small precision. Computing a certificate or its admission stage is classical model-description work; it consumes neither simulated fuel nor an extra quantum register. □

15.5 Dimension, usable entanglement and heat

Theorem 15.7. *For every c.e. set A, integer $r \geq 0$ and integer $q \geq 1$, there is a fixed finite rational BCS game $\widehat{G}_{A,r,q}$ of quantum value one and with no finite-dimensional perfect strategy. With*

$$\beta_{A,r,q} = \frac{4 + r - \theta_A}{4q}, \qquad L_\varepsilon = \log_2(1/\varepsilon),$$

its resource laws are

$$\log_2 D_{\widehat{G}_{A,r,q}}(\varepsilon) = \beta_{A,r,q} L_\varepsilon + o(L_\varepsilon), \tag{15.10}$$

$$d^{\rightarrow}_{\widehat{G}_{A,r,q}}(\varepsilon) = \beta_{A,r,q} L_\varepsilon + o(L_\varepsilon), \tag{15.11}$$

$$w_{\mathrm{local},\widehat{G}_{A,r,q}}(\varepsilon) = 2\beta_{A,r,q} L_\varepsilon + o(L_\varepsilon), \qquad 0 \leq w_{\mathrm{joint},\widehat{G}_{A,r,q}}(\varepsilon) \leq 2\log_2 K_{\mathrm{ans}}. \tag{15.12}$$

The heat statement uses exactly the model and units of Section 13.

Proof. The actual-state flag bound. Moment compactness of Lemma 15.6's flat models supplies a matrix-approximable trace of flag expectation λ_A. Proposition 15.4 gives the reverse inequality, so

$$\min_{\mathcal{T}_{\mathrm{mat}}(\widehat{\mathcal{P}}_A)} \tau(F) = \lambda_A. \tag{15.13}$$

Corollary 14.3 now proves, uniformly on every original weighted precision-δ model,

$$\mathrm{Tr}(\rho F) \geq \lambda_A - \omega_A(\delta), \qquad \omega_A(\delta) \longrightarrow 0. \tag{15.14}$$

Indeed a fixed positive shortfall along $\delta \to 0$ would have a weighted moment limit in $\mathcal{T}_{\mathrm{mat}}$ below (15.13). The bound holds on the original state ρ; compression only identifies the limit class.

Dimension. Every relation used by Lemma 9.3 remains in the presentation. For $t = \log_4(1/\delta)$ and $M = \lfloor t - 30\log_4(2+t) \rfloor$, its joint errors are at most $C(M+1)^6 4^M \delta$. Theorem 8.1 and state transport give

$$\log d \geq 3M + \sum_{i<M} \mathrm{Tr}(\rho F_i) - o(M) \geq (3 + \lambda_A - \omega_A(\delta))M - O(M^2\delta) - o(M),$$

since $|\mathrm{Tr}(\rho F_i) - \mathrm{Tr}(\rho F)| \leq 2i\delta$. This is $(3+\lambda_A)\log_4(1/\delta) - o(\log(1/\delta))$. The compiler sends game loss ε to source precision $C\sqrt{\varepsilon}$, with constant dimension overhead. Lemma 15.6 gives the converse upper loss $C'16^{-N}$. Choose an even depth $N \geq \log_{16}(C'/\varepsilon)$ within a fixed additive constant to match the bound through all small losses. This proves (15.10) for $r = 0, q = 1$.

Append r compulsory clocks with the cross-commutations of Proposition 11.3. Their upper cost is exactly rN, and their digit families add rM to the lower count. Power *every* defining residual, including the new tests and cross-commutators, by (11.10); keep domains and state-commutator requirements. Equation (11.11) turns powered precision η into unpowered precision $\eta^{1/q}$, whereas the explicit operator-norm residuals become at most $3^q 4^{-qN}$. Thus the game lower precision is $\delta = (C\sqrt{\varepsilon})^{1/q}$, upper loss is $C'16^{-qN}$, and the coefficient is $(3 + r + \lambda_A)/(4q)$. The upper losses tend to zero and this positive lower bound diverges, excluding a finite-dimensional perfect strategy.

Usable entanglement on the original mixed state. Proposition 12.2 applies. On a purification of an arbitrary mixed loss-ε strategy, keep the original environment, relabel illegal answers, and dilate locally with pure ancillas. The finite Boolean certificates reconstruct this presentation on the actual Bob marginal with powered precision $C\sqrt{\varepsilon}$; their constants depend on the fixed transducer but not on M or ε. Jensen then gives the same unpowered δ. New data/register variables enlarge a fixed certificate list only. None occurs in the raw translated clock measurements or their controls.

Consequently the same opposite-word construction on Alice, independent local inclusion repairs on both sides, and reverse count order supply the hypotheses of Lemma 12.1, with

$$m = (4+r)M, \qquad \eta = C_{A,r,q}(M+1)^6 4^M \delta.$$

As in Proposition 12.2, only Q_i is repaired, so the activation flags are the raw F_i on the actual marginal.

At the chosen depth $\eta = O(t^{-24})$, and the vector, activation and entropy-continuity errors of Lemma 12.1 tend to zero. Its one-way protocol and hashing therefore give on *every original strategy state*

$$\begin{aligned} E_D^{A\to B}(\rho_{AB}) &\geq (3+r)M + \sum_{i<M} \mathrm{Tr}(\rho_B F_i) - o(1) \\ &\geq (3 + r + \lambda_A - \omega_A(\delta))M - O(M^2\delta) - o(1). \end{aligned}$$

Here (15.14), rather than an all-exact-trace bound, supplies the activation estimate. Since $M = L_\varepsilon/(4q) - O(\log L_\varepsilon)$, the infimum gives the lower side of (15.11). The general inequality $d^{\rightarrow}_G(\varepsilon) \le \log D_G(\varepsilon)$ and the proved dimension upper bound give the other side.

Heat. Substitute the two matching laws into (13.1). Its fixed answer term is $O(1)$, so both endpoints have coefficient $2\beta_{A,r,q}$. The joint bound is the same cited heat theorem. This proves (15.12). □

Corollary 15.8 (Beyond the left-c.e. reals)**.** *The game $\widehat{G}_K := \widehat{G}_{K,0,1}$ has dimension and usable-entanglement coefficient $(4-\theta_K)/4$, and local-heat coefficient $(4-\theta_K)/2$. These are noncomputable right-c.e. reals, not left-c.e., and have the Turing degree of K. Recall the positive dimension spectrum $\mathcal{B}_{\mathrm{BCS}}$ from Section 1. This escape establishes*

$$\{\textit{positive left-c.e. reals}\} \subsetneq \mathcal{B}_{\mathrm{BCS}} \subseteq \{\textit{positive } \Delta^0_2 \textit{ reals}\}. \tag{15.15}$$

Both $\mathcal{A}_{\log}$ and $\mathcal{A}_{\dim}$ also strictly contain the positive left-c.e. reals. The new non-left-c.e. examples accumulate at zero by taking $q \to \infty$.

Proof. The weights $w_e = 4^{-(e+1)}$ have tail at most $w_e/3$ after position e. Given earlier membership bits, absence leaves residual at most $w_e/3$, whereas presence leaves at least w_e. Approximating θ_K within $w_e/6$ distinguishes them at threshold $2w_e/3$. Thus $K \le_T \theta_K$. Conversely K computes the summable series, so $\theta_K \equiv_T K$. Its complement and the displayed positive rational affine images are noncomputable right-c.e.; if also left-c.e., the two rational approximations would compute them. Theorem 15.7 supplies these coefficients, giving an intermediate case of Theorem 1.1. The left-c.e. reader gives the first inclusion, and Lemma 2.4, with the limit argument of Corollary 16.7, the dimension outer class. No Δ^0_2 upper bound for the entire usable-entanglement class is inferred here. □

15.6 Why the trace gap is consistent

Proposition 15.9. *For $\widehat{\mathcal{P}}_K$ the exact and approximable minima differ strictly:*

$$\min_{\mathcal{T}} \tau(F) < \min_{\mathcal{T}_{\mathrm{mat}}} \tau(F) = 1 - \theta_K.$$

An exact minimizing trace has a positive-trace nonhalting unflagged corner carrying $\mathcal{A}_{K,e}$. Such a trace is not matrix approximable.

Proof. The bounded exact-trace hierarchy of Proposition 14.1 makes the exact minimum left-c.e.; compactness supplies a minimizer. It is at most the approximable minimum and cannot equal its non-left-c.e. value. At the minimizer, $\tau(I-F) > \theta_K$. Decoder exhaustion and the total capacity of halting cylinders force $\tau(B_e(I-F)) > 0$ for some $e \notin K$. Testing and packing give its unital tracial model. Left and right multiplication in the tracial GNS representation yield a perfect commuting-operator strategy for $H_{K,e}$, using its two projection families; this is the usual tracial representation construction [35, 22]. Soundness concerns finite-dimensional value, so there is no contradiction. □

This is the matrix-versus-commuting distinction behind the negative solution of Connes' embedding problem in [20], made quantitative in the resource coefficient. Slofstra's nonclosure and undecidability results [40] are essential context for distinguishing finite exact perfection from limiting perfection; the finite exact witnesses used above are the stronger specified input, not a consequence of nonclosure.

There is also an independent consistency argument. The index set $C_{\mathrm{tr}} = \{e : \mathcal{A}_{K,e} \text{ has a unital tracial model}\}$ is Π^0_1. Each bounded tracial moment level, including domain localizers and all visible relation insertions, is a rational semialgebraic feasibility problem

decidable by real quantifier elimination. Infeasibility of some level enumerates the complement; feasibility at every level gives a tracial state by bounded-moment compactness and GNS. Since $K \subseteq C_{\mathrm{tr}}$, equality would make K co-c.e. Hence some nonhalting indices, in fact infinitely many, already admit perfect tracial strategies independently of the clock construction. This argument uses the tracial presentation, rather than assuming arbitrary perfect commuting strategies are tracial. Lin's $\mathrm{MIP}^{\mathrm{co}} = \mathrm{coRE}$ [25] supplies the commuting-complexity context; no equality of commuting and finite-dimensional values is used.

15.7 From a c.e. gate to a convergent approximation

The predicate certified by a finite perfect witness is c.e.: finite exact matrix feasibility can be searched by real quantifier elimination. This limits the predicate at one gate, not the exponent of a whole construction. Section 16 keeps these c.e. continuation gates and adds a different scalar tail test after stopping. A compactly encoded stopping parameter is safe only because a bad address is ruled out both at every finite stop and on an infinite continuation. Thus the right-c.e., d.c.e. and general Δ^0_2 realization questions left in version 2.0.0 are all answered by one construction. No subtraction of games or disjunction of all witness candidates is needed. Appendix B and the ancillary README describe the finite checks of the complementary-construction algebra and of the complete construction.

16 Every positive limit-computable exponent

The reader of Section 11 keeps every accepted address. That monotone rule realizes left-c.e. flag densities. We now allow a model to choose when to stop reading, while checking every later rational approximation. Finite quantum witnesses certify each decision to continue. A scalar computation verifies the tail after a finite stop. Together these two tests handle an arbitrary computable convergent approximation, including one of unbounded total variation.

Theorem 16.1 (A flag for a computable limit)**.** *Let $(a_s)_{s\ge 0}$ be a total computable sequence of rationals in $[0,1]$, with limit θ. There is a finite bounded rational presentation $\mathcal{P}[a]$, retaining the three compulsory clocks and the conditional clock, such that*

$$\min_{\tau\in\mathcal{T}_{\mathrm{mat}}(\mathcal{P}[a])} \tau(F) = 1-\theta. \tag{16.1}$$

For every sufficiently large even N it has flat models whose defining operator-norm residuals are at most $3\cdot 4^{-N}$ and whose dimensions satisfy

$$\log_2 d_N = (4-\theta)N + o(N). \tag{16.2}$$

Every new defining equation is exact in these models.

The presentation is specified by the finite program computing (a_s); its real limit is not a coefficient of any relation. We give the presentation and both sides of (16.1) before deriving the game laws.

16.1 A finite program with two different tests

For a binary word w of length j, with integer binary value $p(w)$ in reader order, put

$$\ell(w) = 2^{-j}p(w), \qquad u(w) = 2^{-j}(p(w)+1).$$

The radical-inverse value of every extending address lies between these endpoints. At the empty word they are zero and one. The continuation predicate is the c.e. statement

$$C(j,w) \iff \exists s\ge j \quad a_s > \ell(w). \tag{16.3}$$

Apply Lemma 15.1 to the search machine for this predicate. Its projection presentation $\mathcal{A}_{j,w}$ has separate Alice and Bob families. A true predicate has an exact finite matrix witness; a false one admits no sequence of flat matrix models with defects tending to zero. Lemma 15.2 packs its complete relation list into D, E, W. Computing that list always terminates: it applies the finite game reduction to a machine description, rather than waiting for the search machine to halt.

A stop at (j,w) instead starts an ordinary counter computation of $a_j, a_{j+1}, \ldots$. It enters an absorbing REJECT control when it finds

$$a_s \le u(w), \qquad s\ge j. \tag{16.4}$$

Impose $q_{\mathrm{REJECT}} \le F$. Absence of rejection is an infinite tail condition, not a finite quantum certificate.

Start with Section 11's reader presentation: delete the old recognition program and its halt-flag equation, retain the reader domains and READ updates, and retain all clock, carry, reset, flag and conditional-clock relations. Thus the decoder L, R of Section 5 is absent, and no decoded-region restriction (15.2) is imposed.

For clarity, the inherited scalar workspace consists of all counter coordinates, their zero-test projections and positive-contraction witnesses, the mutually orthogonal control projections summing to I, and $A_{\mathrm{r}}, C_{\mathrm{r}}, P_{\mathrm{r}}$. They commute pairwise and with Z, A_a, A_f, A_t, F. Retain their domains, zero tests (5.8) and reader equation (11.2). For every ordinary counter, control and the reader impose the initialization and transition forms (5.9)–(5.10), using the new finite instruction table; in particular $ZA_{\mathrm{r}} = ZA_a$. Zero-test and reader witnesses satisfy their domain equations at every time and need no separate updates. The flag still commutes with A_a and the complete fuel/time sets, including U. Add two projection data generators D, E and a unitary W, and represent M, Σ by the four unitaries of (15.3). LOAD maps M to I and Σ to zero without reading data on its outgoing edge; other scalar instructions hold them unless an operator instruction updates them.

Add a positive contraction R_* and a counter c initialized by

$$Zc = ZR_*, \qquad U^*cU = Z_+R_* + (I - Z_+)\,\mathrm{Step}(c), \qquad Z_+ = U^*ZU. \tag{16.5}$$

All ordinary counters are initialized at I, the start control at I, the other controls at zero, and the reader at A_a. Counters retain the zero-test relation (5.8). Counter value zero has encoding $c = 1$; a nonzero decrement doubles c. The choice $R_* = 2^{-m}$ therefore means stop at depth m, while $R_* = 0$ continues at every finite depth.

At depth j the program first zero-tests c. Its zero branch starts the scalar verifier (16.4). Its nonzero branch computes the packed list for (16.3), executes one LOAD, tests that entire list, reads the next address bit, increments j and decrements c. A CHECK never branches on a matrix value. Every gate has *exactly one LOAD, before any data-using instruction*; there is no LOAD between its monomials or relations. After LOAD, initialize the monomial and accumulator registers, then use the signed right-multiplication program of Section 15. All classical arithmetic is ordinary counter computation. This algorithm has a fixed finite instruction table, although its computed relation lists and runtimes are unbounded.

The precise new commutations. Require R_* to commute with A_a, F, Z, every fuel and time clock generator including U, every instantaneous variable of the commutative workspace, the data D, E, W, and the four register-representing unitaries. Each data generator commutes with

$$A_a, A_f, A_t, F, Z, R_* \quad \text{and every instantaneous workspace variable.}$$

The four unitaries representing M, Σ by (15.3) have that same coordinate/workspace commutation list. They need not commute with the data or with one another. These are the complete additional lists: in particular, the new data do not inherit Section 15's commutations with U, fuel/time rotations or fuel/time flips. No address-shift or address-flip commutation is imposed on the new data or registers. Their required square-root-state commutators in approximate models are those of Definition 2.1, as for every declared generator.

Instead of time invariance, for $X \in \{D, E, W\}$ impose

$$(U^*XU - X)(I - Z_+)(I - q_{\mathrm{LOAD}}) = 0. \tag{16.6}$$

The tuple at the next time can change only after a current LOAD or at a next-time reset. Use the register domains, initialization and transition formulas (15.3) and (15.4), with the new finite branch selectors, and impose

$$q_{\mathrm{CHECK}}\Sigma(I - F) = 0, \qquad q_{\mathrm{REJECT}} \le F. \tag{16.7}$$

The normalization of signed monomials bounds every partial accumulator by one, even on flagged addresses. All displayed abbreviations have finite rational polynomial definitions. These equations and commutations complete $\mathcal{P}[a]$.

16.2 Why every completed gate tests one tuple

Write $Y_h = U^{-h} Y U^h$. For a finite address word w, choose k and the initialized unflagged projection

$$Q_0 = B_w C_k D_k (I - F). \tag{16.8}$$

The reader follows the specified address bits while it has read at most $|w|$ of them. We consider a prescribed finite computation shorter than 2^k; Lemma 6.1 gives reset at time zero, and Proposition 4.5 gives no later reset during this interval, as in (6.3).

Lemma 16.2 (A fixed tuple between loads)**.** *Suppose the scalar control flow on Q_0 reaches one continuation gate. From immediately after that gate's LOAD through all its CHECKs, every data generator has the same restriction to Q_0. Every CHECK for a packed polynomial π forces $\pi(d_{\mathrm{load}})Q_0 = 0$. If $\tau \in \mathcal{T}_{\mathrm{mat}}(\mathcal{P}[a])$ and $\tau(Q_0) > 0$, the corresponding continuation predicate is true.*

Proof. Shifted variables reduce the initialized corner. Let Y be any data generator, register unitary or instantaneous workspace variable. It commutes with A_a, A_f, A_t, F. The time increment commutes with the address and fuel sets and with F. Proposition 4.5 says that $U^h D_k U^{-h}$ is a sum of finite time-prefix cylinders, hence a spectral function of A_t. The other factors of $U^h Q_0 U^{-h}$ are unchanged. Consequently

$$[Y_h, Q_0] = 0. \tag{16.9}$$

This proves reduction for every shifted variable needed below without assuming $[Y, U] = 0$.

Conjugate (16.6). Off a programmed LOAD and before the next reset, its right guard acts as I on Q_0, giving

$$X_{h+1} Q_0 = X_h Q_0. \tag{16.10}$$

If LOAD occurs at h_L, put $d_{\mathrm{load}} = (D_{h_L+1}, E_{h_L+1}, W_{h_L+1})$. Its restrictions stay fixed throughout the gate. They have their exact domains on the reducing Q_0 corner. In particular the restricted unitary has the same fixed adjoint.

Register induction. Each register is cleared after LOAD. Suppose $S_h Q_0 = P(d_{\mathrm{load}}) Q_0$. Every polynomial in shifted data commutes with Q_0 by (16.9). For a current data letter d_h and a shifted scalar selector $q_{\sigma,h}$ with scalar action σQ_0, the required identities are

$$\begin{aligned} S_h d_h Q_0 &= S_h Q_0 d_{\mathrm{load}} = P(d_{\mathrm{load}}) d_{\mathrm{load}} Q_0, \\ q_{\sigma,h} P' Q_0 &= q_{\sigma,h} Q_0 P' = \sigma P' Q_0, \\ Z_{h+1} P' Q_0 &= Z_{h+1} Q_0 P' = 0, \qquad [P', Q_0] = 0. \end{aligned}$$

They handle multiplication, branch selection and the absent reset term in the register transition. The scalar dyadic coefficient acts by its programmed value on the same corner. We have not commuted a selector through a data word at a different time. Non-self-adjoint initialization is justified by $[S, Z] = 0$: $SQ_0 = SZQ_0 = ZSQ_0 = Z\,\mathrm{Init}(S)Q_0$. Induction therefore gives at each CHECK

$$\Sigma_h Q_0 = 2^{-b} \pi(d_{\mathrm{load}}) Q_0. \tag{16.11}$$

The CHECK selector equals I there, and F is time invariant. Equation (16.7) forces this polynomial to vanish.

Positive corners. If the packed presentation has m projection generators, the packing relation and traciality give

$$\tau(Q_0 E_{h_L+1}) = \tau(Q_0)/m.$$

The projection $Q_0 E_{h_L+1}$ is a fixed polynomial in the full generators, including the finite time and fuel cylinders. If its trace is positive, it carries the recovered Alice and Bob projections of $\mathcal{A}_{j,w}$.

It also inherits matrix approximation. In flat approximants, self-adjointize its polynomial and round at $1/2$. The pointwise inequality $|1_{[1/2,\infty)}(t) - t| \le 2|t^2 - t|$ makes the normalized Hilbert–Schmidt correction vanish; its rank fraction tends to the positive trace above. Compress and round the recovered projection generators, not the packed unitary itself, into that corner. Bounded-product telescoping preserves all their fixed relations. Renormalizing trace costs a factor tending to the inverse square root of that positive trace. This yields flat approximate models of $\mathcal{A}_{j,w}$. Lemma 15.1 excludes them when the predicate is false. We keep $C_k D_k$ in this corner; no commutation of the new data with fuel/time flips is used. □

16.3 The stopping parameter cannot escape to infinity

Lemma 16.3 (Commutative right actions and stop sectors)**.** *On a fixed finite address itinerary in Q_0, the counter and control actions used by the program are Borel functions of R_* times Q_0. If $Q_0\, 1_{\{2^{-i}\}}(R_*) = 0$ for every $i < j$, so that all of Q_0 has continued through the depths below j, then the depth-j zero test at time h_j satisfies*

$$c_{h_j} Q_0 = 2^j R_* Q_0, \qquad Q_0 z_{c,h_j} = Q_0\, 1_{\{2^{-j}\}}(R_*). \tag{16.12}$$

The right-hand projection is the polynomial projection on the left. Every shifted program or data variable reduces it.

Proof. The reader action is $s^i(A_a)Q_0$ after i reads, as in (11.5); its next-bit action is scalar on the specified itinerary. Ordinary counter initialization and updates are unchanged. The exception is c: using $[c, Z] = 0$, $ZQ_0 = Q_0$ and $[R_*, Q_0] = 0$, its initialization gives $cQ_0 = R_* Q_0$.

The scalar induction must now take place in the commutative algebra of functions of R_* on Q_0. If $AQ_0 = f(R_*)Q_0$ and $BQ_0 = g(R_*)Q_0$, then

$$ABQ_0 = AQ_0 g(R_*) = f(R_*)g(R_*)Q_0. \tag{16.13}$$

Thus the polynomial transition maps preserve these right actions. For a counter with such a self-adjoint right action, taking adjoints shows that Q_0 reduces it. The zero-test relation forces $z = 1_{\{1\}}(c)$, since $c = 1$ on z and $0 \le c \le 1/2$ on $I - z$. Functional calculus on Q_0 supplies the next zero-test projection as a Borel function of R_*. Only $(I - z)s$ of the auxiliary witness is used; the unconstrained block zs is never read by the program. All right actions entering the induction are therefore functions of R_*. Control projections may be such functions while different stop choices coexist; they are not assumed scalar at that point.

Conjugating the transition of c in (16.5) kills its reset term on the no-reset interval. On a surviving nonzero branch, c is held fixed until its decrement and then doubled. After j such decrements it acts as $2^j R_*$. The spectral zero-test formula gives (16.12).

Every shifted variable reduces Q_0 by (16.9). It commutes with R_* because its instantaneous variable does and $[U, R_*] = 0$. It therefore reduces each stop sector. On the depth-j sector the stop counter has value one, and the subsequent verifier is the ordinary scalar computation prescribed by that prefix.

No discrete spectrum was assumed for R_*. On a continuing sector the zero-test domain itself forces R_* into $[0, 2^{-(j+1)}]$ after the depth-j atom has been removed. The value zero always continues. These conclusions follow at every finite depth and cover any continuous spectral part as well. □

Proposition 16.4 (The approximable lower bound)**.** *Every $\tau \in \mathcal{T}_{\mathrm{mat}}(\mathcal{P}[a])$ satisfies $\tau(F) \geq 1 - \theta$.*

Proof. Call a word w of length J bad when

$$\ell(w) > \theta, \qquad a_s < \ell(w) \quad (s \geq J). \tag{16.14}$$

Every address with $v(x) > \theta$ has a bad prefix: choose a gap between its value and the limit, then take J large enough both for the prefix lower endpoint to lie inside that gap and for the sequence's entire remaining tail to lie below it.

Fix a bad word. Each possible stop at an ancestor of depth $j \leq J$ has $u(w|j) \geq \ell(w) > \theta$, so its verifier eventually finds a rejection. There are finitely many such stop depths. Their first rejecting runtimes, including preceding continuation work, are finite. Computing each gate list and its instruction stream through depth J also has a fixed finite runtime independent of the data and of R_*. Choose one k with 2^k above all these runtimes.

If $\tau(B_w(I - F)) > 0$, the retained fair fuel/time cylinders give

$$\tau(Q_0) = 2^{-(2k+1)}\tau(B_w(I - F)) > 0. \tag{16.15}$$

Only $B_w(I - F)$, not the new data, must commute with the fuel/time flips for this equality.

Eliminate the stop sectors in increasing order $j = 0, \ldots, J$. Once the sectors of smaller depth are known to vanish, Lemma 16.3 identifies the next one as $Q_j = Q_0 z_{c,h_j}$. On that reducing projection the scalar verifier reaches REJECT before reset. Since $q_{\mathrm{REJECT}} \leq F$ and $Q_j \leq I - F$, it follows that $Q_j = 0$. The continue branch is then forced on the entire remaining Q_0. At the next depth the same argument applies. Intermediate continuation CHECKs do not branch, so their finite control schedule is fixed. After all these stop sectors vanish, the computation must execute the gate for $C(J, w)$.

Condition (16.14) makes that predicate false. Lemma 16.2 contradicts (16.15). Thus $\tau(B_w(I - F)) = 0$ for every bad word, including on a never-stopping sector $R_* = 0$.

For a finite measure argument, fix $\zeta > 0$. The compact set $\{v \geq \theta + \zeta\}$ is covered by finitely many bad cylinders. Refine them to a common address depth. Their orthogonal sum has zero unflagged trace, and fairness bounds the trace of the remaining cylinders by $\theta + \zeta$ plus an arbitrarily small dyadic endpoint error. Let that error and then ζ decrease to zero. We obtain $\tau(I - F) \leq \theta$. The possibly enormous runtime chosen above is used only for exact tracial forcing, not in the quantitative measurement depth. □

16.4 Matched finite models with a fixed witness budget

Lemma 16.5 (Finite threshold models)**.** *For each dyadic $0 < b < \theta$ there are integers m and $\ell_b \geq 1$, independent of N, such that for every even $N \geq m$ the presentation has flat models with all new equations exact, old residuals at most $3 \cdot 4^{-N}$ in operator norm, and*

$$\mathrm{Tr}(F)/d_{N,b} = 1 - b, \qquad d_{N,b} = 2^{3N+k_N}\ell_b, \qquad k_N = (1 - b)N + O(\log(N + 2)). \tag{16.16}$$

Proof. Choose m with $2^m b$ integral and $a_s > b$ for every $s \geq m$. Convergence guarantees this fixed choice. On the N-bit address cycle set

$$I - F = 1_{\{v_N(x) < b\}}, \qquad R_* = 2^{-m} I.$$

Every unflagged address stops after m reads. Its final prefix has $u(w_m) \leq b < a_s$ for all $s \geq m$, so its scalar verifier never rejects. At every preceding depth $j < m$, $\ell(w_j) \leq v_N(x) < b < \theta$, so the continuation predicate is true.

Only finitely many prefix nodes occur on these paths. Choose an exact matrix witness at each, pack both players' families, and tensor their spaces. The resulting common data factor has some finite dimension ℓ_b depending on b, m, not on N. Use the three compulsory N-qubit registers and the optional-slot construction of Theorem 7.1 for this Boolean mask.

For fuel with k initial ones use the actual initialized history at time t mod 2^k; all-ones fuel uses $k = N$. On each address/fuel/time block assign the commutative workspace its scalar history. On unflagged blocks, after a gate's single LOAD assign that gate's packed tuple on its tensor factor and identities on the others, holding it fixed until the next LOAD or reset. From a reset until the first LOAD after it, use the default tuple $(0, I, I)$. On flagged addresses take one default tuple $(D, E, W) = (0, I, I)$ *constant in time*, even at LOAD. CHECK is vacuous there and scalar rejection is permitted.

Assign M, Σ their actual bounded operator histories. Within each block choose their average-of-two-unitary representations as in Section 15; this adds no tensor factor. All declared coordinate/workspace commutations hold because these assignments are block diagonal in the existing address, fuel and time registers. The scalar R_* has every required commutation. We impose no new commutation with a fuel/time rotation or flip.

At an ordinary non-LOAD transition the data are unchanged, so (16.6) is exact. At LOAD its final guard vanishes. At the next-time reset its reset guard vanishes. Since every fuel period divides 2^N, the cyclic boundary is also a reset. Register transitions follow the assigned history, including resets that cut a gate before completion. Each completed unflagged CHECK uses its one exact witness tuple; every unflagged tail comparison passes. Hence all new equations hold exactly at every block and boundary. Address dressing acts only on address and optional slots and commutes with these added assignments. The old residual bounds are therefore unchanged.

In an aligned block of 2^h consecutive address integers, the radical-inverse grid count below b differs from $2^h b$ by at most one. Decompose a length-N interval into at most two aligned blocks of each relevant size. Uniformly over starts, including cyclic intervals,

$$\sum_{i=1}^{N} F(x - i) = (1 - b)N + O(\log(N + 2)).$$

Taking one more optional slot than the largest such count gives (16.16). The finite grid has exactly a fraction b of unflagged addresses because $N \geq m$. □

Proof of Theorem 16.1. Choose dyadics $b_i \uparrow \theta$ and associated m_i, ℓ_i from Lemma 16.5. Choose strictly increasing even thresholds N_i with

$$N_i \geq m_i, \qquad N_i \geq i^2 \max\{1, \log_2 \ell_i\}.$$

For $N_i \leq N < N_{i+1}$ use the b_i model. Its witness overhead divided by N is at most i^{-2}; the uniform discrepancy bound and $b_i \to \theta$ give (16.2). Its flag trace tends to $1 - \theta$. A moment limit attains that value in $\mathcal{T}_{\mathrm{mat}}$, and Proposition 16.4 gives the reverse inequality. When $\theta = 0$, use $F = I$, $R_* = I$ and constant default data directly. The case $\theta = 1$ uses the same thresholds below one. The diagonal choices are existential: the finite presentation is still specified by the program for (a_s), and no effective convergence modulus or uniform witness bound is claimed. □

16.5 The game laws and the positive classification

Theorem 16.6 (Matching dimension and usable entanglement)**.** *For every total computable rational sequence $a_s \to \theta \in [0, 1]$, integers $r \geq 0$ and $q \geq 1$, there is a fixed finite rational*

BCS game $G^{\mathrm{stop}}_{a,r,q}$ of quantum value one, without a finite-dimensional perfect strategy, such that, with $L_\varepsilon = \log_2(1/\varepsilon)$ and $\beta = (4 + r - \theta)/(4q)$,

$$\begin{aligned} \log_2 D_{G^{\mathrm{stop}}_{a,r,q}}(\varepsilon) &= \beta L_\varepsilon + o(L_\varepsilon), \\ d^{\rightarrow}_{G^{\mathrm{stop}}_{a,r,q}}(\varepsilon) &= \beta L_\varepsilon + o(L_\varepsilon). \end{aligned} \tag{16.17}$$

Proof. By Corollary 14.3 and Theorem 16.1, uniformly on every original weighted precision-δ model,

$$\mathrm{Tr}(\rho F) \geq 1 - \theta - \omega(\delta), \qquad \omega(\delta) \longrightarrow 0. \tag{16.18}$$

Otherwise a moment limit with a fixed positive shortfall would contradict (16.1). Append r compulsory clocks and power every defining residual, including all new equations and cross-commutators, as in Proposition 11.3. Retain the exact domains and square-root-state requirements. Jensen gives unpowered precision $\delta = (C_{\mathcal{P},r,q}\sqrt{\varepsilon})^{1/q}$ from a loss-ε strategy.

Set $t = \log_4(1/\delta)$ and $M = \lfloor t - 30\log_4(2 + t) \rfloor$. The retained raw clock measurements have error

$$\eta = C_{\mathcal{P},r,q}(M + 1)^6 4^M \delta = O(t^{-24}).$$

The new data, the stopping parameter R_*, the stop counter and the registers occur in none of those measurements or controls. The same count as in Theorem 8.1, with the extra clocks, gives

$$\begin{aligned} \log_2 d &\geq (3 + r)M + \sum_{i<M} \mathrm{Tr}(\rho F_i) - o(M) \\ &\geq (4 + r - \theta - \omega(\delta))M - O(M^2\delta) - o(M). \end{aligned}$$

The second line uses (16.18) on this same marginal and $|\mathrm{Tr}(\rho F_i) - \mathrm{Tr}(\rho F)| \leq 2i\delta$. Since $M = L_\varepsilon/(4q) - O(\log L_\varepsilon)$, this proves the dimension lower coefficient.

For the other direction, the flat models of Theorem 16.1 gain exactly rN logarithmic dimension from the extra clocks. Every powered operator residual is at most $3^q 4^{-qN}$. The compiler gives loss at most $C'16^{-qN}$ and constant dimension overhead. Neighboring even depths change logarithmic precision by a bounded amount. Their dimension law is $(4 + r - \theta)N + o(N)$, proving the matching upper coefficient at every sufficiently small loss. The upper losses tend to zero and the positive lower bound diverges, so the game has value one and no finite perfect strategy.

For usable entanglement, Proposition 12.2 applies on the original mixed strategy state and its original purifying environment. Its Boolean reconstruction is generic in the finite presentation; time invariance of the additional data is not a hypothesis. Opposite words, independent local inclusion repairs and reverse count order supply Lemma 12.1 with $m = (4 + r)M$ and the same η. Only Q_i is repaired, leaving the actual activation flags. The vector, mean and entropy-continuity errors vanish at this depth. The one-way protocol and hashing yield on every original state

$$E_D^{A\to B}(\rho_{AB}) \geq (3 + r)M + \sum_{i<M} \mathrm{Tr}(\rho_B F_i) - o(1) \geq (4 + r - \theta)M - o(M).$$

Take the infimum, and use $d^{\rightarrow}_G(\varepsilon) \leq \log_2 D_G(\varepsilon)$ for the upper bound. Compression in (16.18) identifies limiting traces; it is not a replacement of the strategy used for extraction. □

Corollary 16.7 (Complete positive dimension spectrum)**.**

$$\mathcal{B}_{\mathrm{BCS}} = \{\alpha > 0 : \alpha \text{ is } \Delta^0_2\}. \tag{16.19}$$

Every real on the right is also the original-state usable-entanglement coefficient of a game in this class. In particular all positive right-c.e. and all positive d.c.e. reals occur.

Proof. For $\alpha > 0$ computable in $0'$, choose a fixed q with $4q\alpha \geq 4$ and put

$$m = \lceil 4q\alpha \rceil, \qquad r = m - 4, \qquad \theta = m - 4q\alpha \in [0, 1).$$

Shoenfield's limit lemma [38] supplies a total computable rational sequence converging to θ; clipping it to $[0, 1]$ preserves convergence. Theorem 16.6 realizes $(4 + r - \theta)/(4q) = \alpha$, for both resources. These fixed integer choices need not be uniform in a limit name for α.

Conversely, Lemma 2.4 makes $D_G(2^{-n})$ computable for a rational finite value-one game. Approximate $\log_2 D_G(2^{-n})/n$ by computable rationals with error tending to zero. If its finite exponent exists, that sequence converges. A $0'$ oracle can find a term with no later deviation larger than a given rational tolerance; convergence guarantees one. Its distance from the limit has that bound. Thus the limit is Δ^0_2. This proves Theorem 1.1. □

Real strategies and question weights. The distributed real simulation of McKague–Mosca–Gisin [27, Section II.C] adds one rebit per party and preserves all measurement probabilities. Real strategies are also complex strategies, so

$$D_{\mathbb{C},G}(\varepsilon) \leq D_{\mathbb{R},G}(\varepsilon) \leq 2D_{\mathbb{C},G}(\varepsilon).$$

The two logarithms differ by at most one. Their exponents therefore coincide whenever either exists, and the dimension classification holds verbatim for real quantum strategies. By the reweighting lemma [12, Lemma 11.12, Section 11.6], exponents and logarithmic resource coefficients are unchanged by any strictly positive reweighting on the same finite support; the dimension classification thus also covers real, even noncomputable, positive question weights.

16.6 Why this passes the earlier barriers

Each continuation gate still tests a c.e. predicate. What changes is the choice of a finite stopping depth followed by a universal scalar tail condition:

$$\exists j \; \forall s \geq j \quad a_s > u(w_j).$$

The arbitrary-length finite histories verify that condition without a finite quantum certificate of the whole tail. An infinite continuation cannot evade a false gate above the limit, by Proposition 16.4. This is the additional mechanism missing from the fixed finite-witness rule of Section 15.

The exact-trace barrier remains intact. If $1 - \theta$ is not left-c.e., Propositions 14.1 and 14.4 give

$$\min_{\mathcal{T}(\mathcal{P}[a])} \tau(F) < \min_{\mathcal{T}_{\mathrm{mat}}(\mathcal{P}[a])} \tau(F) = 1 - \theta.$$

Exact traces can contain tracial models of false continuation gates; matrix approximation excludes precisely those positive corners. Thus even when the quantum and commuting values of the compiled game both equal one, the failure of Connes' embedding conjecture changes the attainable computability class in this construction: its non-left-c.e. coefficients require different exact and matrix-approximable flag minima.

Convergence is the matching hypothesis. For a bounded nonconvergent sequence, the same bad-prefix argument gives a flag lower bound $1 - \limsup_s a_s$, whereas safe finite-stop models approach $1 - \liminf_s a_s$. These bounds need not agree, and no equality or exponent realization for an arbitrary oscillating sequence follows from this argument. The unequal-limit problem is stated separately below. Appendix B and the ancillary README describe finite checks of the new matrix algebra and histories on known-answer toy games.

17 Open spectra and sharper resource laws

The dimension classification concerns a positive finite limit and a fixed rational finite game. It leaves the following questions, including the distinction between dimension and operational entanglement.

Problem 17.1 (Zero without perfection)**.** Can a finite rational value-one BCS game with no finite-dimensional perfect strategy have dimension exponent zero? Can its logarithmic usable coefficient be zero with diverging sublogarithmic usable demand?

Finite-dimensional perfection forces a zero dimension exponent, but this does not settle the converse. The positive lower bounds in our constructions exclude that case without addressing subpolynomial dimension growth in other games. The present architecture forces coefficient at least $3/(4q)$ for every fixed residual power q, from its three compulsory clocks alone. A zero example needs a verifier cheaper than those clocks or a different mechanism; changing flags or taking one larger fixed power cannot produce zero.

Problem 17.2 (The full usable spectrum)**.** Which positive numbers occur as limits $d_G^{\rightarrow}(\varepsilon)/\log_2(1/\varepsilon)$ over finite rational games of value one?

Theorem 1.1 includes every positive Δ_2^0 real. It does not give the opposite inclusion. One-way distillable entanglement is a regularized operational quantity; the dimension feasibility algorithm in Lemma 2.4 does not compute it. The full usable spectrum could be wider than Δ_2^0; compare [12, Theorem 11.14].

Problem 17.3 (Linear systems and groups)**.** Does the positive Δ_2^0 classification hold for linear-system games, with matched usable-entanglement laws? What is the corresponding classification of finite hyperlinear-profile exponents for finitely presented groups?

The present compiler gives Boolean contexts. Converting them to linear systems or solution groups must preserve simultaneous finite upper models and the original-state lower bound, in addition to an algebraic embedding.

Problem 17.4 (Power-law usable demand)**.** Which positive exponents γ occur in $d_G^{\rightarrow}(\varepsilon) = \varepsilon^{-\gamma+o(1)}$ for one fixed finite rational game?

This is an exponent of usable entanglement itself. It is different from the logarithmic coefficients classified for our matched family, and from a power law for local dimension.

Conjecture 17.5 (Upper and lower dimension exponents)**.** Over finite rational value-one BCS games, the positive finite upper dimension exponents are exactly the positive Π_2^0 reals, and the positive finite lower dimension exponents are exactly the positive Σ_2^0 reals, where

$$\overline{\alpha}_G = \limsup_{n\to\infty} \frac{\log_2 D_G(2^{-n})}{n}, \qquad \underline{\alpha}_G = \liminf_{n\to\infty} \frac{\log_2 D_G(2^{-n})}{n}.$$

Here the real-number hierarchy uses the Zheng–Weihrauch convention [46]: a finite limsup of computable rationals is Π_2^0, and a finite liminf is Σ_2^0. Lemma 2.4 makes $D_G(2^{-n})$ computable; approximate its logarithm with rational error at most $1/n$ to obtain the sequence giving the automatic outer bounds. The realization directions are conjectural. Running Section 16 on a nonconvergent approximation only gives potentially different upper and lower flag bounds and does not prove these equalities.

The conjectured upper class differs from Meyerovitch's Π_3 class for upper entropy dimensions of multidimensional SFTs [29, Theorem 5.1(1)]. The distinction is the computability of each finite-precision dimension here; the SFT entropy-dimension input has a different effective approximation. The common limit case is Δ_2^0 (Meyerovitch's Δ_2), as in Theorem 1.1 and Meyerovitch's Theorem 5.1(3).

Problem 17.6 (The full local-heat spectrum)**.** Which positive finite local-reset heat coefficients occur in the block thermodynamic model of Section 13?

Equation (13.7) places this spectrum between the positive Δ_2^0 reals and the positive reals left-c.e. in $0'$. Neither the dimension classification nor the arithmetic upper bound proves equality for heat.

How this work was done

Roles. This paper is the work of its two authors together with current frontier AI models from more than one developer. The authors set the direction throughout: they chose the questions, fixed the scope and the standards of proof and review, decided which lines to pursue and which to stop, and take full responsibility for every claim. The models did most of the exploration, derivation, computation and drafting, in many short working sessions. The written construction was preceded by separate proofs of the finite clock, guarded increment, and obstructions to exact periodic verification. A refutation-first audit then expanded the weighted carry estimates, the joint measurement count, and the simultaneous finite-model bounds. Those written proofs are incorporated here.

Validation. The evidence for the results is of two kinds at this stage: complete written proofs and computations by the scripts in `anc/`, which the reader can check. The review served to find errors; it does not replace the proofs.

- *Proofs.* The new construction and its quantitative estimates are proved in the text and appendices. The Boolean correspondence is invoked from [12, Theorem 4.9], with its local-dimension precision ingredients recorded in Section 10. The conjugacy gadgets are credited in Section 3. Results used for comparison are cited.
- *Proof audit.* A written refutation-first audit read the construction and its preceding clock and guard proofs. It found no fatal step or required change to the defining relations. It expanded the polynomial translation bound, the absence of a product of digit-depth losses, the all-branch measurement estimate and the entropy decoder, and checked the finite models with a disclosed test program.
- *Computation.* The archived finite-model script builds the operators at $N = 4, 6, 8$ with a fixed one-counter recognizer that halts exactly on even inputs. It records every defining relation, required commutation, domain and state condition, and the derived decoding and reset identities. Its defect tables are in `anc/audit/recorded/`. A separate script multiplies the two conjugacy gadgets on finite matrices. No inequality in these tests was violated. Finite checks support the proofs and do not replace them; the arbitrary-state lower bound rests on the written proof.
- *Independent proof check.* A second refutation-first proof check was made by a model of another developer. It wrote its findings on the construction before reading the audit above, and then compared the two. It found no false step and no counterexample. It identified three repairs, each with a proof, and all three are applied here. The orthogonality context of the compiler now gives the projection E its own Boolean variable, defined by an exact AND context (Section 3). The list of joint relations in Lemma 9.3 now states explicitly the commutators between different compulsory clocks and between conditional and fuel or time measurements, and bounds them through the raw conjugated digits. The translation bound (9.4) is now linear in the number of translations, and the compulsory measurements are fixed explicitly. The check also recorded the two structural facts in Remark 9.4. Its

two numerical scripts, for the gadgets, the selector identities and the orthogonality repair, are in `anc/crosscheck/`.

- *Independent computation.* A separately written program, built from the construction without the archived script or its output, evaluates the finite models of Section 7 at $N = 4, 6, 8$. It stores each operator as a sum of weighted basis permutations and uses its own three-counter test program. It evaluates every defining relation of Sections 4 and 5, every required commutation and domain condition, and the identities (6.1): 911, 915 and 919 relations. It computes operator norms, by exact block decomposition, as well as flat norms. Every relation stated to be exact has defect zero. The same ten relations as in the archived script have nonzero defects. The nine that do not depend on the program have the same flat ratios, and the conditional-tail ratio agrees at $N = 4, 8$, where the two test programs have the same K_N. The operator-norm defects are $3 \cdot 4^{-N}$ for the clocks, $\sqrt{2}\,4^{-N}$ for the guarded increments, 4^{-N} for the carry step and at most $0.012 \cdot 4^{-N}$ for (5.15), and the dimension is exactly 2^{3N+k_N}. Its code and recorded output are in the repository directory `verify/`. Like the archived script, it tests the finite models, not the arbitrary-state lower bound or the compiled game.

A Lean formalization of the version 1.0.0 core is in progress and is not complete. The newer sections are not yet formalized. No completed Lean verification is claimed for this paper. The proofs, computations and review are offered for checking as they stand.

Version 2.0.0. The new results came from the same process of model-assisted research and drafting. A refutation-first spectrum audit expanded the reader induction, zero-tail histories, moving-window count, residual-power precision, and the tracial moment and compression arguments. A separate refutation-first usable-entanglement audit checked the inactive-outcome protocol, its original-mixed-state bridge and its operational rate. An independent usable-entanglement check by a model of another developer re-derived those steps and supplied sharper constants. An independent spectrum check by a model of another developer also found no false step; its clarifications on early halting, monotone approximations, domain encoding, arithmetic and synchronous hardness are incorporated. These expansions are in Sections 11–14. The programs and their recorded runs are in `anc/spectrum/` and `anc/extraction/`. They test finite reader models, window counts, compression moments and Jensen, and the actual extraction protocol, including noncommuting controls and mixed inputs. They do not prove the lower bounds. The ancillary README records the commands, archived outputs and limitations of these checks.

Complementary construction. Section 15 was checked by a refutation-first audit and by an independent check by a model of another developer. The latter required one repair, the use of separate projection families for the two players, which is incorporated here. The checks covered the tester induction, its non-self-adjoint register initialization, corner approximation, simultaneous finite models and the address-wraparound gauge. The original-state bridge and extraction proof apply to the new presentation, yielding its usable-entanglement and heat laws in Theorem 15.7. The exact external $\mathrm{MIP}^* = \mathrm{RE}$ theorem statements are cited there. The ancillary README specifies the commands, archived evidence and scope of the complementary-construction checks.

Version 3.0.0. The complete construction in Section 16 underwent a refutation-first proof audit and an independent check by a model of another developer. Both required repairs, incorporated in the written proof. Lemma 16.2 now derives its data identities from the fact that every time-shifted variable commutes with the initialized corner, instead of assuming

that selectors and data at different times commute. Lemma 16.3 extends scalar induction to functions of R_*. The presentation specifies the exact new commutation list, one LOAD per gate before every data use, and time-constant default data on flagged addresses. The threshold models and original-state extraction are proved with these requirements. Four archived scripts in `anc/classification/` test finite histories, stop/continue logic and the new matrix algebra on known-answer toys. They test neither the MIP* games nor a full universal compiler and do not prove the asymptotic lower bounds. Their records, rerun commands and limitations are given in the ancillary README. Meyerovitch's exact entropy-dimension precedent was checked against its primary text and is quoted in Section 1.

Further consequences. The projection-minimum range and recovery consequences were checked against the complete dynamic-presentation list and initialized-corner proof. The scalar domains, control partition, reset transitions, data domains and outgoing LOAD initialization are explicit in Section 16.1. The heat consequence uses the companion paper's general block-recovery theorem; only its demand estimates are supplied here. The entropy-infimum proof treats feasible sets of empty interior by algebraic-point density. The normal form for approximable minima, the real simulation and the precise reweighting citation were checked against their primary texts. These are written arguments and citation checks, not additional numerical validation of a universal compiler.

Priority. For each headline result, bounded searches of arXiv and the general literature compared the statements with the closest work; the credits they found are given at the results. They found no exact preemption of the composed fixed-game exponent, which is not an exhaustive priority verdict. The search included dimension witnesses and hyperlinear profiles, noncomputable exponents, symbolic entropy and frequency realization, computability and asymptotic density, radical-inverse discrepancy, almost synchronous compression, tracial moment hierarchies, one-way hashing, bounded polynomial and Boolean encodings, Cantor adding machines, and computation in finite presentations. The conjugacy gadgets are credited to Fanizza and coauthors; the measurement information bound to Holevo and the random-access-code literature; the operator-assignment interpretation to the constraint-game literature; and the state-preserving Boolean correspondence to the cited companion work. The finite presentation, its simultaneous models, and the matched same-state estimates are the specific construction proved here. For version 2.0.0 the searches also covered realization of computability classes as resource exponents, the MIP* = RE witness and commuting-value literature, and synchronous approximation.

A Why exact periodic verification is insufficient

This appendix records two obstructions that explain the use of approximate clocks and vanishing guard amplitudes. Neither is a general impossibility theorem for finite presentations. For the distinction between locally invariant finite marginals and globally extendible measures, see Chazottes, Gambaudo, Hochman and Ugalde [3]; the compactness argument below uses all box sizes, not extension of one finite marginal.

Proposition A.1 (Periodic extrema)**.** *Let X be a nonempty shift of finite type on $\mathbb{Z}^r$ and f a rational local observable. Put $\alpha = \min_\mu \int f \, d\mu$, where μ ranges over invariant probability measures. If averages of periodic configurations approach α, then α is computable.*

Proof. On each finite box, consider probability distributions on locally admissible patterns and impose every translation-invariance equation whose two supports stay in the box. This is a finite rational polytope. Minimize the expected value of f at the origin, obtaining a computable

rational lower bound ℓ_N. Restriction from larger boxes makes these bounds monotone. A convergent subsequence of minimizers has consistent marginals on every finite support, gives an invariant measure on X, and has expectation $\lim \ell_N$. Thus $\ell_N \uparrow \alpha$. Periodic configurations and their finite fundamental domains are enumerable; local validity and their rational averages are decidable. Their successive minimum averages give upper bounds converging to their infimum. By hypothesis that infimum is α. Search until the upper and lower bounds differ by the desired error. □

Hochman–Meyerovitch's noncomputable extremal frequencies [17] therefore cannot be combined with arbitrarily good exact periodic models of those minima. In the present upper construction the workspace histories close as labels, but the complete clocks remain approximate. Proposition A.1 does not apply to those clocks.

Theorem A.2 (Extraction under a fixed rejection gap)**.** *Suppose $\mathbb{C}^d = \bigoplus_{v\in\mathcal{V}} \mathcal{H}_v$, with $M = |\mathcal{V}|$ nonzero sector projections P_v. Unitaries $U_1, \dots, U_s$ permute the sectors exactly and satisfy $\|[U_i, \xi]\|_2 \le \delta$. For forbidden sets $B_j \subseteq \mathcal{V}$, $1 \le j \le r$, suppose*

$$D_j^* D_j \ge \kappa P_{B_j}, \qquad |D_j|_\rho \le \delta, \qquad P_{B_j} = \sum_{v\in B_j} P_v,$$

where $\kappa > 0$. Let $F = \sum_v c_v P_v$, $0 \le c_v \le 1$, and $a = M\delta$. If $2(s + r/\kappa)a^2 < 1$, a transition orbit avoiding all forbidden sets has average cost at most

$$\mathrm{Tr}(\rho F) + \sqrt{2s}\, a + 2(s + r/\kappa)a^2. \tag{A.1}$$

Proof. Let $p_v = \mathrm{Tr}(\rho P_v)$ and $q_v = \sqrt{p_v}$. If π_i is the sector permutation of U_i, the reverse triangle inequality applied to the row blocks $P_v\xi$ and $P_v U_i \xi U_i^*$ gives

$$\sum_v (q_v - q_{\pi_i(v)})^2 \le \delta^2.$$

The union B of forbidden sets has $p(B) \le (r/\kappa)\delta^2$. For an orbit O, let $E_O = \sum_i \sum_{v\in O}(q_v - q_{\pi_i(v)})^2$, so $\sum_O E_O \le s\delta^2$. A simple path in the connected orbit graph has at most $|O| - 1$ edges; Cauchy–Schwarz gives

$$(q_v - q_w)^2 \le (|O| - 1)E_O.$$

Choose a bad vertex w_O in each invalid orbit. Then

$$\sum_{v\in O} q_v^2 \le 2M q_{w_O}^2 + 2M^2 E_O.$$

Summing shows that the mass b of invalid orbits is at most $2(s + r/\kappa)a^2$.

Average p uniformly within each orbit, obtaining $\bar{p}$, and put $u_v = \sqrt{\bar{p}_v}$. The variance identity and the path bound give

$$\sum_{v\in O} (q_v - \bar{q}_O)^2 \le M^2 E_O,$$

where $\bar{q}_O$ is the ordinary average of q on O. If the orbit has mass $p_O > 0$, set $t_O = (\sum_{v\in O} q_v)/\sqrt{|O| p_O}$. Then

$$\sum_{v\in O} (q_v - u_v)^2 = 2p_O(1 - t_O) \le 2p_O(1 - t_O^2) = 2\sum_{v\in O} (q_v - \bar{q}_O)^2.$$

Consequently $\|q - u\|_2 \le \sqrt{2s}\, M\delta$. Since $\mathrm{TV}(p, \bar{p}) \le \frac{1}{2}\|q - u\|_2 \|q + u\|_2 \le \|q - u\|_2$, orbit averaging changes expected cost by at most this amount. Conditioning $\bar{p}$ on valid orbits changes total variation by $b < 1$. The result is a mixture of uniform valid-orbit laws, one of which has cost at most (A.1). □

Corollary A.3. *Assume valid finite orbits are effectively enumerable, have computable rational costs, and every such cost is at least a left-c.e. real θ. If near-minimizing models have flag cost tending to θ and $M\delta \to 0$, then θ is computable. If θ is uncomputable and flat models have flag cost tending to θ, their dimensions eventually satisfy $d \geq \kappa\delta^{-2}$.*

Proof. Theorem A.2 makes the infimum of valid finite-orbit costs equal to θ under the first hypothesis. Enumerating those costs gives computable upper bounds tending to θ; left semicomputability gives lower bounds. This computes θ.

More generally, for uncomputable θ the infimum θ_{fin} of valid finite-orbit costs must be strictly greater than θ, or there are no valid finite orbits: equality would already make θ computable by the same enumeration. If a flat model had no nonzero forbidden projection, every orbit would be valid. Exact sector permutation makes ranks constant on an orbit, so its flat cost would be a mixture of uniform orbit costs, at least θ_{fin}. Thus near-minimizing flat models eventually have a nonzero P_{B_j}. The rejection gap gives

$$\delta^2 \geq d^{-1} \operatorname{Tr}(D_j^* D_j) \geq \kappa/d.$$

This proves the claimed dimension bound. □

The semantic assumption about all valid finite-orbit costs is essential. A program orbit may fail to extend to the full clock system, or may have a cheap flag cost that the extra clock relations exclude. The corollary does not apply automatically to every operator presentation. Its flat bound explains the amplitude issue: the target dimensions here are below order δ^{-2}, so a construction satisfying these hypotheses could not use fixed-gap bad sectors. Our increment defect instead has amplitude 4^{-N} at overflow, as Proposition 4.5 proves. The initialization proof uses exact tracial cancellation and compactness, not a fixed-gap finite-sector extraction.

B Finite computations and reproducibility

The code in `anc/audit/check_models.py` represents an operator as a finite sum of weighted basis permutations. For each defining relation it enumerates all basis columns of the tensor factors that occur in that relation. The absent factors are identities, so averaging squared column norms computes exactly the full-space normalized Hilbert–Schmidt expression, before floating-point rounding. Equal output rows are combined before squaring, avoiding cancellation of large squared terms. The periodic coordinates in (7.6) are summed with integer numerators and denominator $4^{\ell} - 1$ for the permutation period ℓ, then rounded once to floating point.

The test program has one counter and five controls. From the starting control, zero halts, while nonzero decrements and moves to an odd-parity control. There zero enters an absorbing nonhalting control, while nonzero decrements and returns to start. Idle and halt are absorbing. Hence input e halts exactly when it is even, at step $e + 1$. This is a test of the finite-model mechanism, not a computation of the universal halting set.

The remaining defining defects are zero to within 10^{-14} in these tests. Complete per-relation values and their reference scales are archived, not just the maxima in Table 1. The two guarded increments have identical ratios in this example. The theorem's operator-norm bounds are proved in Section 7; a script's maximum column norm is not generally an operator norm. The displayed dimensions are represented without dense matrices. The original $N = 8$ run had an observed peak working set about 1.16 GB.

The gadget script `anc/gadgets/gadget_check.py` tests self-adjointness, squares, all pairwise commutations within each context, context products, and the auxiliary conjugacies on random finite instances. It is a numerical check of the explicit extension, not a proof of the converse; that converse is proved in Section 3.

Table 1: Finite-model checks for the disclosed test program. Each ratio is a flat Hilbert–Schmidt defect divided by 4^{-N}. The first two rows apply separately to all three compulsory clocks.

Quantity	$N = 4$	$N = 6$	$N = 8$
Clock affine ratio	$3/\sqrt{2}$	$3/\sqrt{2}$	$3/\sqrt{2}$
Clock tail-flip ratio	3	3	3
Guarded increment ratio	0.353553	0.176777	0.088388
Carry-step ratio	0.25	0.125	0.0625
Conditional tail ratio	0.00585946	0.000160228	0.00256366
K_N	$\{0\}$	$\{0, 2\}$	$\{0, 2\}$
k_N	2	4	4
Dimension	16 384	4 194 304	268 435 456
Recorded checks	390	394	398

The ancillary README gives commands, dependencies, recorded outputs and limitations. All computations are run with one numerical-library thread. The finite-model script caps its exhaustive workspace implementation at $N = 8$ to stay within the stated memory budget. The independent engine described in the methods section is in the repository directory `verify/`, with its recorded output, and the two scripts of the independent proof check are in `anc/crosscheck/`.

Version 2 checks. The spectrum program in `anc/spectrum/` builds 24 finite reader models at $N = 2, 4, 6$ and eight rational thresholds, including both endpoints. It tests reset boundaries, reads beyond N, halt inclusion, slot collisions, dressing and defining operator residuals on the nontrivial kernels. Its other tests cover radical-inverse windows, compression with domain repairs, finite word moments and noncommuting-state Jensen. The protocol programs in `anc/extraction/` evaluate the stage inequality, full exact and perturbed protocols with mixed inputs and noncommuting controls, small perturbations with positive lower bounds, bounded adversarial searches, and one-sided inclusion repair. Their recorded outputs are preserved in the respective `recorded/` directories. The ancillary README states each file's scope and distinguishes the original allowances from the constants of Lemma 12.1.

Complementary-construction checks. The script `anc/gate/toy_gate.py` retains its original recorded results in `gate/recorded/` and writes a new result beside the script. It executes the signed packed polynomial instructions for three-colouring of K_3, a pair of noncommuting binary bases, and a flagged bad three-colouring of K_4. It checks contraction-register decompositions, initialized finite histories through every distinct reset period, cyclic boundaries, simultaneous data factors, decoder and reset recursions, complementary window counts and address dressing. This script was written for an earlier presentation with a single projection family for both players; its history, reset and window tests do not depend on that choice and cover 16,893 history transitions, 35 cyclic reset boundaries and 2,396,744 mask/start cases across 254 masks. The separate two-family suite in `anc/gate/crosscheck/` tests the presentation of Section 15.1, with separate families for the two players: K_3 colouring has 18 generators, and the magic-square witness has 42 generators and packed dimension 168. It also checks the reduced-state eigenspaces, an anti-synchronous PCC control, 53,248 history points, sparse address models and windows through depth 22. Original logs and rerun commands are specified in the ancillary README. These are tests of the new matrix algebra and finite histories on known-answer toy games, not the MIP* games themselves. They do not certify the halting reduction, arbitrary-state lower bound or compiled game.

Complete-classification checks. The four scripts in `anc/classification/` test the new stop-or-continue algebra on finite known-answer instances. They cover oscillating rational approximations, finite stopping parameters, tail rejection, per-gate LOAD boundaries, matrix-valued registers, noncommuting witness tuples, reset periods, flagged default data and finite threshold/window budgets. The operator-level check evaluates the new equations as sparse operators built from a toy history, with negative controls for an omitted LOAD factor and imposed time invariance. The matrix-interface check displays the shifted-corner and stop-sector identities on a hand-built instance, in which the stop-sector identity holds by construction, with a negative control for dropped coordinate commutation. The toy scripts add negative controls that change data between the checks of a false gate (`toy_classification.py`) or one step before a LOAD (`toy_attack.py`). The two corrected register checks initialize $M = I$ and $\Sigma = 0$ on the outgoing LOAD edge. The operator-level script also checks a synthetic LOAD with incoming $M = 0$, $\Sigma = I$, testing both the instruction history and the polynomial selectors against that transition; holding the incoming registers gives nonzero negative controls. Their recorded outputs are archived and hashed. They do not instantiate the MIP* reduction, synthesize a universal counter compiler, certify a noncomputable limit, or test the full compiled game. The lower bound and the nonuniform slow-diagonal upper bound are written proofs in Section 16, not conclusions of these finite tests.

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